terrasphere_csem: validation

October 16, 2026

Validation of the series solution of this package against two independent codes: the 1-D code empymod for the flat model, and the 3-D finite-element code elfe3D for Gaussian topography (hills and depressions of ±30 m and ±90 m, width 150 m) and for a conductive sphere under the 90 m hill. Four sources (grounded wire in line and broadside, offset loop, large loop around the hill), three layers with interfaces at 200 m and 600 m depth, receivers on the x- and y-profile from −600 m to 600 m every 20 m, frequencies 0.1 Hz, 10 Hz and 1 kHz; the flat model also in the time domain (switch-off, 10 μs to 10 s, every 300 m). The four reports follow in full; each ends with its table of deviations.

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Conventions

Time dependence of this package: $e^{-i\omega t}$; frame (x, y, z) right-handed with z up; quasi-static (no displacement currents). The conversions below are applied in all comparisons of this report and of the paper.

empymod

empymod includes displacement currents by default (relative permittivity 1 in every layer), whereas this package is quasi-static; the two then differ by about $\omega\epsilon_0/\sigma$ (10^{-5} at 1 kHz in 0.005 S/m), which first showed up as a frequency-proportional residual of the first-order null test. All comparisons use empymod with `epermH = epermV = 0`. In that mode its air layer is a near-perfect insulator, and a source or receiver lying exactly on an interface, which empymod assigns to the layer above (for $z = 0$: the air), is moved 1 μm into the layer below, where this package places it; the displacement itself changes the fields by about 10^{-8}, which sets the level of the 1-D comparisons with surface sources (fields in the ground from a source in the air are unaffected, so the Born test uses no displacement). Further, empymod uses $e^{+i\omega t}$ and a depth frame (z down) that is left-handed with respect to our z-up frame, follows Hunziker et al. (2015) in normalising magnetic sources to a unit magnetic current moment[1], and defines its vertical magnetic dipole (`dip=90`) pointing down. Established against closed-

[1] Hunziker, Thorbecke & Slob (2015), Geophysics 80(1), F1–F18, p. F2: "The configuration has an electric or magnetic current type source located in a layer of finite thickness"; the Green's functions are the fields of a unit point source of either type.

form full-space dipole fields and used in all tests:

$$\mathbf{E}_{\text{ours}} = f\,\overline{\mathbf{E}}_{\text{emp}}\circ(1,1,-1), \qquad \mathbf{H}_{\text{ours}} = -f\,\overline{\mathbf{H}}_{\text{emp}}\circ(1,1,-1), \qquad f = \begin{cases} 1 & \text{HED} \\ i\omega\mu_0 & \text{HMD} \\ -i\omega\mu_0 & \text{VMD (our moment up)} \end{cases}$$

where the bar is the complex conjugate and $\circ$ the component-wise product (polar vectors flip their z-component under the frame change, axial vectors their horizontal components, which gives the extra sign of $\mathbf{H}$). For loop sources at the surface empymod places the wire in the air; for a horizontal electric dipole exactly on the surface we place it in the ground, and the two agree to 10^{-6}. A receiver on the wire of a coincident loop cannot be evaluated by empymod (the segment ends produce a static near field), and its Fourier route loses accuracy at late times; Key's 401-point Hankel filter with exact frequencies was used for the comparisons at the 10^{-4} level. With its default settings (displacement currents, grounded wire in the air layer) empymod describes a different problem from the quasi-static grounded wire used here: its galvanic electric field differs below 0.1 Hz (by a factor 20 at 10^{-4} Hz) and its switch-off response is not comparable; quasi-static and with the wire in the ground it agrees with ours to 10^{-6} down to 10^{-4} Hz.

elfe3D

elfe3D was used in the version v1.0.1 plus 16 commits, commit 58528bd7 of its public repository (`git describe`: v1.0.1-16-g58528bd), compiled with gfortran against MUMPS 5.8.2 and meshed with TetGen 1.5.0, the version recommended by its authors. The build reproduces the example shipped with the repository (line source over a half-space with a conductive block, three frequencies, 73 receivers) to 4×10^{-6} of the profile maximum. The repository example uses "z positive upwards"[2], which coincides with our component frame; receiver positions are therefore entered as $z = -d$. The time convention was determined by running the same example with the block set to the host resistivity, a homogeneous half-space that this package computes exactly: the ratio of the two electric fields is real and independent of frequency (0.965 at 1, 10 and 100 Hz on the unrefined example mesh), whereas the ratio to the complex conjugate acquires a phase that grows with frequency (19° at 100 Hz). The output of elfe3D thus follows $e^{-i\omega t}$, as this package. This agrees with the system matrix in the code, $\mu^{-1}K - i\omega\rho^{-1}M$ with right-hand side $+i\omega$ times the source term[3], i.e. $\nabla \times \mu^{-1}\nabla \times \mathbf{E} - i\omega\sigma\mathbf{E} = i\omega\mathbf{J}$, although the thesis of the main author formulates the method with $e^{+i\omega t}$[4]. The magnetic field, however, is formed with the $e^{+i\omega t}$ form of Faraday's law, $\mathbf{H} = -(i\omega\mu)^{-1}\nabla \times \mathbf{E}$[5], so that the output $\mathbf{H}$ carries the opposite sign of the field belonging to the output $\mathbf{E}$; in the half-space test all three magnetic components have ratio -1 and all electric components $+1$, which rules out a frame reflection (that would flip E_z and H_z against the horizontal components). Our own $\mathbf{H}$ satisfies $\nabla \times \mathbf{E} = i\omega\mu_0\mathbf{H}$ to 10^{-4} in a finite-difference check. Finally, elfe3D distributes the source current as source moment divided by the total length of the source edges; with the moment set to current times wire length (or loop perimeter) both codes carry the same current. Hence

$$\mathbf{E}_{\text{elfe3D}} = \mathbf{E}_{\text{ours}}, \qquad \mathbf{H}_{\text{elfe3D}} = -\mathbf{H}_{\text{ours}},$$

without complex conjugation and without reflection.

Two properties of the source definition matter for loops. elfe3D marks a mesh edge as a source edge if both of its end points lie on a segment of the source polygon to within the machine epsilon, 2.2×10^{-16} m in absolute terms[6]. With the optimisation switches of the distributed makefile (`-Ofast`

[2] elfe3D, `elfe3D/in/README.md` (commit 58528bd7): "modelling domain: 30 x 30 x 30 km / z positive upwards".

[3] elfe3D commit 58528bd7, `elfe3d.f90`, line 594: `AA = cmplx((1/mu(l))*K, -w*(1/rho(l))*MM, kind=dp)`; line 659: `Bg = cmplx(0.0_dp, (edl*source*w*p)/(source_length),kind=dp)`.

[4] Rulff (2023), PhD thesis, Uppsala University, p. 27: "Note, that the BVP and all subsequent equations are formulated with a time dependency of $e^{-i\omega t}$ in Paper I, whereas the implementation had to be changed to a time dependency of $e^{i\omega t}$ at a later stage (Papers III & IV) for compatibility reasons."

[5] elfe3D commit 58528bd7, `mod_calculate_tf.f90`, line 129: `factor = cmplx(0.0,(1/(w*mu(rec1_el))), kind=dp)`, applied to $\sum_j E_j \nabla \times \mathbf{N}_j$; the comment reads "H_rec1 = -1/(iwmu)sum(Ej curl (Nj)".

[6] elfe3D commit 58528bd7, `mod_calculate_global_source.f90`, function `dist2segment` (distance via a cross product compared with `eps_dp`, then the exact test `p1p2 .ne. p1cp + p2cp`) and `mod_constant.f90`, line 50: `eps_dp = abs(epsilon(1.0_dp))`.

`-mfma`) this test is not exact for oblique segments: a copy of the routine compiled with these switches finds only 52 of the 64 sides of a 64-gon, compiled with `-O2` all of them. Our sources are therefore axis-parallel chains of 10 m segments with integer corner coordinates, for which all products are exact: the wires and the two rectangular loops of the validation cases. Further, for the horizontal segmented loop (source type 7) the code runs over the segments 1 to $n-1$ only; the closing segment from the last to the first corner is added for the vertical loop (type 9) alone[7]. We therefore repeat the first corner at the end of the corner list; an open loop would carry charges at its gap whose galvanic field dominates the horizontal components. With both measures elfe3D reports source lengths of 1600 m and 10 400 m in all runs, the perimeters of the two loops.

References

empymod (flat model; version 2.6.0, quasi-static, digital filter key_201_2009; time domain: Fourier filter key_601_2009)

- Werthmüller, D. (2017). An open-source full 3D electromagnetic modeler for 1D VTI media in Python: empymod. *Geophysics*, 82(6), WB9–WB19. doi:10.1190/geo2016-0626.1
- Key, K. (2009). 1D inversion of multicomponent, multifrequency marine CSEM data: Methodology and synthetic studies for resolving thin resistive layers. *Geophysics*, 74(2), F9–F20. doi:10.1190/1.3058434 (Hankel filters)

elfe3D (topography and sphere; `https://github.com/emsig/elfe3d`, v1.0.1 plus 16 commits, commit 58528bd7 of 21 July 2026; Zenodo concept DOI 10.5281/zenodo.13309721)

- Rulff, P. (2025). elfe3D v1.0.1: Modelling with the total electric field approach using finite elements in 3D. *Journal of Open Source Software*, 10(110), 7949. doi:10.21105/joss.07949
- Rulff, P., Buntin, L. M., Kalscheuer, T. (2021). Efficient goal-oriented mesh refinement in 3-D finite-element modelling adapted for controlled source electromagnetic surveys. *Geophysical Journal International*, 227, 1624–1645. doi:10.1093/gji/ggab264

TetGen (mesh generator used by elfe3D; version 1.5.0)

- Si, H. (2015). TetGen, a Delaunay-based quality tetrahedral mesh generator. *ACM Transactions on Mathematical Software*, 41(2), 11:1–11:36. doi:10.1145/2629697

MUMPS (direct solver used by elfe3D; version 5.8.2, sequential, with the orderings METIS 5.1.0 and SCOTCH 7.0.11 and OpenBLAS 0.3.34)

- Amestoy, P. R., Duff, I. S., L'Excellent, J.-Y., Koster, J. (2001). A fully asynchronous multifrontal solver using distributed dynamic scheduling. *SIAM Journal on Matrix Analysis and Applications*, 23(1), 15–41. doi:10.1137/S0895479899358194
- Amestoy, P. R., Buttari, A., L'Excellent, J.-Y., Mary, T. (2019). Performance and scalability of the block low-rank multifrontal factorization on multicore architectures. *ACM Transactions on Mathematical Software*, 45(1), 2:1–2:26. doi:10.1145/3242094

[7] elfe3D commit 58528bd7, `mod_calculate_global_source.f90`, subroutine `segmented_source`: case(7) loops `do i = 1,num_segments-1` with no closing segment; case(9) ends with "! last segment to close the loop".

terrasphere_csem: flat model against empymod

October 14, 2026

Series solution of this package for the flat model (lines) against the 1-D code empymod (circles) for the four sources of the elfe3D comparison. Three layers with interfaces at 200 m and 600 m depth. Receivers on the x- and y-profile from −600 m to 600 m every 20 m: series on the surface (earth side), empymod sources and receivers 1 mm below it. empymod quasi-static (no displacement currents), digital filter key_201_2009, finite bipoles (40 Gauss points each; a loop as its four sides). Fields per 1 A source current, E in mV/m, H in μA/m, time dependence $e^{-i\omega t}$. Components that vanish on a profile by symmetry are plotted as they come out (noise below 10^{-10} of the field maximum in both codes; a power of ten of an axis is part of its label).

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1 T1_wire_inline, x-profile

empymod comparison (flat model) T1: grounded wire in line with the x-profile, 500 m, centre 1500 m left of the hill.

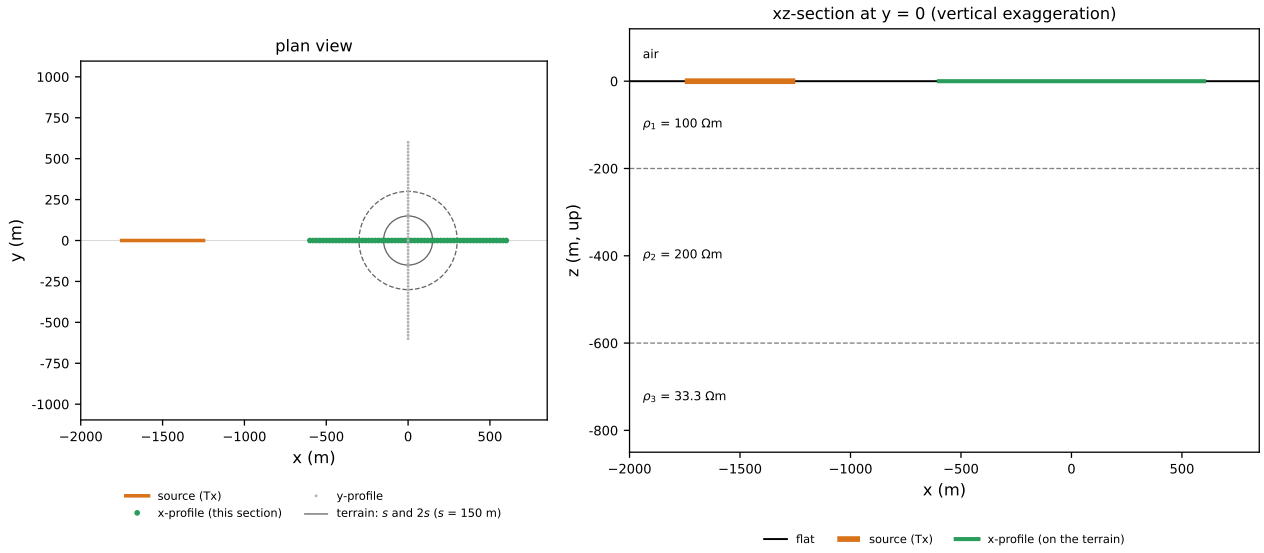


Figure 1: Setup of T1_wire_inline, x-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

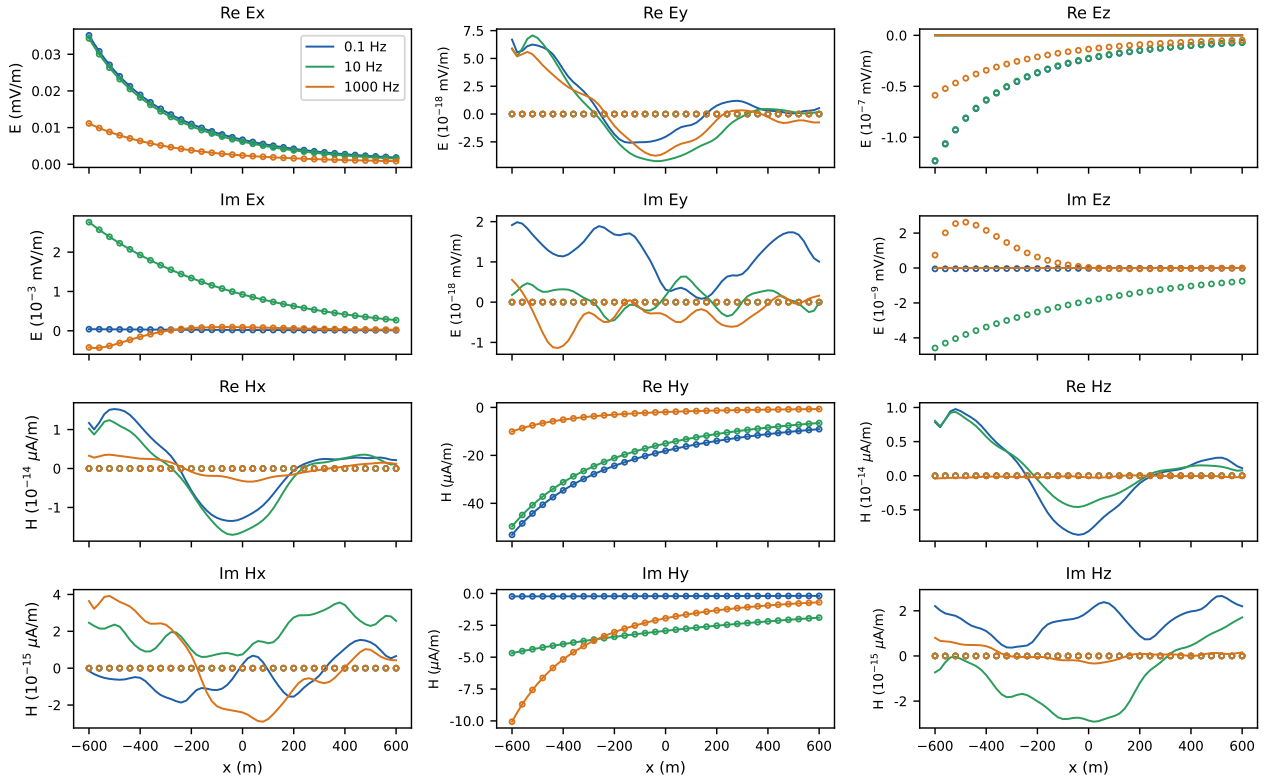


Figure 2: T1_wire_inline, x-profile, flat model: series (lines) and empymod (circles, every second receiver).

2 T1_wire_inline, y-profile

empymod comparison (flat model) T1: grounded wire in line with the x-profile, 500 m, centre 1500 m left of the hill.

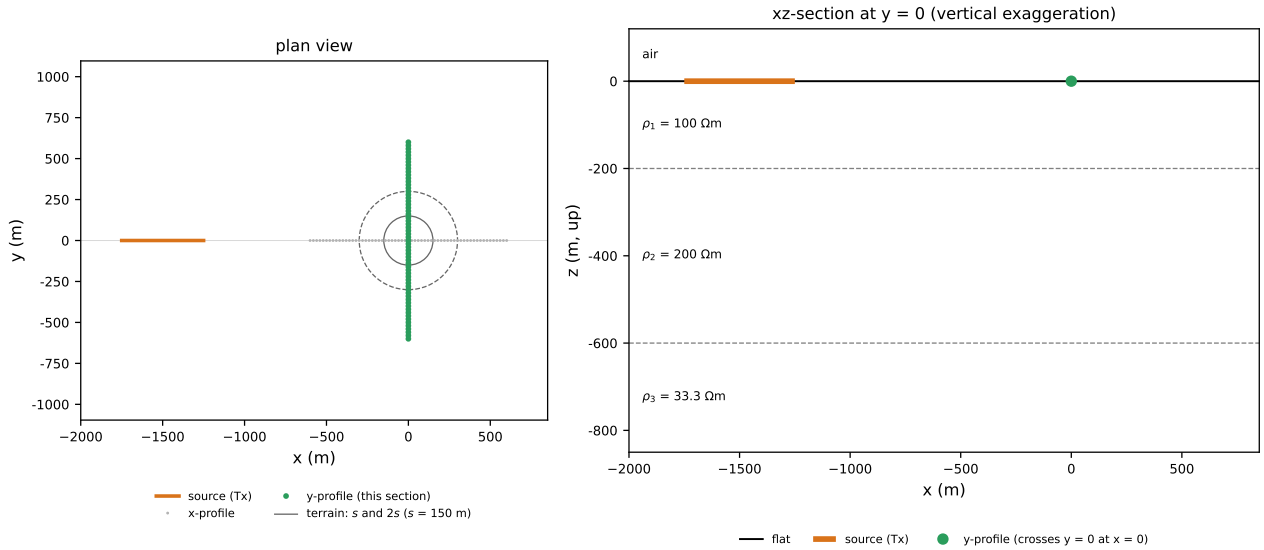


Figure 3: Setup of T1_wire_inline, y-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

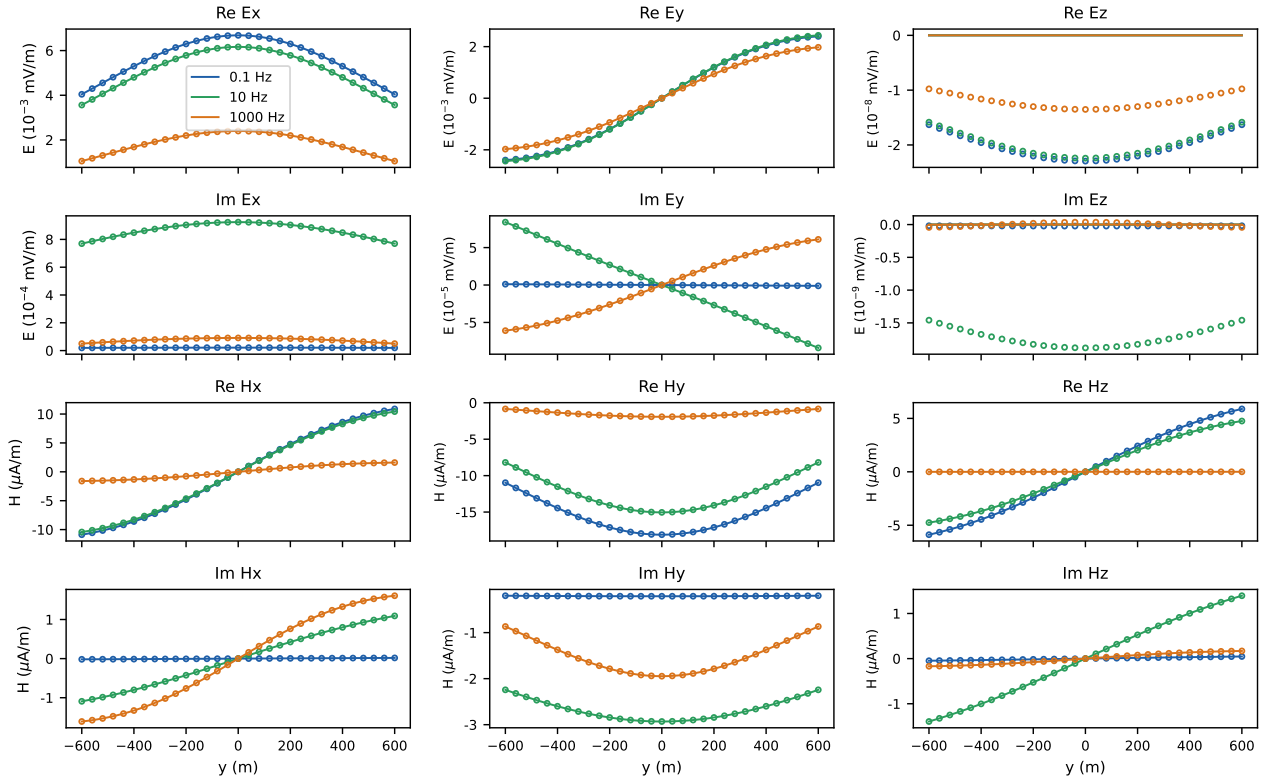


Figure 4: T1_wire_inline, y-profile, flat model: series (lines) and empymod (circles, every second receiver).

3 T2_wire_broadside, x-profile

empymod comparison (flat model) T2: grounded wire broadside (along y), 500 m, centre 1500 m left of the hill.

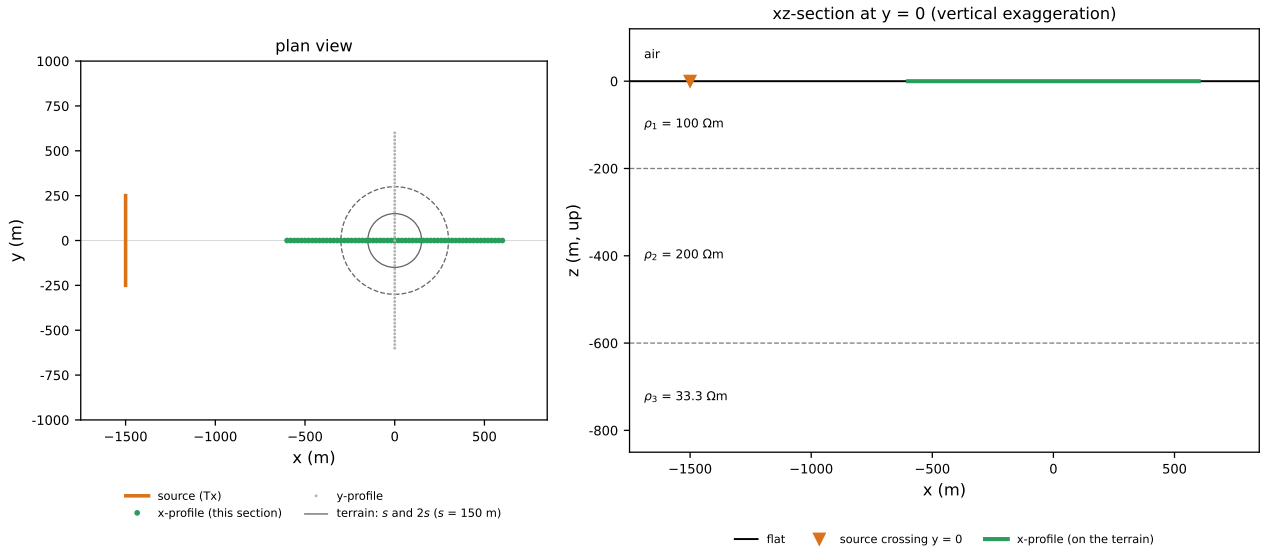


Figure 5: Setup of T2_wire_broadside, x-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

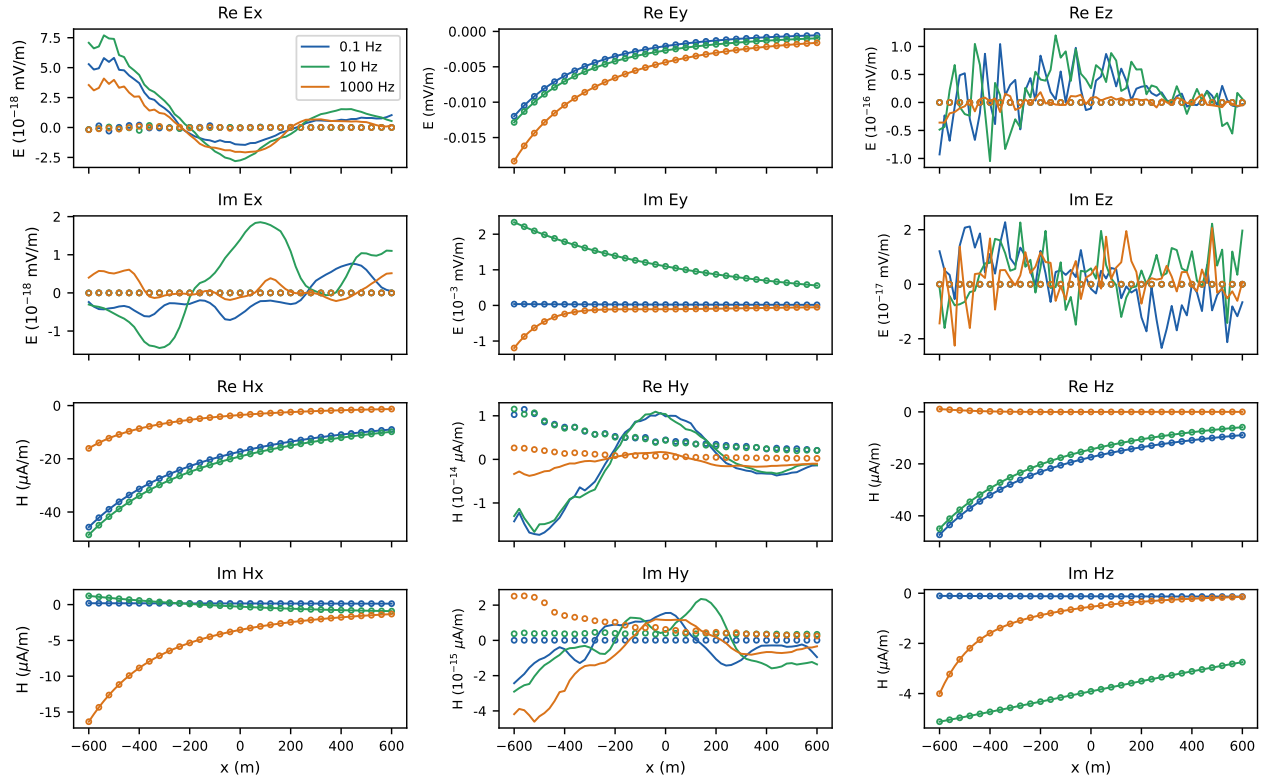


Figure 6: T2_wire_broadside, x-profile, flat model: series (lines) and empymod (circles, every second receiver).

4 T2_wire_broadside, y-profile

empymod comparison (flat model) T2: grounded wire broadside (along y), 500 m, centre 1500 m left of the hill.

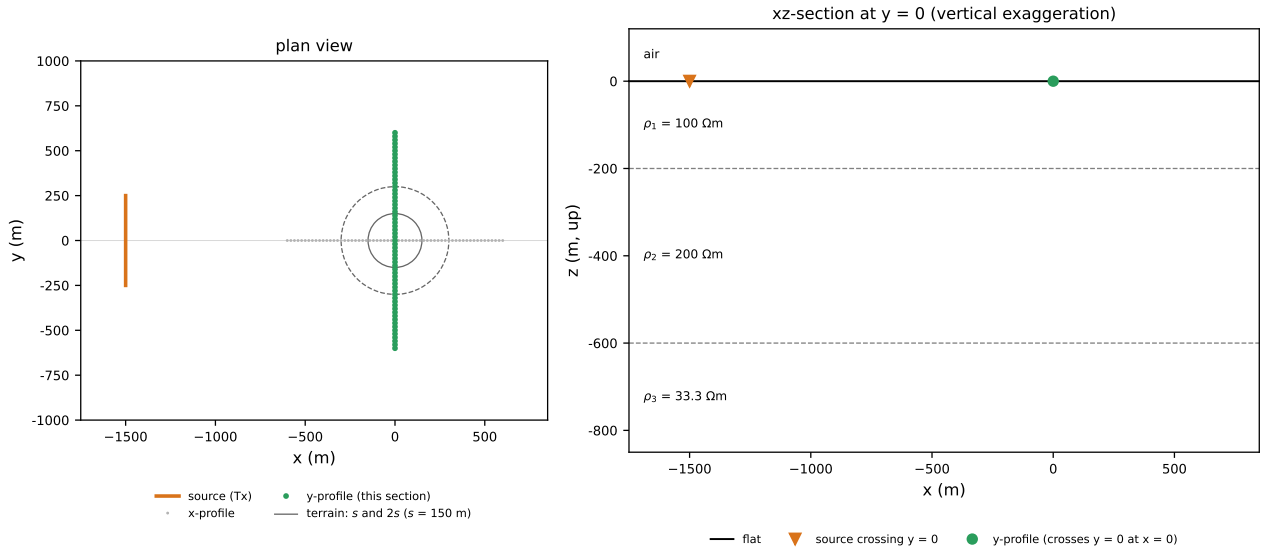


Figure 7: Setup of T2_wire_broadside, y-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

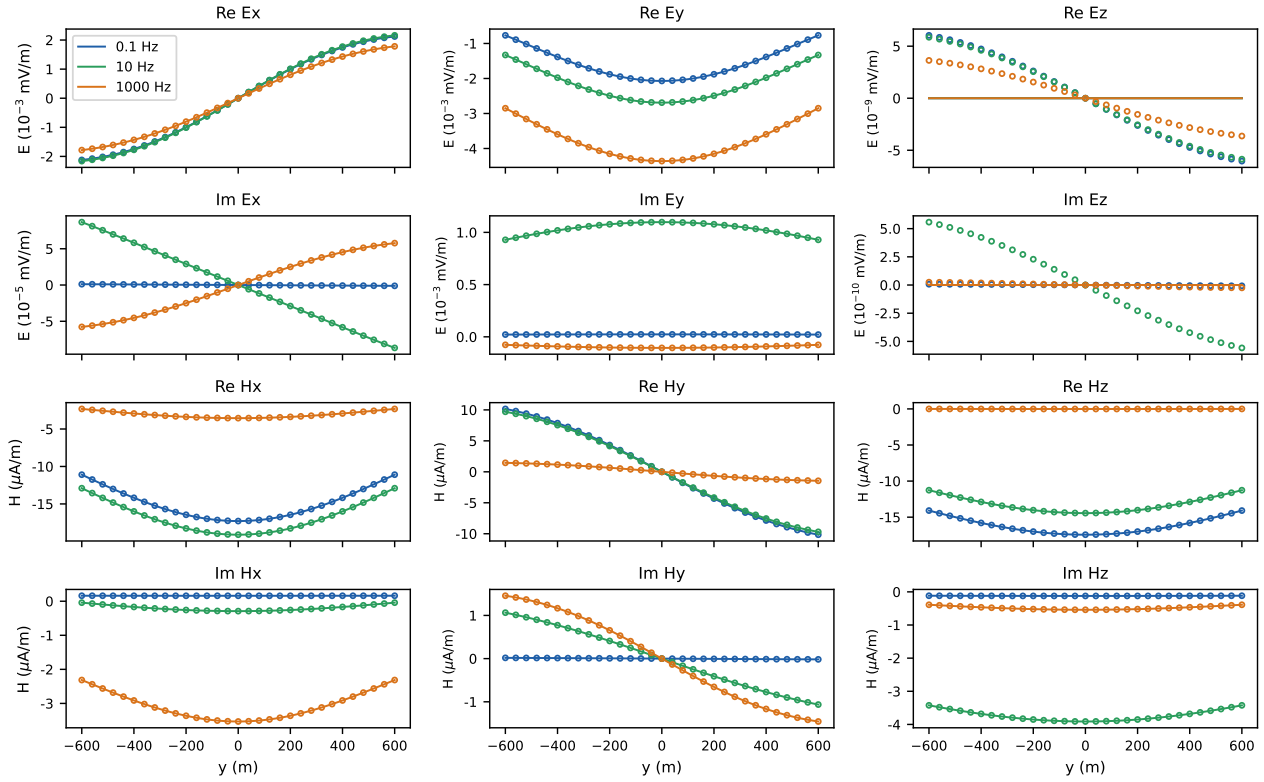


Figure 8: T2_wire_broadside, y-profile, flat model: series (lines) and empymod (circles, every second receiver).

5 T3_loop_offset, x-profile

empymod comparison (flat model) T3: square loop 400 x 400 m, centre (-1500, 0), counter-clockwise seen from above.

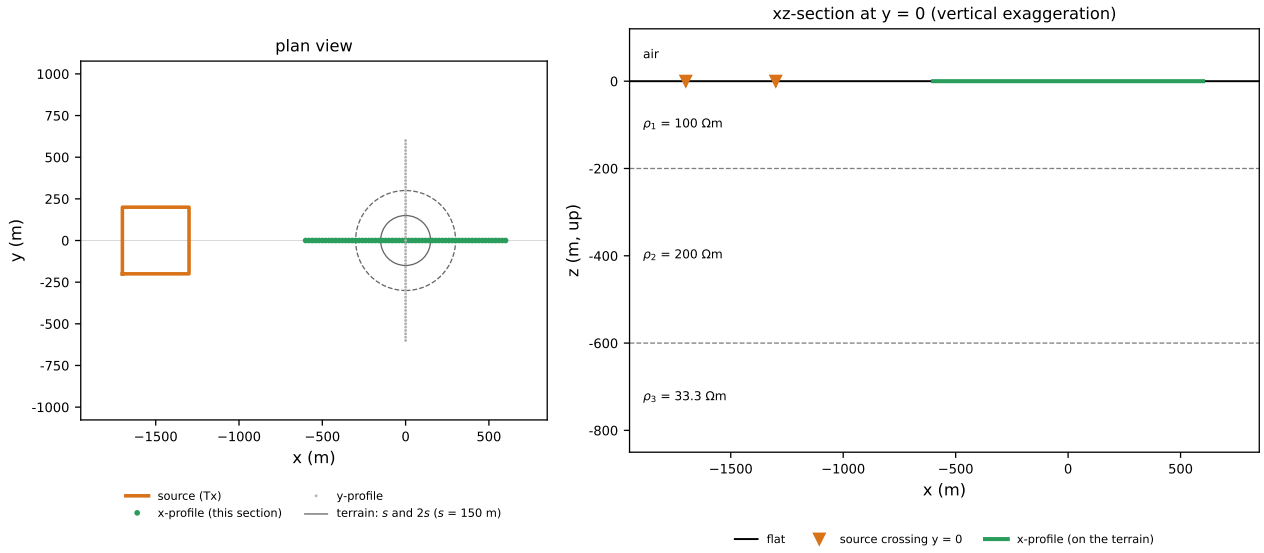


Figure 9: Setup of T3_loop_offset, x-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

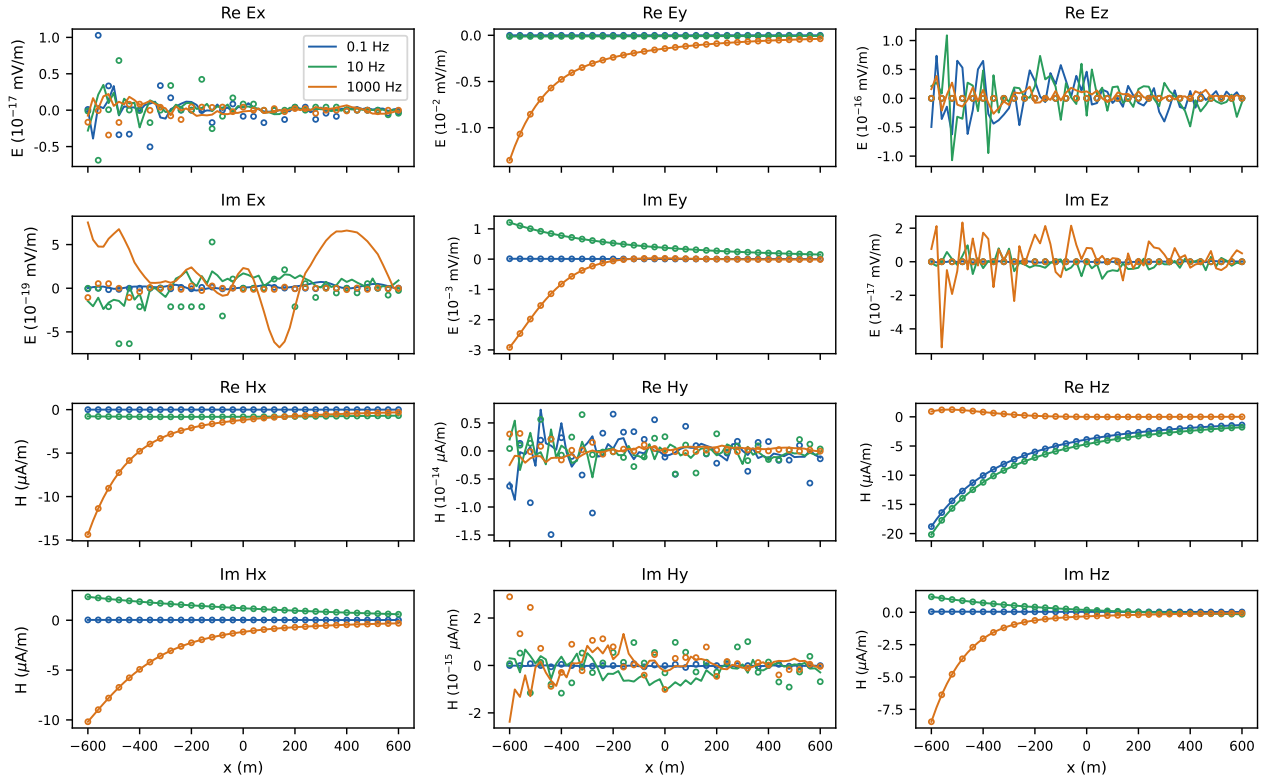


Figure 10: T3_loop_offset, x-profile, flat model: series (lines) and empymod (circles, every second receiver).

6 T3_loop_offset, y-profile

empymod comparison (flat model) T3: square loop 400 x 400 m, centre (-1500, 0), counter-clockwise seen from above.

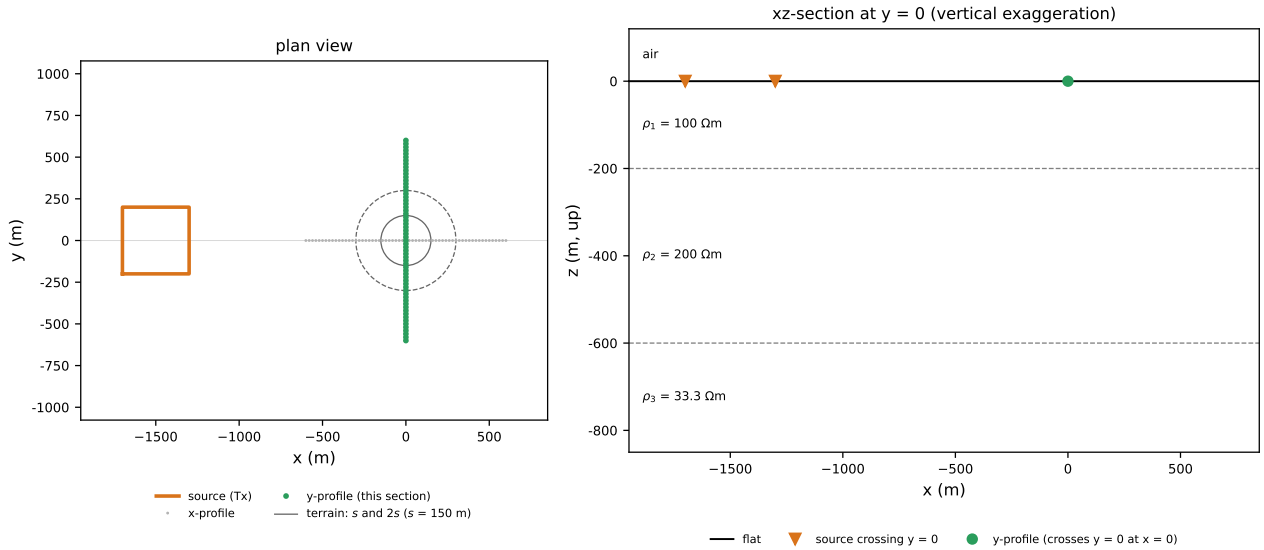


Figure 11: Setup of T3_loop_offset, y-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

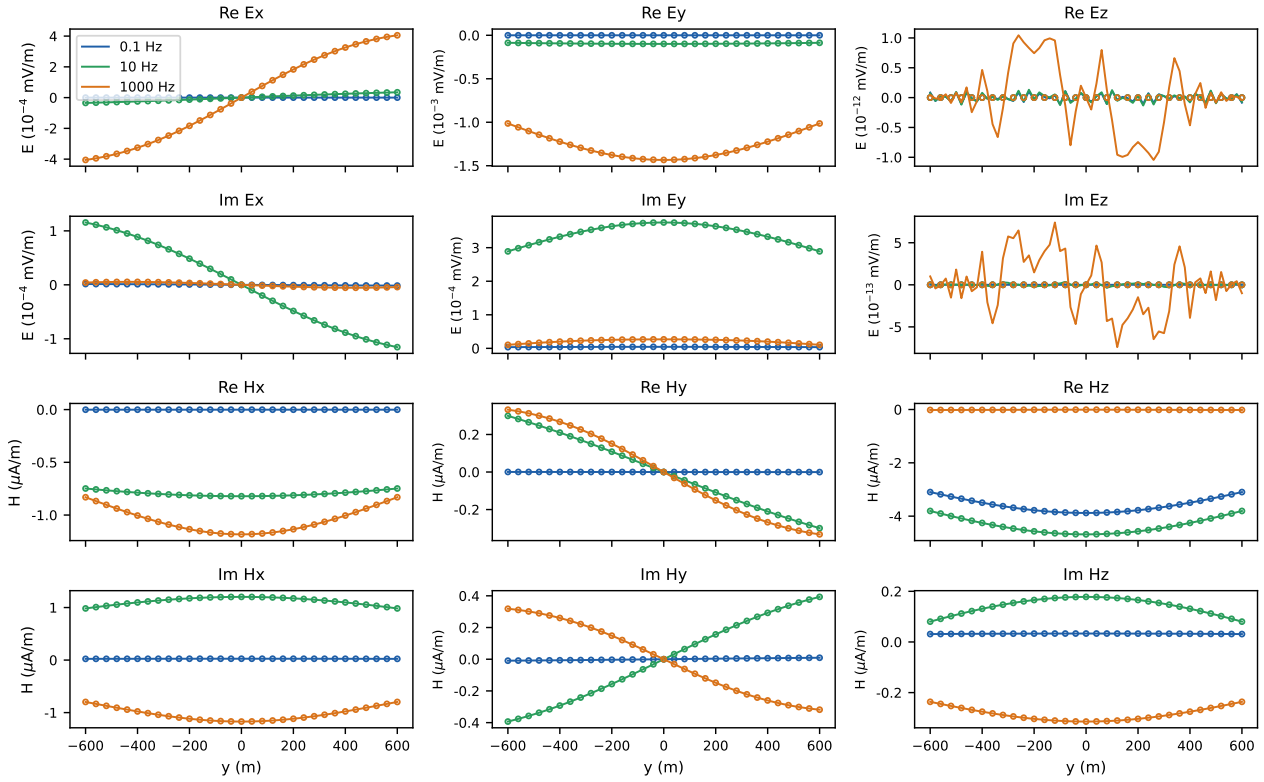


Figure 12: T3_loop_offset, y-profile, flat model: series (lines) and empymod (circles, every second receiver).

7 T4_loop_large, x-profile

empymod comparison (flat model) T4: square loop 2.6 x 2.6 km around the hill, counter-clockwise seen from above.

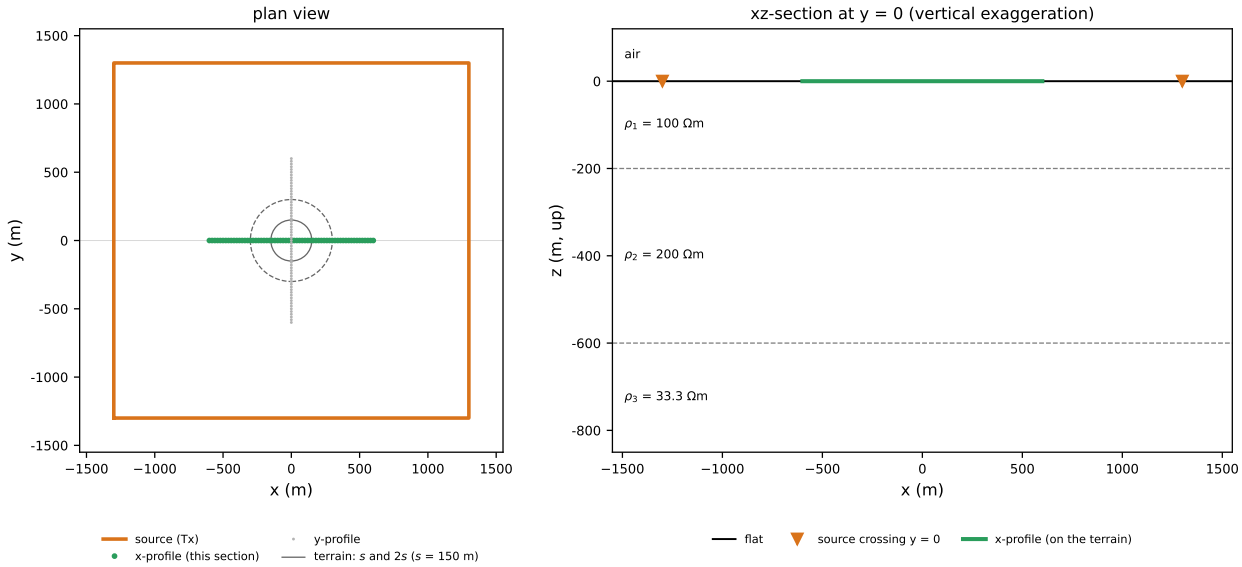


Figure 13: Setup of T4_loop_large, x-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

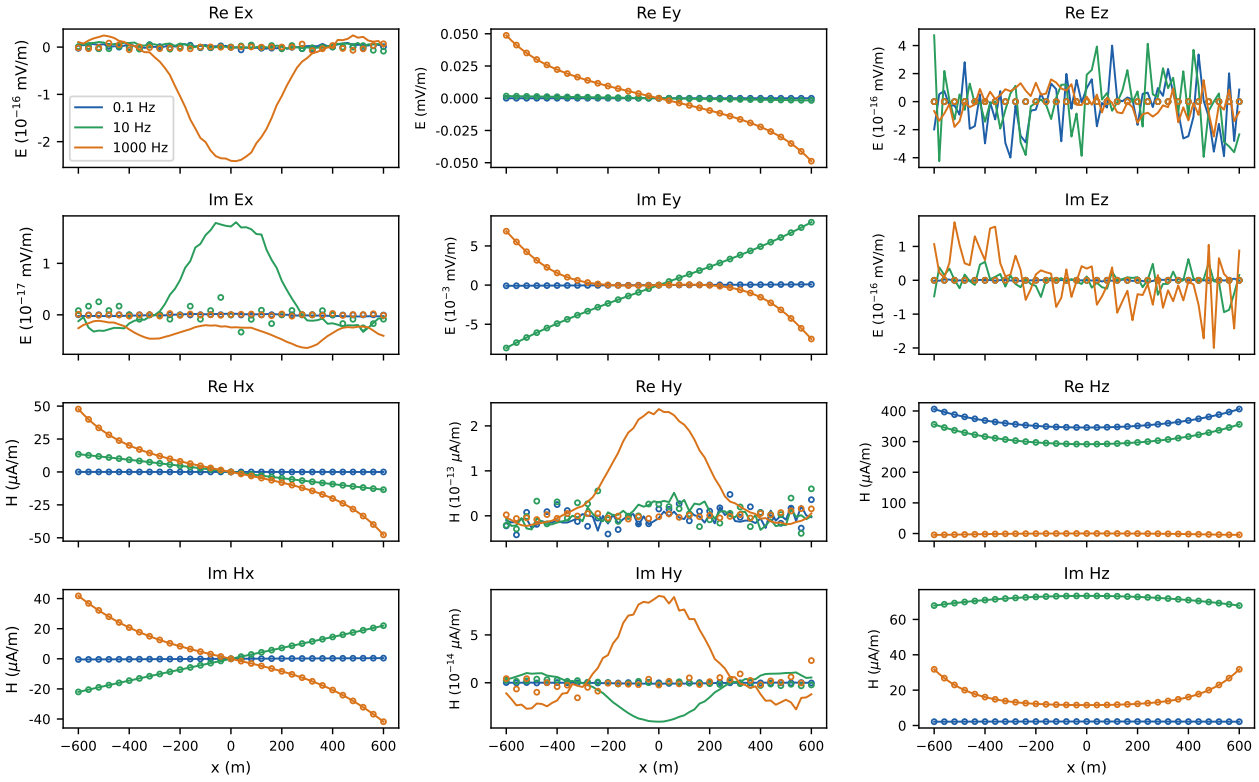


Figure 14: T4_loop_large, x-profile, flat model: series (lines) and empymod (circles, every second receiver).

8 T4_loop_large, y-profile

empymod comparison (flat model) T4: square loop 2.6 x 2.6 km around the hill, counter-clockwise seen from above.

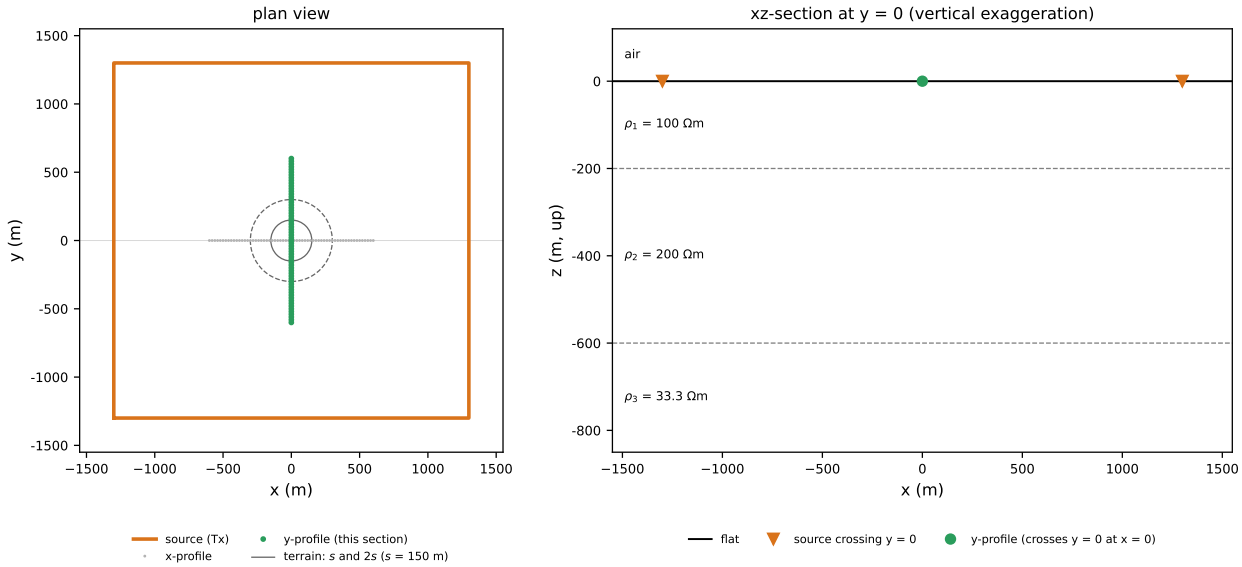


Figure 15: Setup of T4_loop_large, y-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

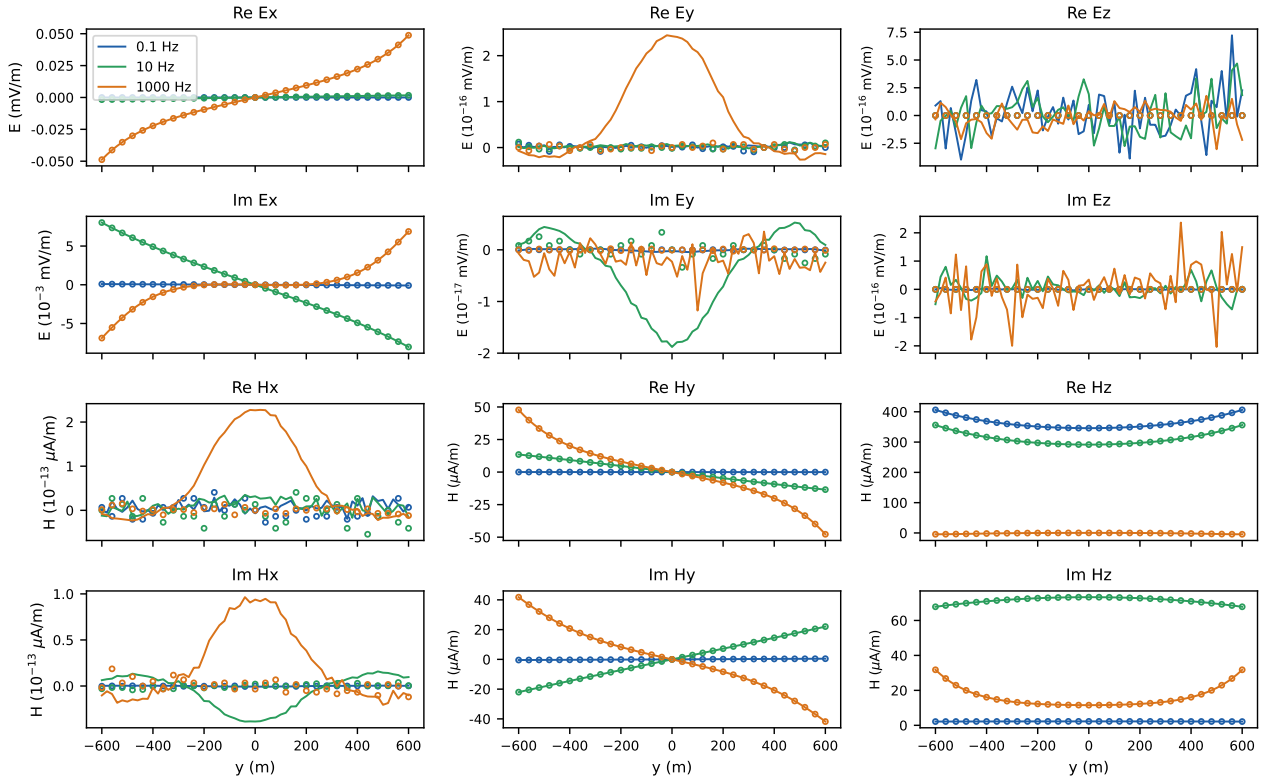


Figure 16: T4_loop_large, y-profile, flat model: series (lines) and empymod (circles, every second receiver).

9 Deviations

$|V_{\text{series}} - V_{\text{empymod}}| / \max |V_{\text{empymod}}|$ over both profiles, median / maximum; E_z: $\max |E_{z,\text{empymod}}| / \max |E|$ (zero on the flat surface; empymod 1 mm below it).

case	f (Hz)	E_x	E_y	E_z	H_x	H_y	H_z
T1_wire_inline	0.1	2e-07 / 1e-06	3e-15 / 9e-07	4e-06	1e-15 / 2e-06	1e-06 / 7e-06	2e-15 / 1e-08
T1_wire_inline	10	2e-07 / 1e-06	3e-15 / 1e-06	4e-06	2e-15 / 2e-06	1e-06 / 7e-06	2e-15 / 6e-07
T1_wire_inline	1000	4e-06 / 2e-05	3e-15 / 2e-05	5e-06	2e-15 / 2e-05	3e-06 / 2e-05	5e-15 / 2e-05
T2_wire_broadside	0.1	3e-15 / 9e-07	1e-08 / 7e-08	5e-07	4e-07 / 3e-06	3e-15 / 2e-06	4e-09 / 5e-09
T2_wire_broadside	10	4e-15 / 1e-06	2e-07 / 6e-07	5e-07	6e-07 / 3e-06	3e-15 / 2e-06	2e-07 / 3e-07
T2_wire_broadside	1000	2e-15 / 2e-05	4e-06 / 2e-05	2e-07	4e-06 / 2e-05	4e-15 / 2e-05	2e-06 / 2e-05
T3_loop_offset	0.1	6e-12 / 4e-04	1e-04 / 1e-03	4e-11	4e-06 / 7e-05	2e-12 / 2e-05	4e-09 / 7e-08
T3_loop_offset	10	7e-14 / 5e-06	1e-06 / 1e-05	4e-13	8e-07 / 3e-06	3e-14 / 1e-06	7e-08 / 2e-07
T3_loop_offset	1000	1e-14 / 2e-05	2e-06 / 2e-05	2e-14	2e-06 / 1e-05	1e-14 / 2e-05	7e-07 / 2e-05
T4_loop_large	0.1	1e-13 / 6e-04	2e-13 / 6e-04	4e-19	1e-13 / 1e-05	1e-13 / 1e-05	7e-09 / 9e-09
T4_loop_large	10	2e-15 / 8e-06	3e-15 / 8e-06	4e-21	3e-15 / 2e-06	3e-15 / 2e-06	5e-07 / 5e-07
T4_loop_large	1000	5e-15 / 2e-05	5e-15 / 2e-05	4e-22	4e-15 / 2e-05	4e-15 / 2e-05	9e-06 / 2e-05

terrasphere_csem: flat model against empymod in the time domain

October 14, 2026

Transients of the series solution of this package for the flat model (lines) against the 1-D code empymod (circles) for the four sources of the elfe3D comparison. Three layers with interfaces at 200 m and 600 m depth. Switch-off of 1 A, 101 times from 9.19e-06 s to 10 s (the lagged time grid of this package). Receivers on the x- and y-profile from −600 m to 600 m every 300 m: series on the surface (earth side), empymod sources and receivers 10^{-6} m below it. Components E_x, E_y, $\partial_t B_x$, $\partial_t B_y$, $\partial_t B_z$; E_z vanishes on the earth side of the flat surface and is not compared. empymod 2.6.0, quasi-static (no displacement currents); Hankel transform the digital filter key_201_2009 in its standard form, Fourier transform the digital filter key_601_2009 in its standard form (every time its own frequencies, no interpolation); every bipole an explicit sum of 40 Gauss–Legendre point dipoles, a loop as its four sides. E in mV/m, $\partial_t B$ in V/m^2 (= T/s), all per 1 A.

Contents

1 T1_wire_inline, x-profile **2**

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5 T3_loop_offset, x-profile **6**

6 T3_loop_offset, y-profile **7**

7 T4_loop_large, x-profile **8**

8 T4_loop_large, y-profile **9**

9 Deviations **10**

1 T1_wire_inline, x-profile

empymod comparison in the time domain (flat model) T1: grounded wire in line with the x-profile, 500 m, centre 1500 m left of the hill.

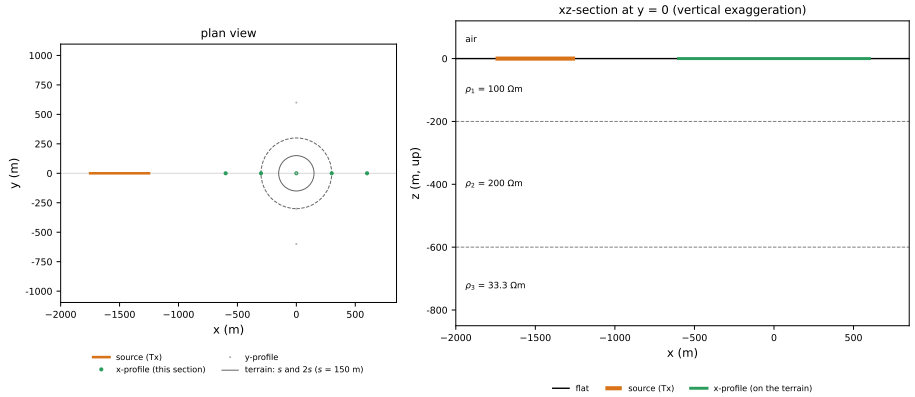


Figure 1: Setup of T1_wire_inline, x-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

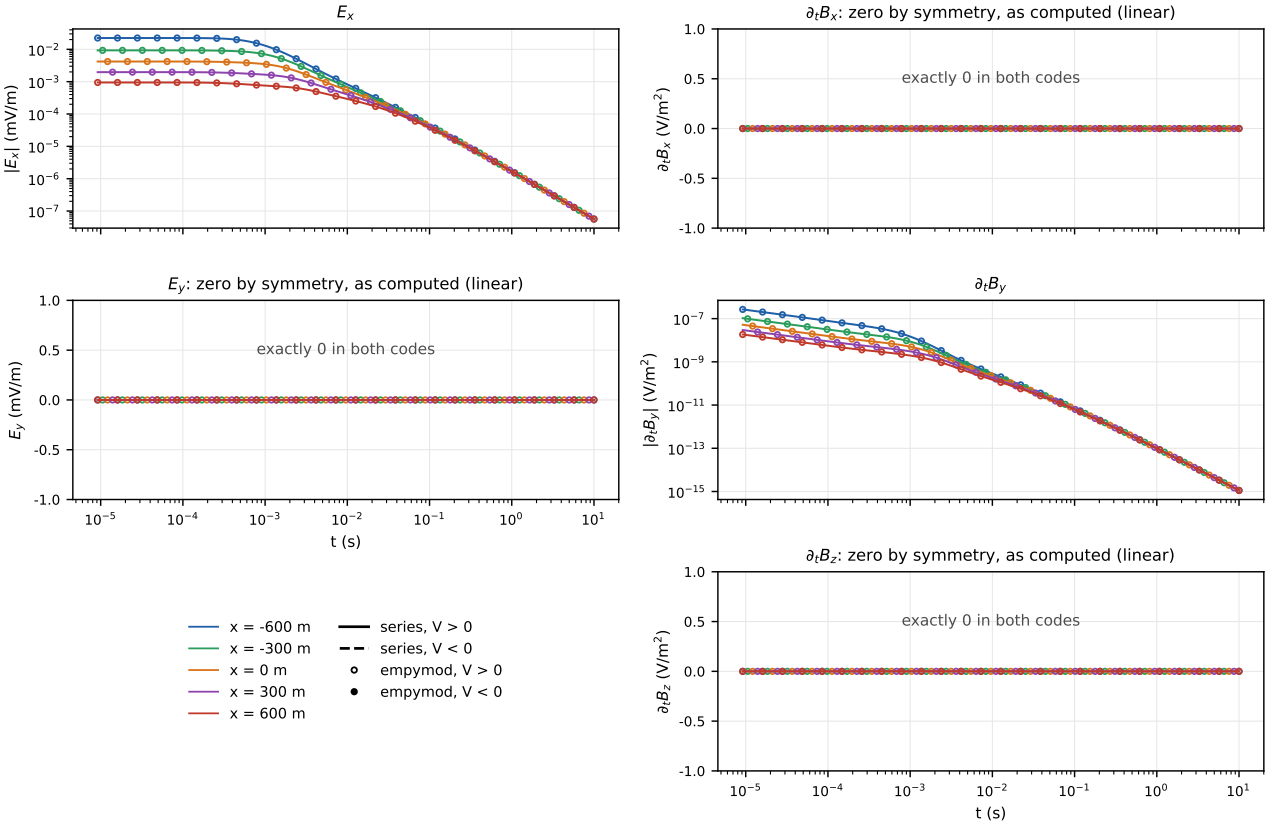


Figure 2: T1_wire_inline, x-profile, flat model, switch-off of 1 A: series (lines) and empymod (circles, every fourth time), one colour per station; left E, right $\partial_t B$. A component that vanishes on the whole profile by symmetry is shown as computed: V itself, linear. Deviations: section 9.

2 T1_wire_inline, y-profile

empymod comparison in the time domain (flat model) T1: grounded wire in line with the x-profile, 500 m, centre 1500 m left of the hill.

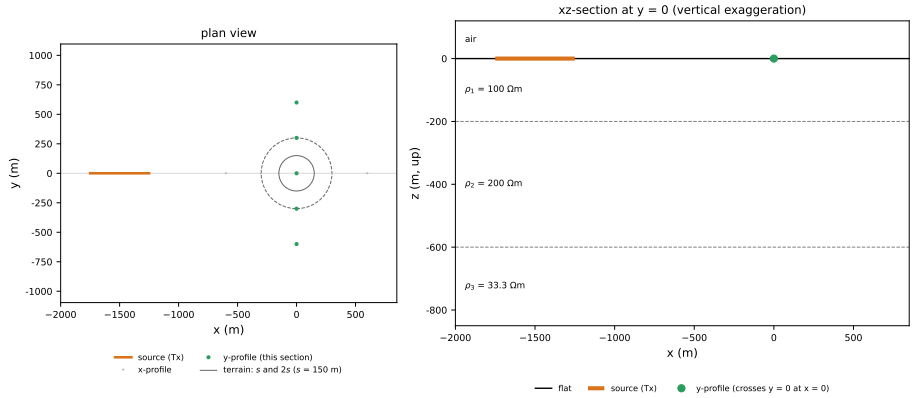


Figure 3: Setup of T1_wire_inline, y-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

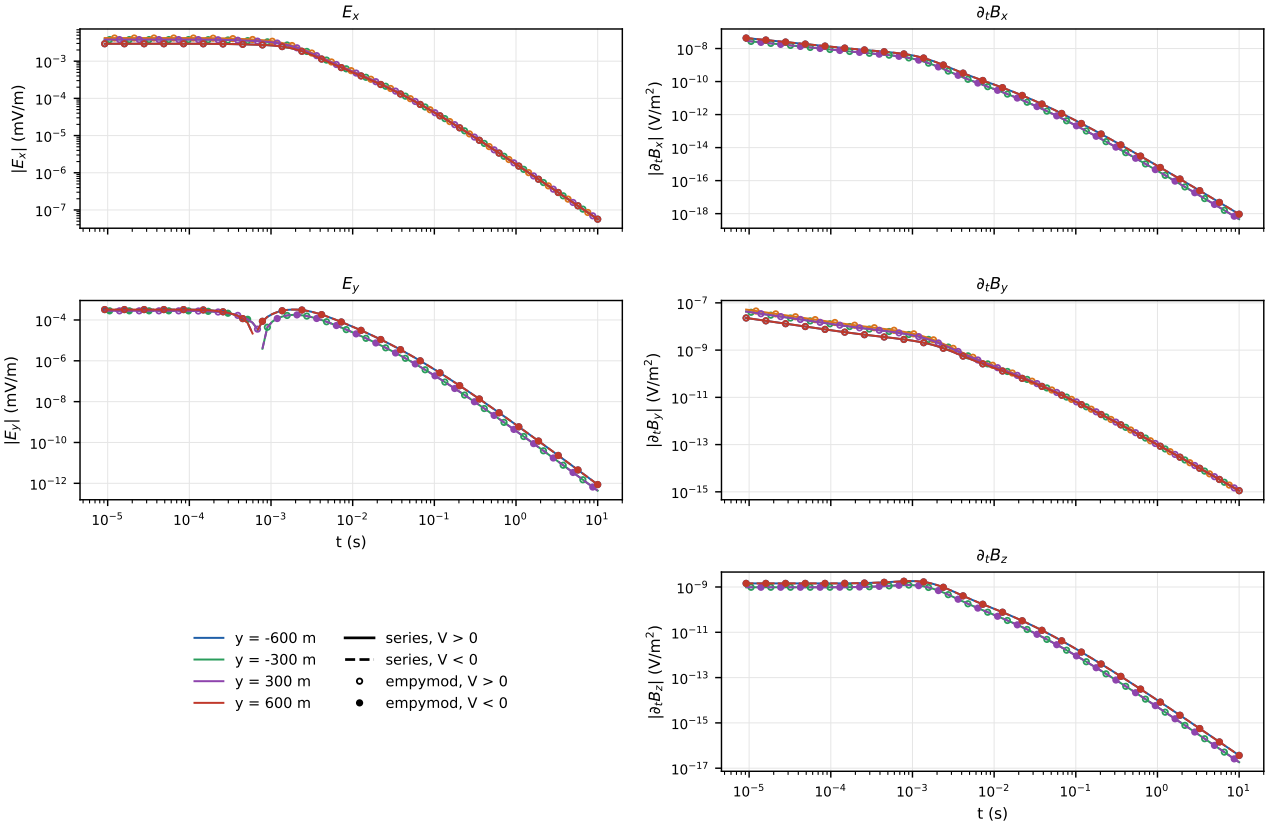


Figure 4: T1_wire_inline, y-profile, flat model, switch-off of 1 A: series (lines) and empymod (circles, every fourth time), one colour per station; left E, right $\partial_t B$. A component that vanishes on the whole profile by symmetry is shown as computed: V itself, linear; single stations left out (zero by symmetry): E_y at y = 0 m, $\partial_t B_x$ at y = 0 m, $\partial_t B_z$ at y = 0 m. Deviations: section 9.

3 T2_wire_broadside, x-profile

empymod comparison in the time domain (flat model) T2: grounded wire broadside (along y), 500 m, centre 1500 m left of the hill.

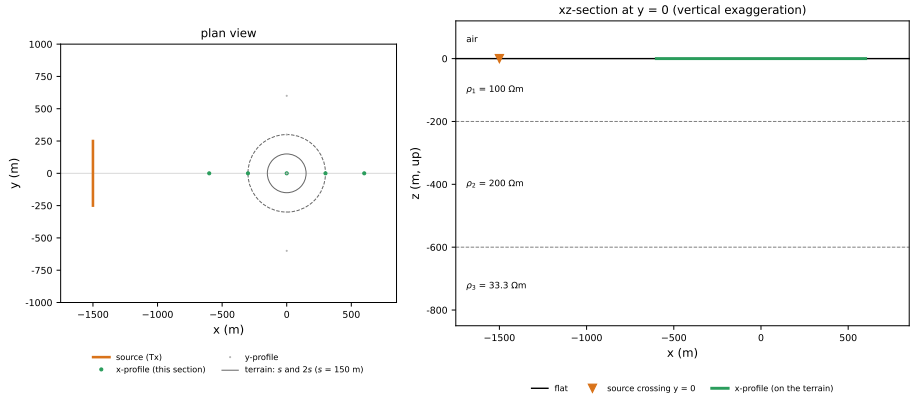


Figure 5: Setup of T2_wire_broadside, x-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

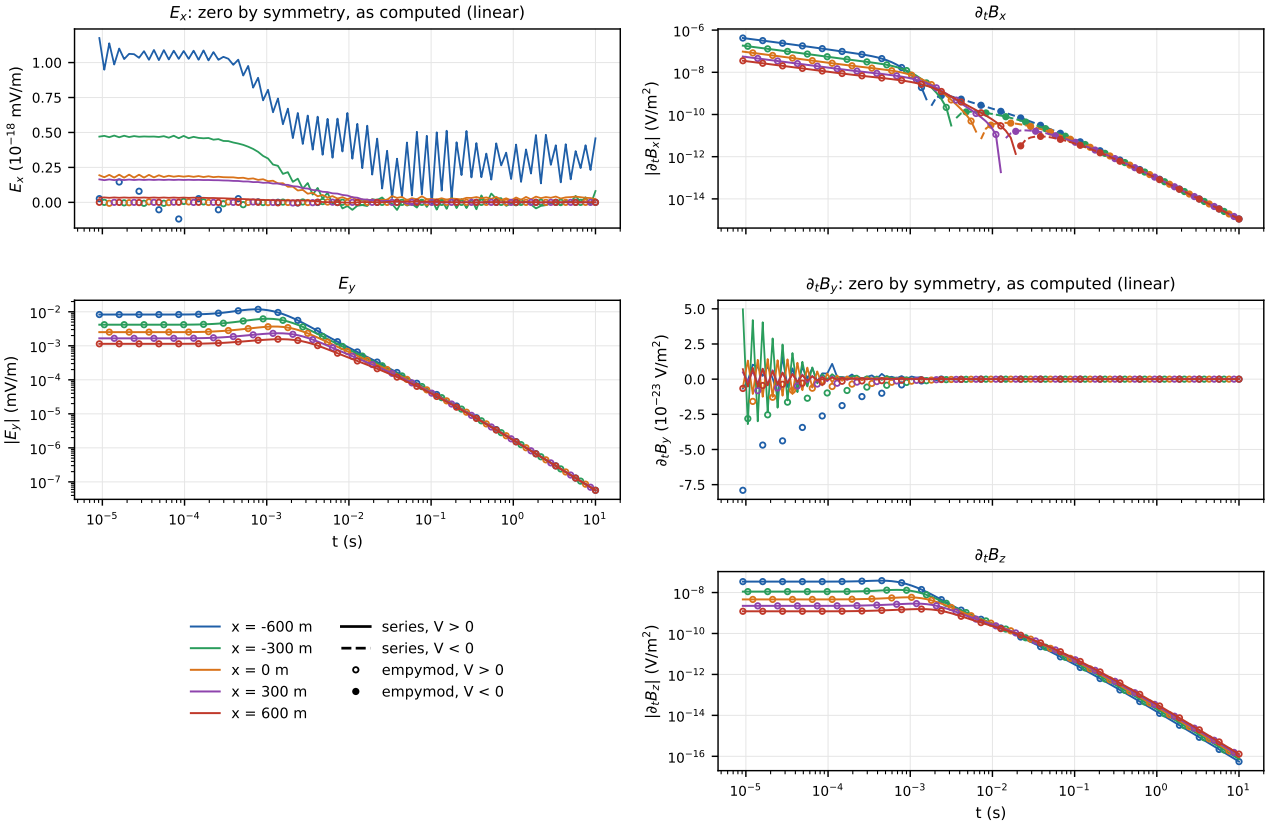


Figure 6: T2_wire_broadside, x-profile, flat model, switch-off of 1 A: series (lines) and empymod (circles, every fourth time), one colour per station; left E, right $\partial_t B$. A component that vanishes on the whole profile by symmetry is shown as computed: V itself, linear. Deviations: section 9.

4 T2_wire_broadside, y-profile

empymod comparison in the time domain (flat model) T2: grounded wire broadside (along y), 500 m, centre 1500 m left of the hill.

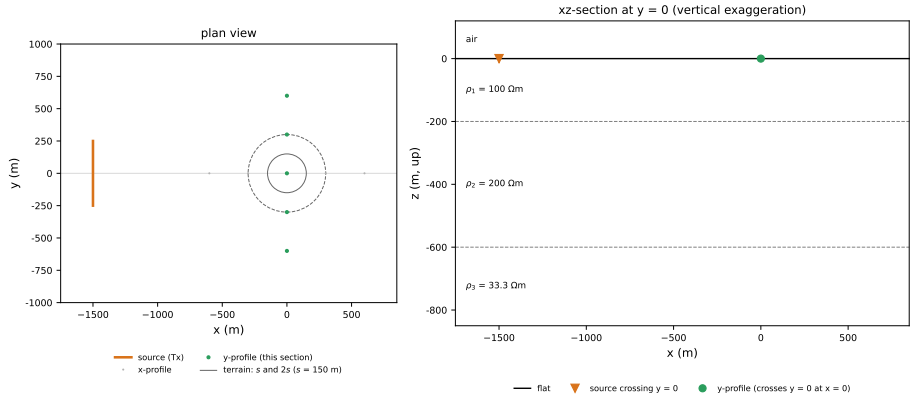


Figure 7: Setup of T2_wire_broadside, y-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

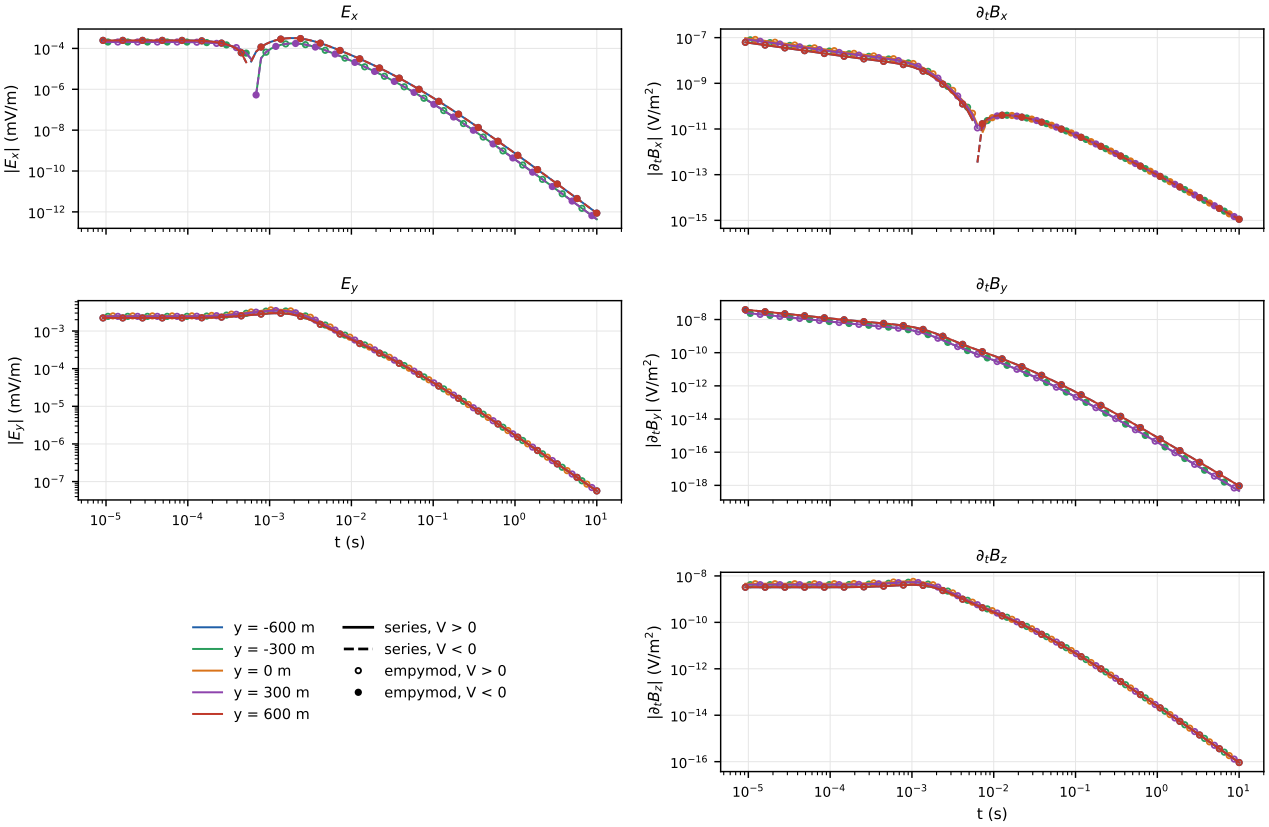


Figure 8: T2_wire_broadside, y-profile, flat model, switch-off of 1 A: series (lines) and empymod (circles, every fourth time), one colour per station; left E, right $\partial_t B$. A component that vanishes on the whole profile by symmetry is shown as computed: V itself, linear; single stations left out (zero by symmetry): E_x at y = 0 m, $\partial_t B_y$ at y = 0 m. Deviations: section 9.

5 T3_loop_offset, x-profile

empymod comparison in the time domain (flat model) T3: square loop 400 x 400 m, centre (-1500, 0), counter-clockwise seen from above.

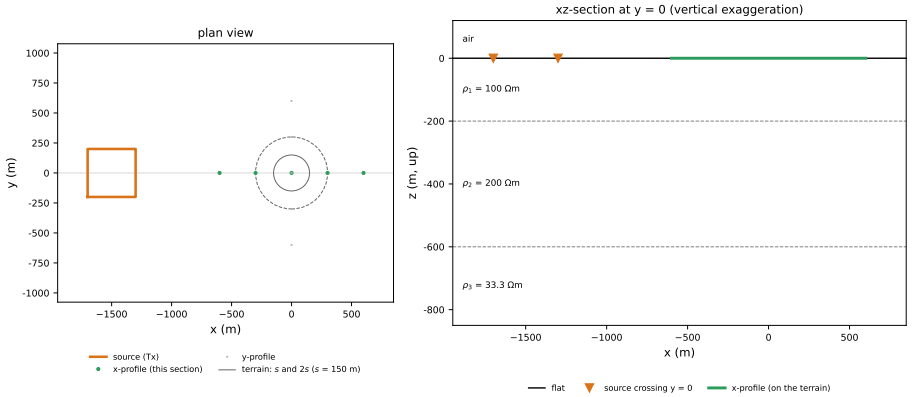


Figure 9: Setup of T3_loop_offset, x-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

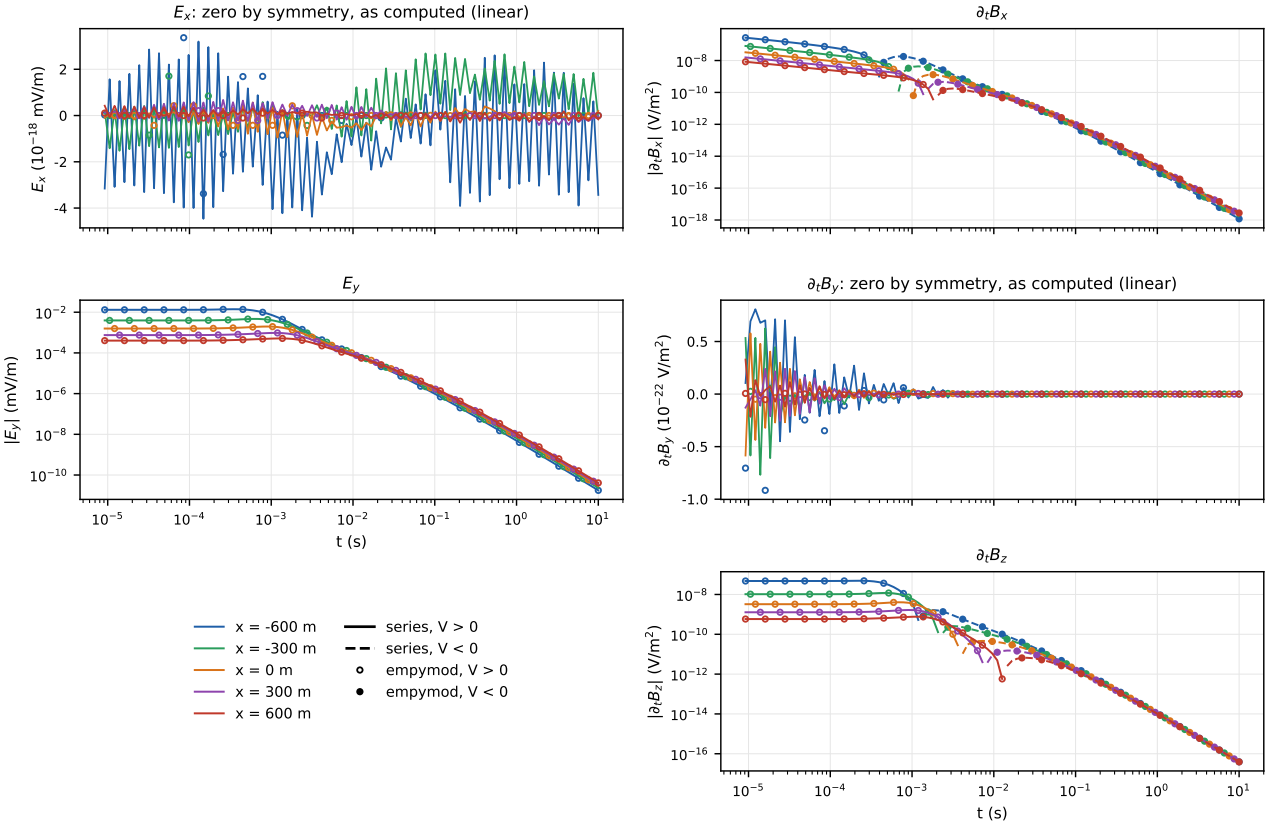


Figure 10: T3_loop_offset, x-profile, flat model, switch-off of 1 A: series (lines) and empymod (circles, every fourth time), one colour per station; left E, right $\partial_t B$. A component that vanishes on the whole profile by symmetry is shown as computed: V itself, linear. Deviations: section 9.

6 T3_loop_offset, y-profile

empymod comparison in the time domain (flat model) T3: square loop 400 x 400 m, centre (-1500, 0), counter-clockwise seen from above.

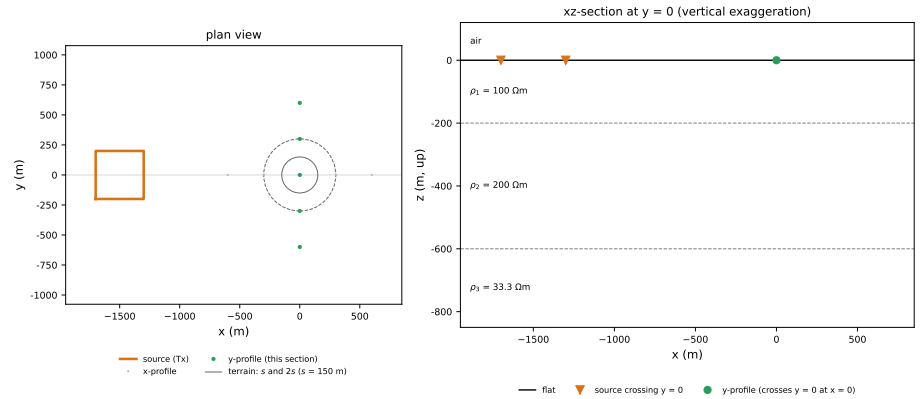


Figure 11: Setup of T3_loop_offset, y-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

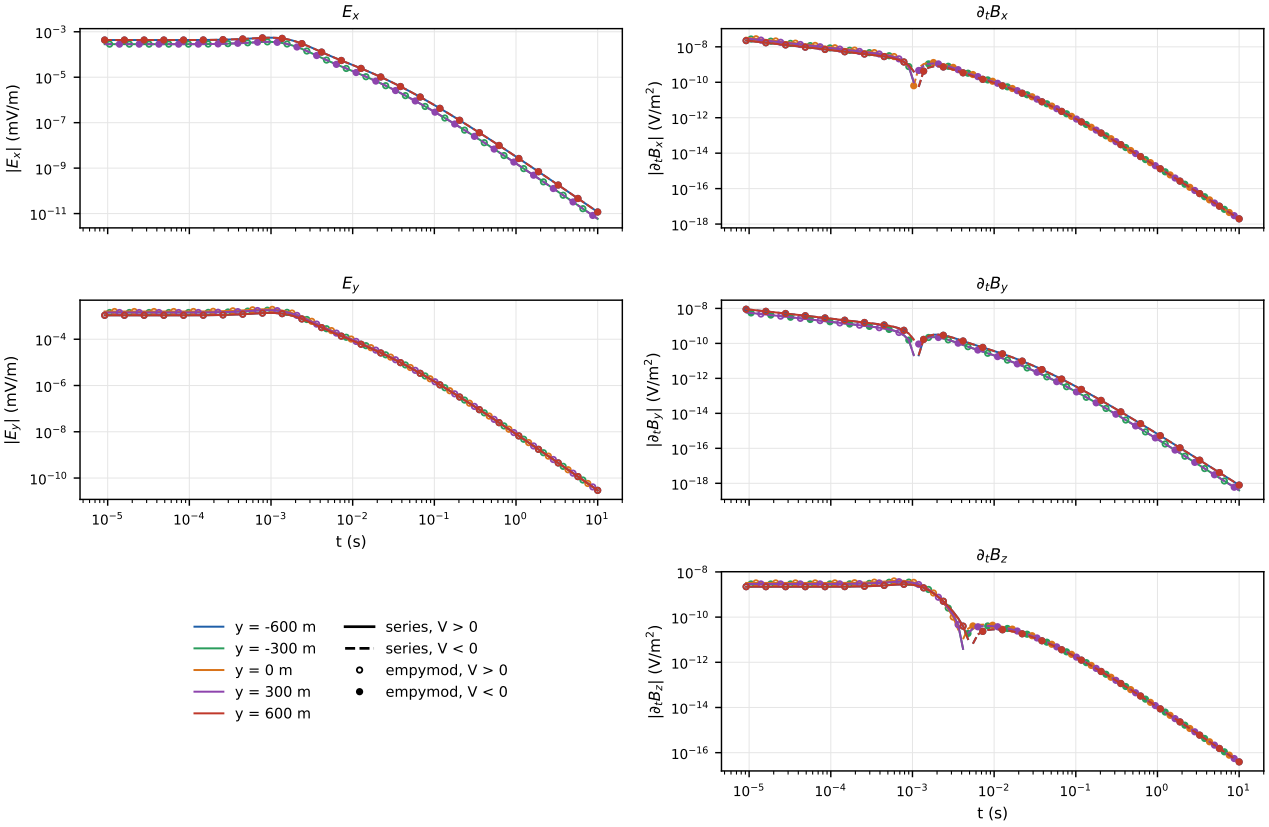


Figure 12: T3_loop_offset, y-profile, flat model, switch-off of 1 A: series (lines) and empymod (circles, every fourth time), one colour per station; left E, right $\partial_t B$. A component that vanishes on the whole profile by symmetry is shown as computed: V itself, linear; single stations left out (zero by symmetry): E_x at y = 0 m, $\partial_t B_y$ at y = 0 m. Deviations: section 9.

7 T4_loop_large, x-profile

empymod comparison in the time domain (flat model) T4: square loop 2.6 x 2.6 km around the hill, counter-clockwise seen from above.

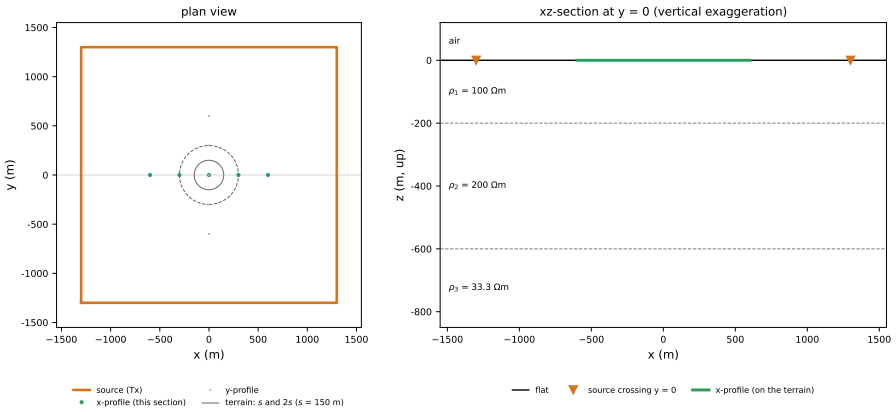


Figure 13: Setup of T4_loop_large, x-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

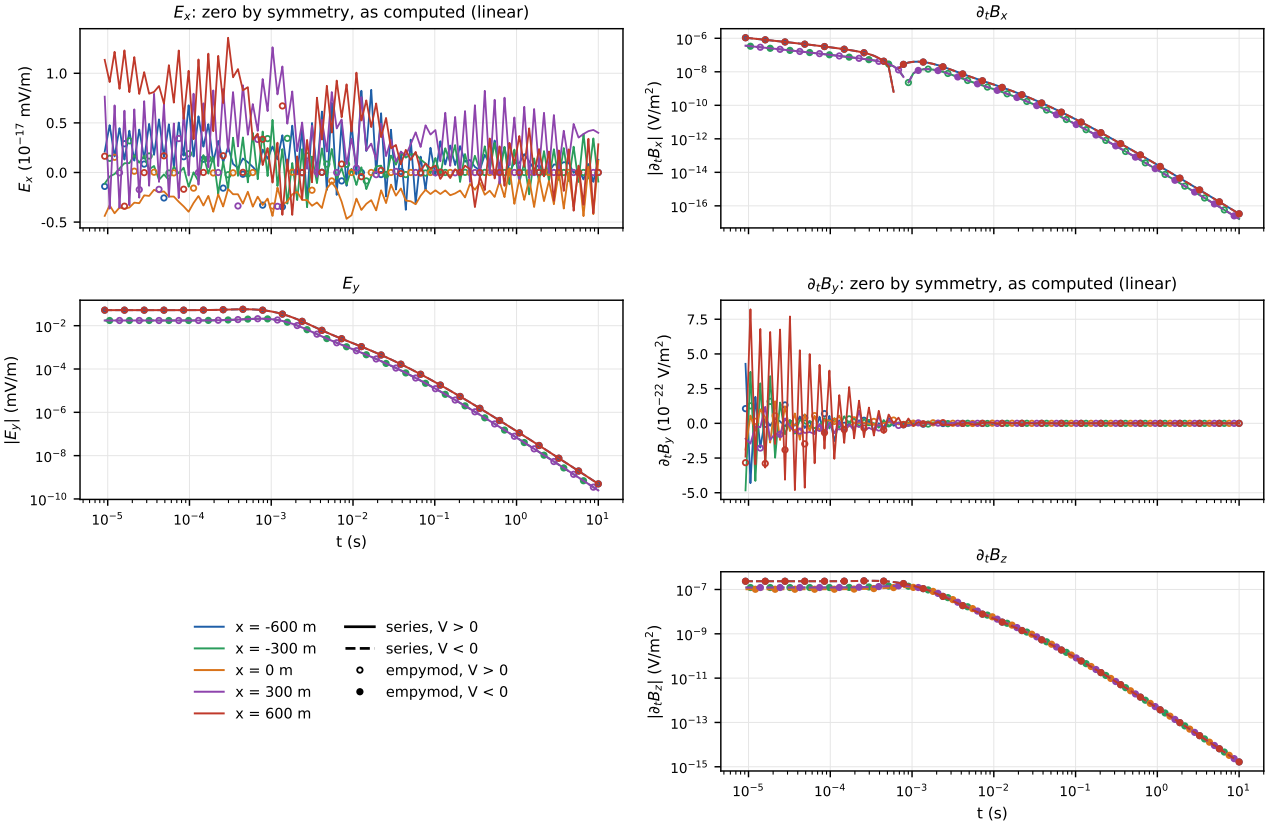


Figure 14: T4_loop_large, x-profile, flat model, switch-off of 1 A: series (lines) and empymod (circles, every fourth time), one colour per station; left E, right $\partial_t B$. A component that vanishes on the whole profile by symmetry is shown as computed: V itself, linear; single stations left out (zero by symmetry): E_y at x = 0 m, $\partial_t B_x$ at x = 0 m. Deviations: section 9.

8 T4_loop_large, y-profile

empymod comparison in the time domain (flat model) T4: square loop 2.6 x 2.6 km around the hill, counter-clockwise seen from above.

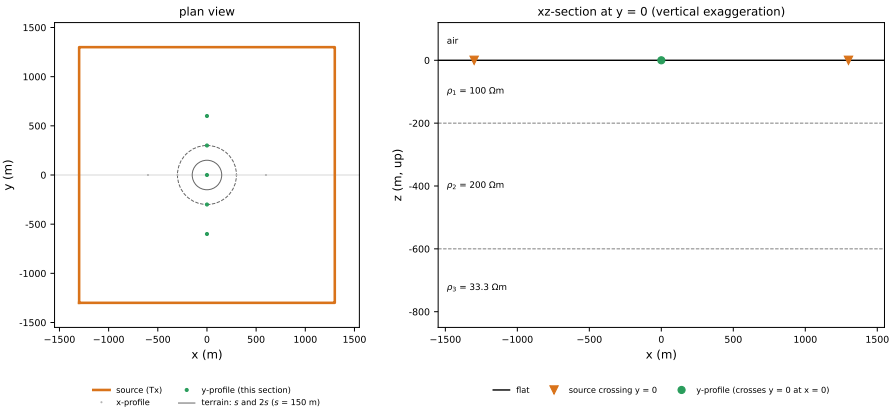


Figure 15: Setup of T4_loop_large, y-profile, flat model: plan view (left) and xz-section at $y = 0$ (right).

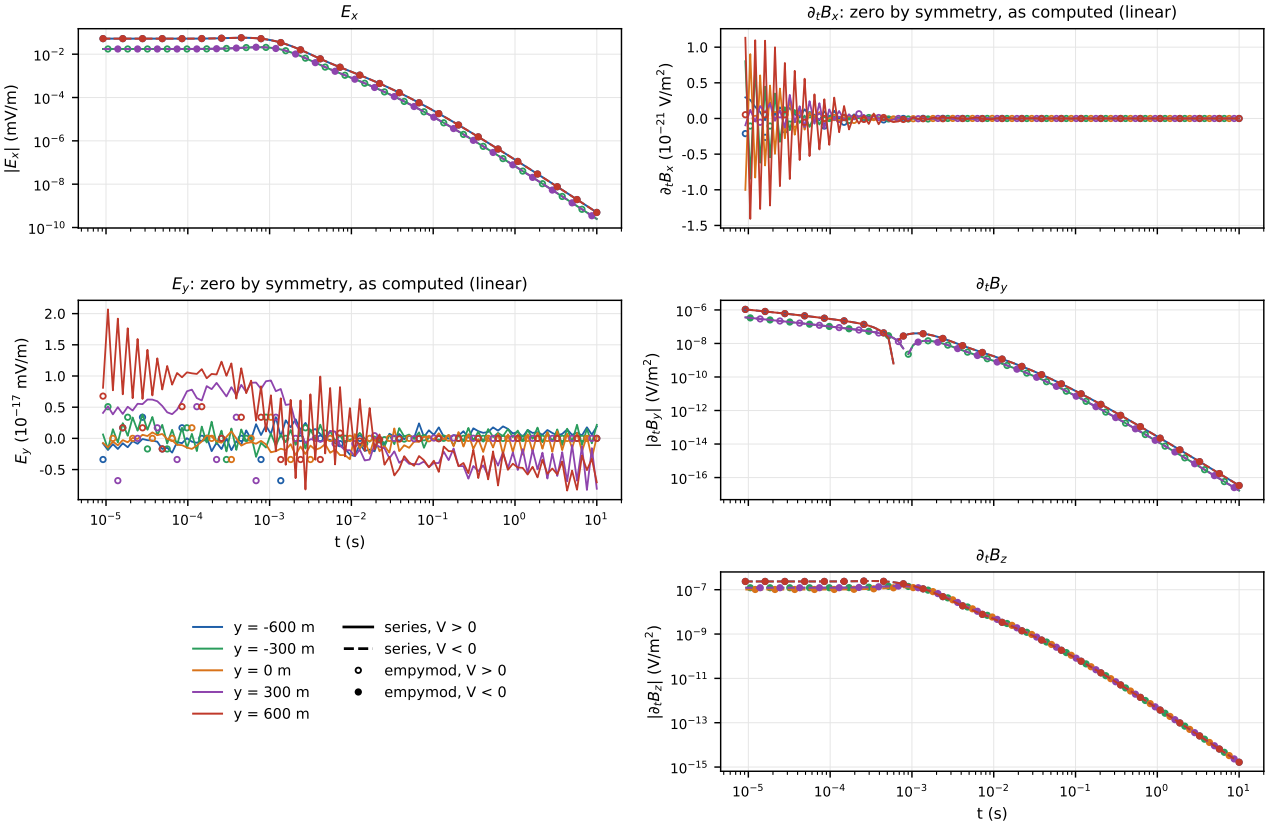


Figure 16: T4_loop_large, y-profile, flat model, switch-off of 1 A: series (lines) and empymod (circles, every fourth time), one colour per station; left E, right $\partial_t B$. A component that vanishes on the whole profile by symmetry is shown as computed: V itself, linear; single stations left out (zero by symmetry): E_x at y = 0 m, $\partial_t B_y$ at y = 0 m. Deviations: section 9.

9 Deviations

Per receiver: curve = $\max_t |V_{\text{series}} - V_{\text{empymod}}| / \max_t |V_{\text{empymod}}|$, median and maximum over the n receivers of both profiles; pointwise = $\max |\Delta V| / |V_{\text{empymod}}|$ over the times where $|V_{\text{empymod}}|$ exceeds 10^{-6} of its maximum; zero: largest $\max_t |V|$ of both codes on the receivers where the component vanishes by symmetry (below 10^{-6} of the largest E resp. $\partial_t B$ component), relative to that component; these receivers are left out of the other measures.

case	component	n	curve median	curve max	pointwise	zero
T1_wire_inline	E_x	10	2.3e-08	3.9e-08	8.1e-08	–
T1_wire_inline	E_y	4	2.4e-07	2.7e-07	1.5e-06	0e+00
T1_wire_inline	$\partial_t B_x$	4	3.5e-10	3.5e-10	6.0e-09	0e+00
T1_wire_inline	$\partial_t B_y$	10	5.2e-10	1.3e-09	7.3e-09	–
T1_wire_inline	$\partial_t B_z$	4	3.3e-08	3.3e-08	4.1e-08	0e+00
T2_wire_broadside	E_x	4	2.4e-07	2.4e-07	1.1e-05	1e-16
T2_wire_broadside	E_y	10	4.9e-08	7.1e-08	1.0e-07	–
T2_wire_broadside	$\partial_t B_x$	10	3.4e-10	7.8e-10	7.4e-06	–
T2_wire_broadside	$\partial_t B_y$	4	3.4e-10	3.4e-10	5.7e-09	2e-16
T2_wire_broadside	$\partial_t B_z$	10	3.3e-08	3.7e-08	4.1e-08	–
T3_loop_offset	E_x	4	3.3e-08	3.3e-08	6.2e-07	3e-16
T3_loop_offset	E_y	10	3.3e-08	3.9e-08	4.7e-07	–
T3_loop_offset	$\partial_t B_x$	10	7.8e-10	2.2e-09	5.8e-07	–
T3_loop_offset	$\partial_t B_y$	4	7.6e-10	7.8e-10	2.6e-07	3e-16
T3_loop_offset	$\partial_t B_z$	10	3.3e-08	4.0e-08	1.5e-07	–
T4_loop_large	E_x	4	3.6e-08	3.8e-08	4.5e-07	2e-16
T4_loop_large	E_y	4	3.6e-08	3.8e-08	4.5e-07	4e-16
T4_loop_large	$\partial_t B_x$	4	1.2e-09	1.5e-09	1.2e-06	1e-15
T4_loop_large	$\partial_t B_y$	4	1.2e-09	1.5e-09	1.2e-06	8e-16
T4_loop_large	$\partial_t B_z$	10	3.5e-08	3.9e-08	4.1e-08	–

Curve measure, maximum over the receivers, per decade of time:

case	component	10^{-6} s	10^{-5} s	10^{-4} s	10^{-3} s	10^{-2} s	10^{-1} s	10^{0} s	10^{1} s
T1_wire_inline	E_x	4e-08	4e-08	1e-08	4e-09	3e-10	2e-11	5e-13	3e-13
T1_wire_inline	E_y	3e-07	2e-07	8e-08	2e-08	3e-10	3e-12	2e-13	1e-13
T1_wire_inline	$\partial_t B_x$	4e-10	3e-10	2e-10	2e-10	6e-12	3e-14	2e-15	7e-16
T1_wire_inline	$\partial_t B_y$	1e-09	9e-10	4e-10	3e-10	9e-12	1e-13	3e-15	7e-16
T1_wire_inline	$\partial_t B_z$	3e-08	3e-08	9e-09	1e-09	1e-10	2e-12	3e-15	6e-17
T2_wire_broadside	E_x	2e-07	2e-07	7e-08	2e-08	3e-10	3e-12	7e-13	3e-13
T2_wire_broadside	E_y	7e-08	7e-08	2e-08	4e-09	1e-10	1e-11	5e-13	4e-13
T2_wire_broadside	$\partial_t B_x$	8e-10	8e-10	7e-10	3e-10	2e-11	9e-14	3e-15	7e-16
T2_wire_broadside	$\partial_t B_y$	3e-10	3e-10	2e-10	2e-10	6e-12	3e-14	2e-15	7e-16
T2_wire_broadside	$\partial_t B_z$	4e-08	3e-08	9e-09	2e-09	2e-10	6e-12	1e-14	3e-16
T3_loop_offset	E_x	3e-08	3e-08	9e-09	1e-09	1e-10	2e-12	2e-12	1e-12
T3_loop_offset	E_y	4e-08	4e-08	9e-09	2e-09	2e-10	5e-12	5e-13	5e-13
T3_loop_offset	$\partial_t B_x$	2e-09	2e-09	2e-09	6e-10	4e-12	2e-13	2e-15	8e-16
T3_loop_offset	$\partial_t B_y$	8e-10	7e-10	6e-10	6e-10	4e-12	5e-14	2e-15	8e-16
T3_loop_offset	$\partial_t B_z$	4e-08	4e-08	9e-09	4e-09	1e-10	2e-12	9e-15	8e-16
T4_loop_large	E_x	4e-08	4e-08	9e-09	2e-09	8e-11	6e-13	5e-13	4e-13
T4_loop_large	E_y	4e-08	4e-08	9e-09	2e-09	8e-11	6e-13	5e-13	4e-13
T4_loop_large	$\partial_t B_x$	2e-09	2e-09	1e-09	6e-10	3e-12	2e-14	2e-15	8e-16
T4_loop_large	$\partial_t B_y$	2e-09	2e-09	1e-09	6e-10	3e-12	2e-14	2e-15	8e-16
T4_loop_large	$\partial_t B_z$	4e-08	4e-08	9e-09	1e-09	9e-11	1e-12	2e-15	2e-16

terrasphere_csem: validation against elfe3D

October 15, 2026

Series solution of this package (lines) against the 3D finite-element code elfe3D (circles) for four sources, Gaussian hills and depressions (height ±30 m and ±90 m, width $s = 150$ m); the flat model is validated against empymod. Three layers with interfaces at 200 m and 600 m depth. Receivers 1 cm below the surface on the x- and y-profile from −600 m to 600 m every 20 m. Fields per 1 A source current, E in mV/m, H in μA/m, time dependence $e^{-i\omega t}$, in the convention of this package: the series exactly as it writes them, elfe3D converted ($H = -H_{\mathrm{elfe3D}}$).

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1 T1_wire_inline, x-profile

elfe3D comparison T1: grounded wire in line with the x-profile, 500 m, centre 1500 m left of the hill.

1.1 hill 30 m

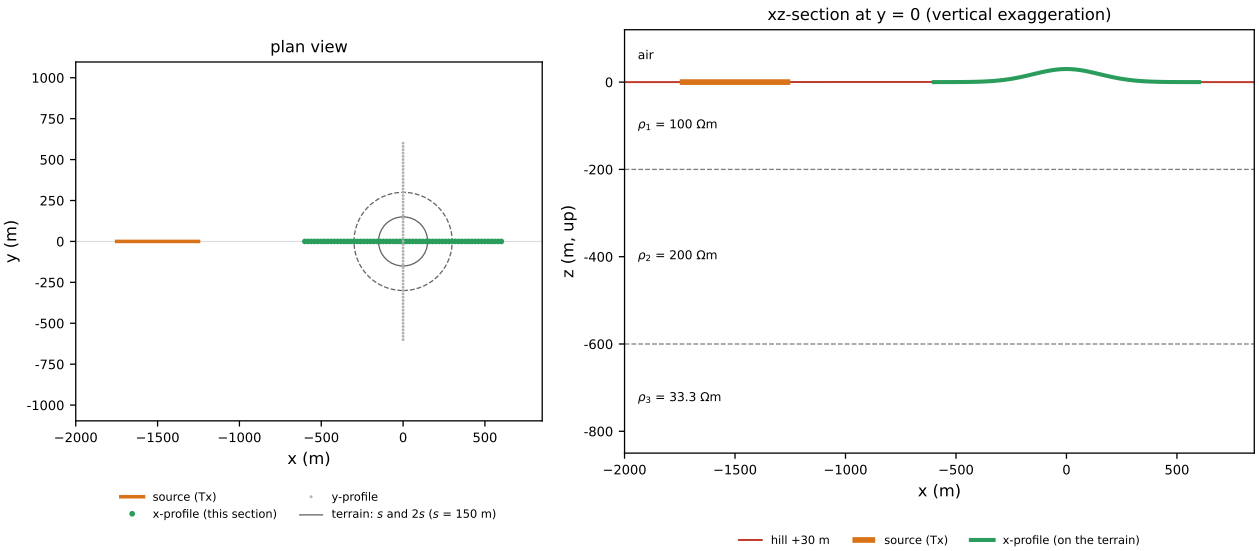


Figure 1: Setup of T1_wire_inline, x-profile, hill 30 m: plan view (left) and xz-section at $y = 0$ (right).

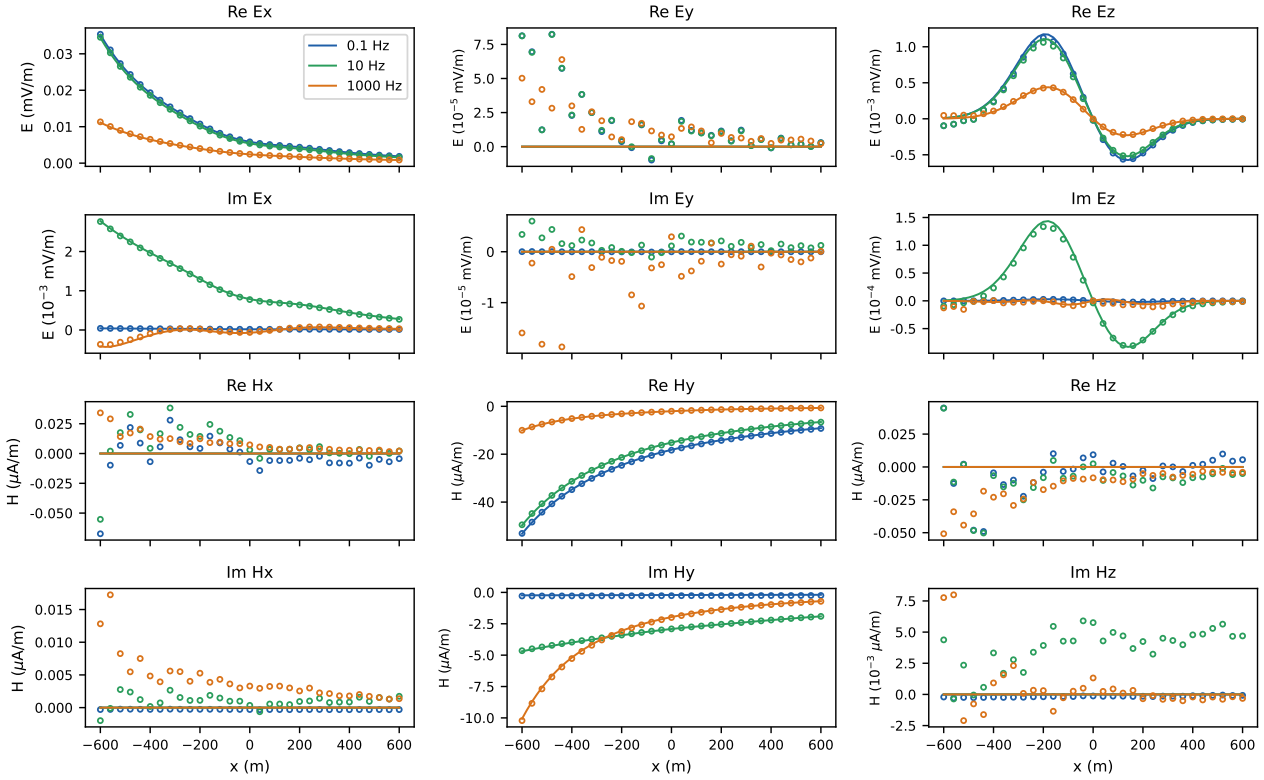


Figure 2: T1_wire_inline, x-profile, hill 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

1.2 depression 30 m

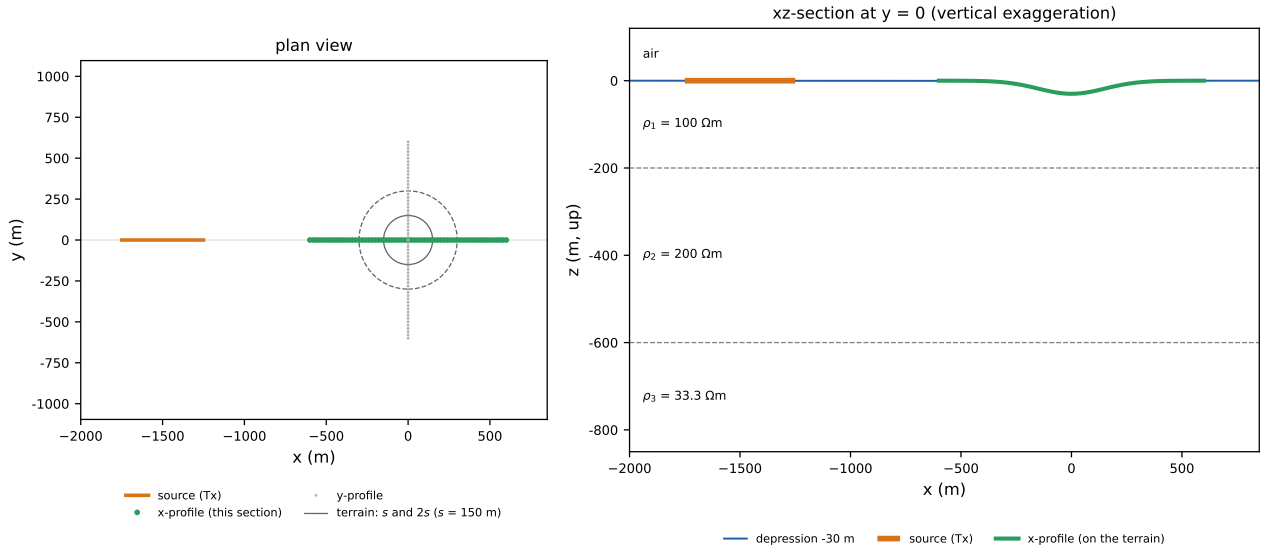


Figure 3: Setup of T1_wire_inline, x-profile, depression 30 m: plan view (left) and xz-section at $y = 0$ (right).

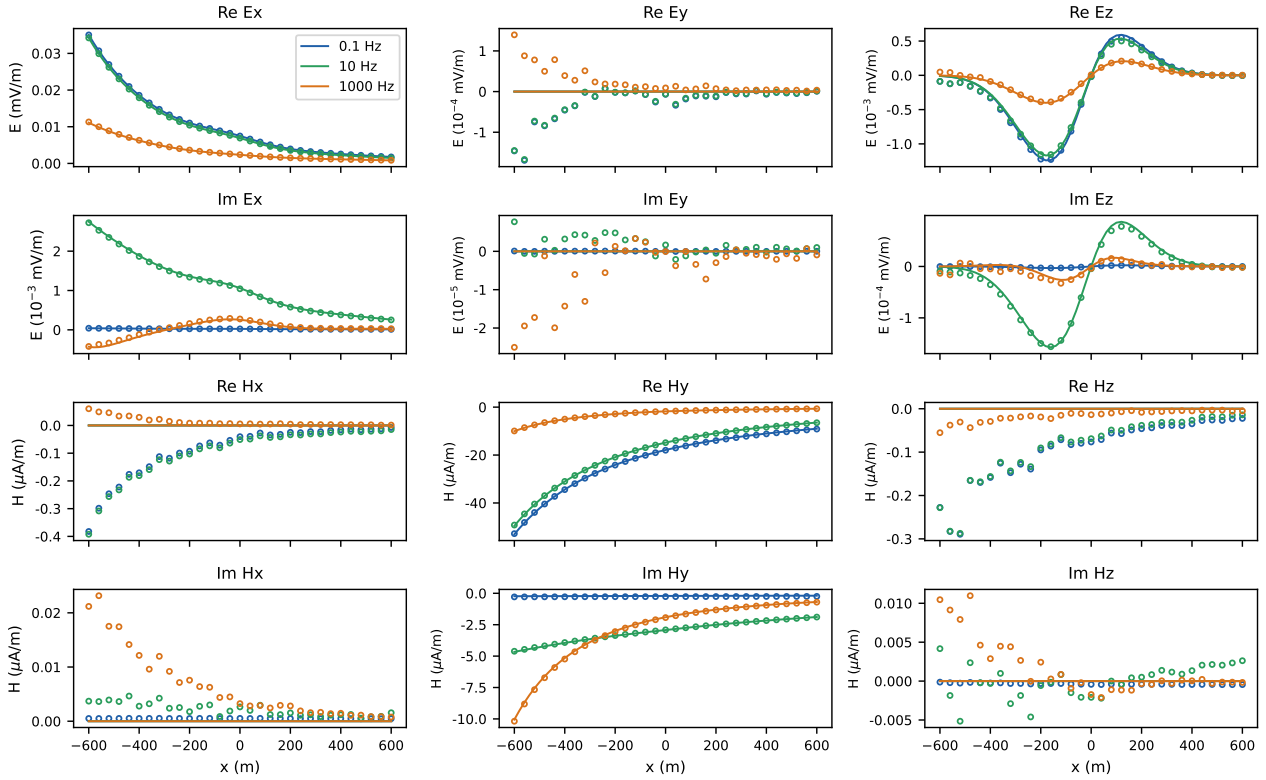


Figure 4: T1_wire_inline, x-profile, depression 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

1.3 hill 90 m

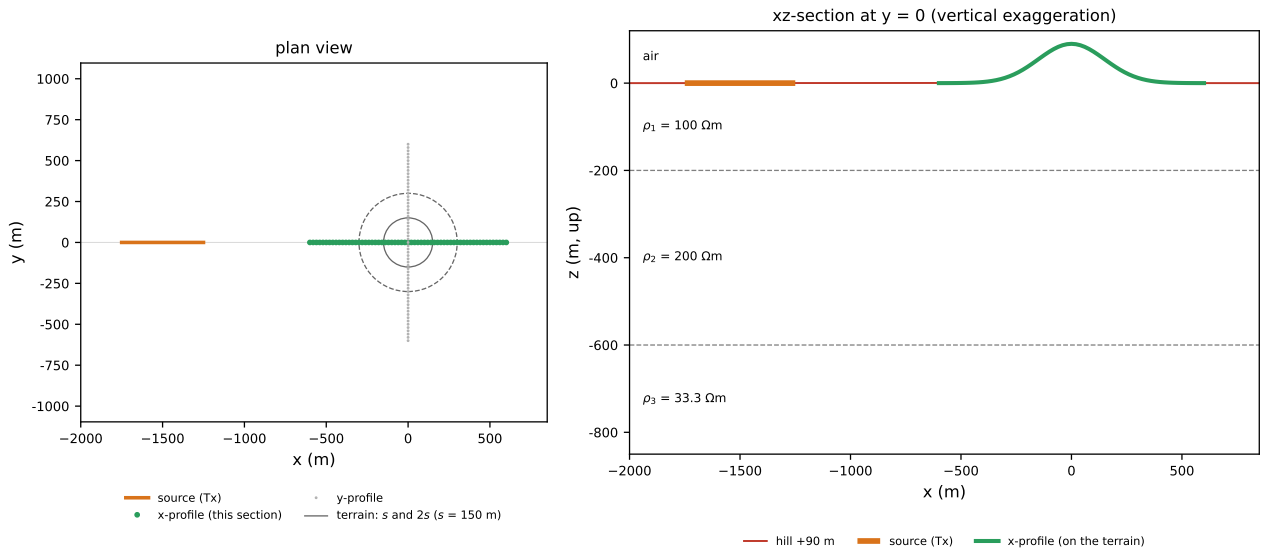


Figure 5: Setup of T1_wire_inline, x-profile, hill 90 m: plan view (left) and xz-section at $y = 0$ (right).

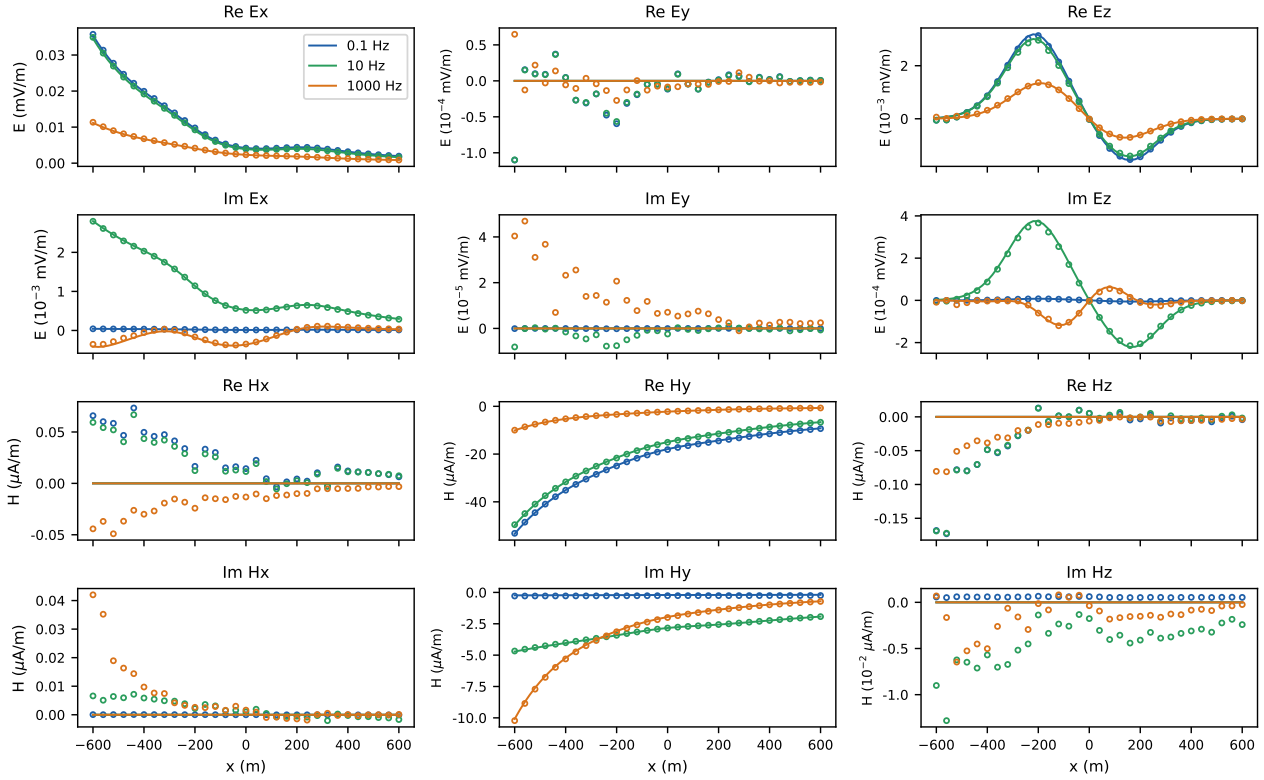


Figure 6: T1_wire_inline, x-profile, hill 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

1.4 depression 90 m

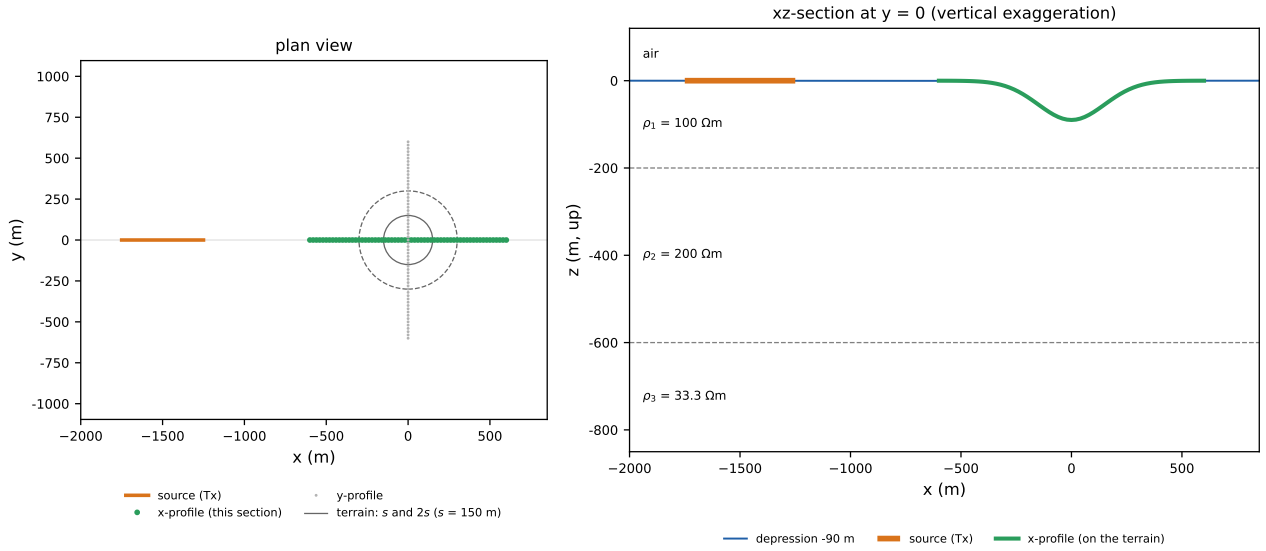


Figure 7: Setup of T1_wire_inline, x-profile, depression 90 m: plan view (left) and xz-section at $y = 0$ (right).

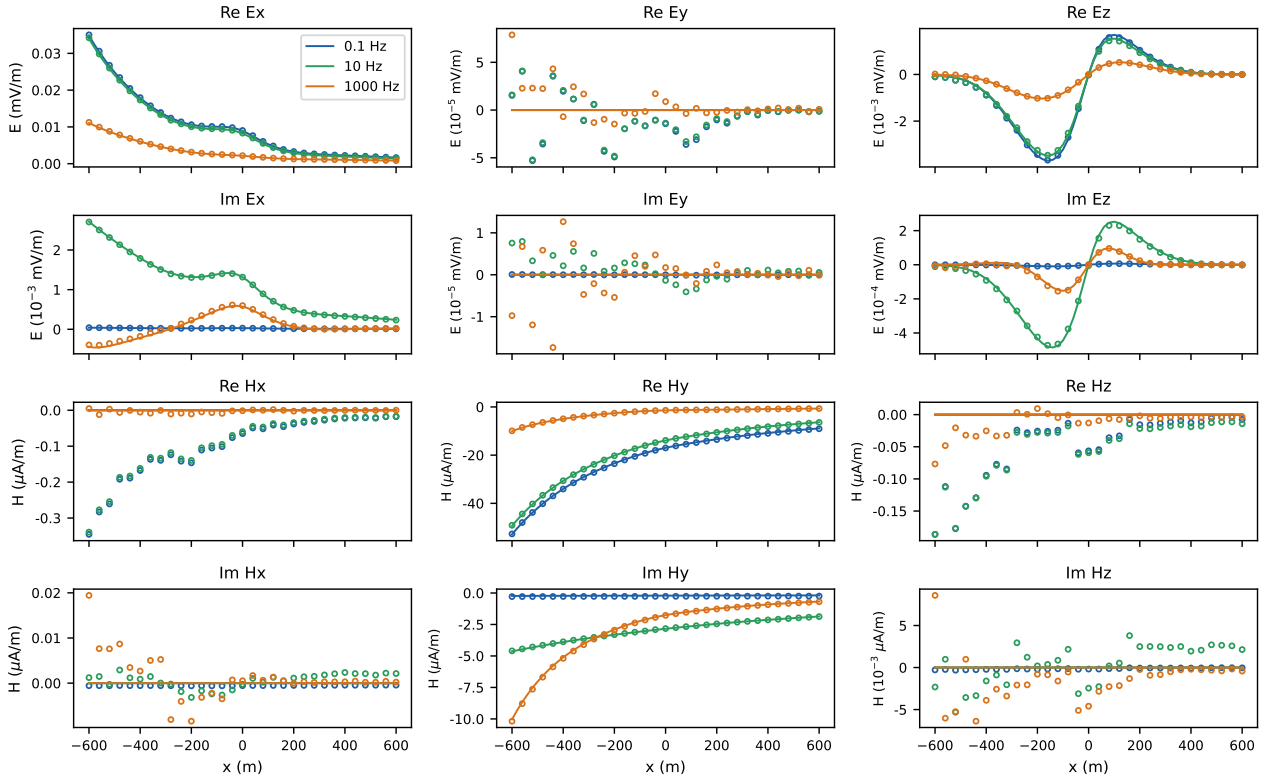


Figure 8: T1_wire_inline, x-profile, depression 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

2 T1_wire_inline, y-profile

elfe3D comparison T1: grounded wire in line with the x-profile, 500 m, centre 1500 m left of the hill.

2.1 hill 30 m

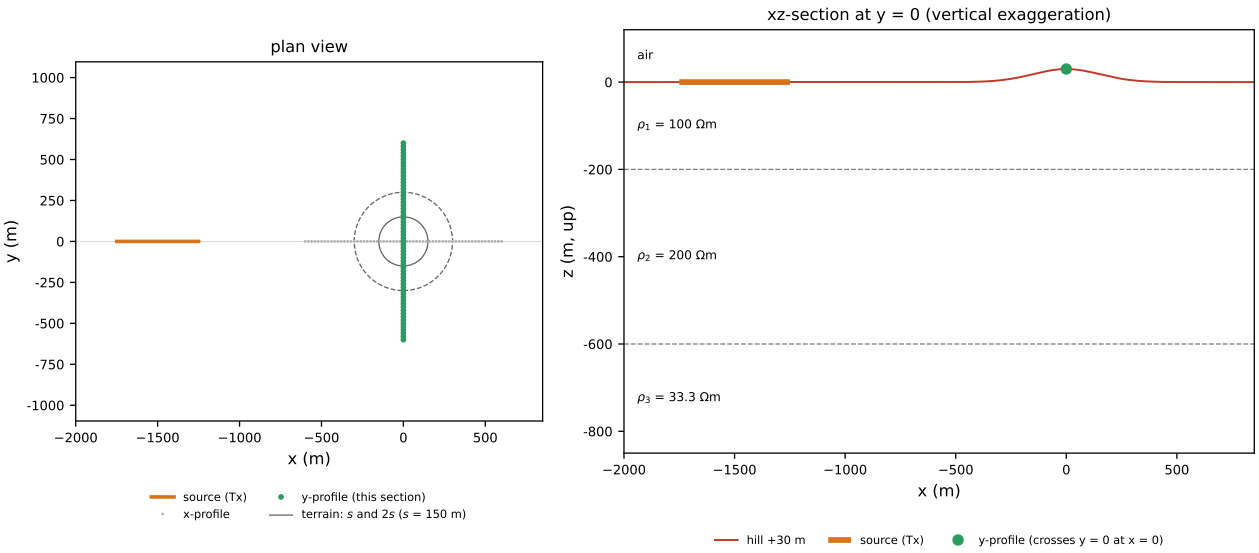


Figure 9: Setup of T1_wire_inline, y-profile, hill 30 m: plan view (left) and xz-section at $y = 0$ (right).

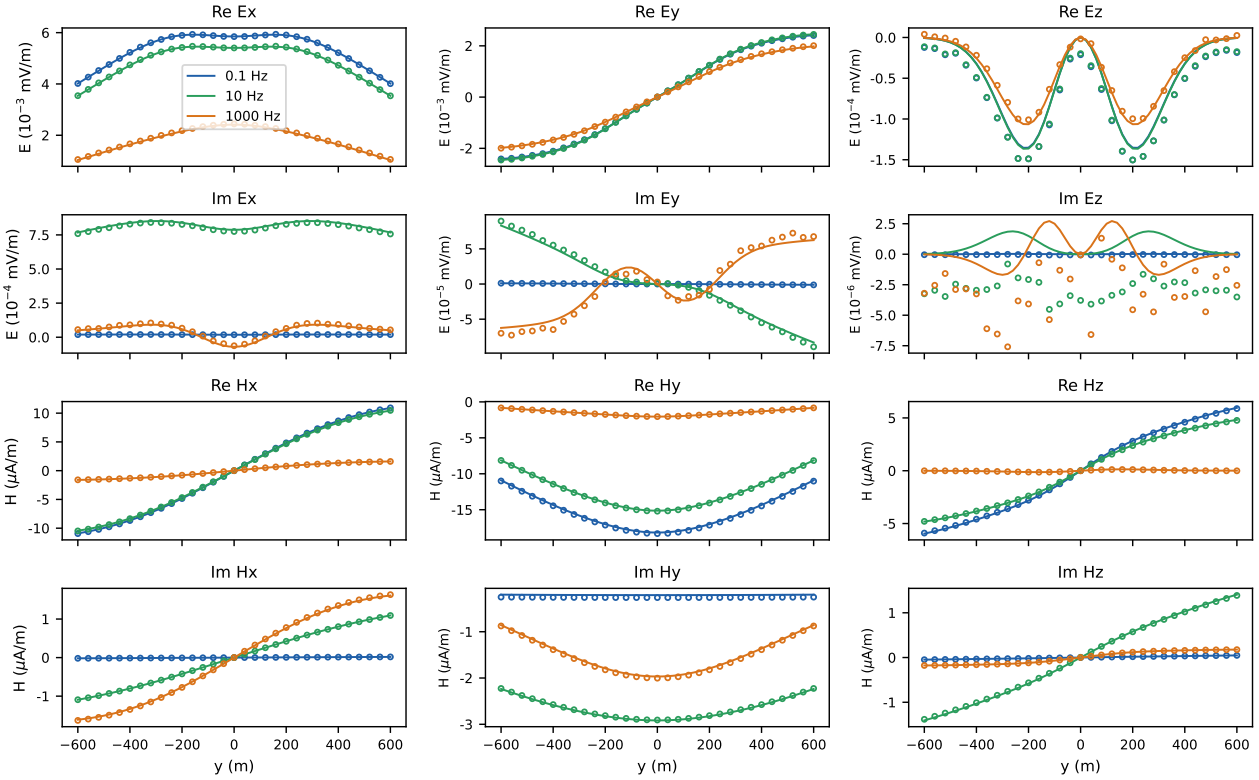


Figure 10: T1_wire_inline, y-profile, hill 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

2.2 depression 30 m

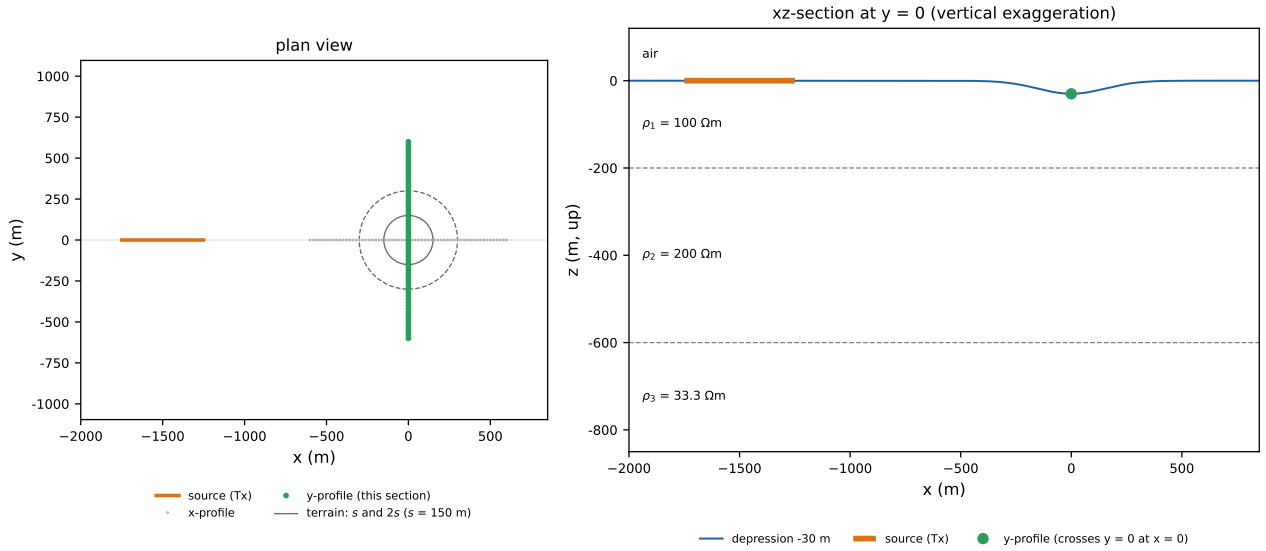


Figure 11: Setup of T1_wire_inline, y-profile, depression 30 m: plan view (left) and xz-section at $y = 0$ (right).

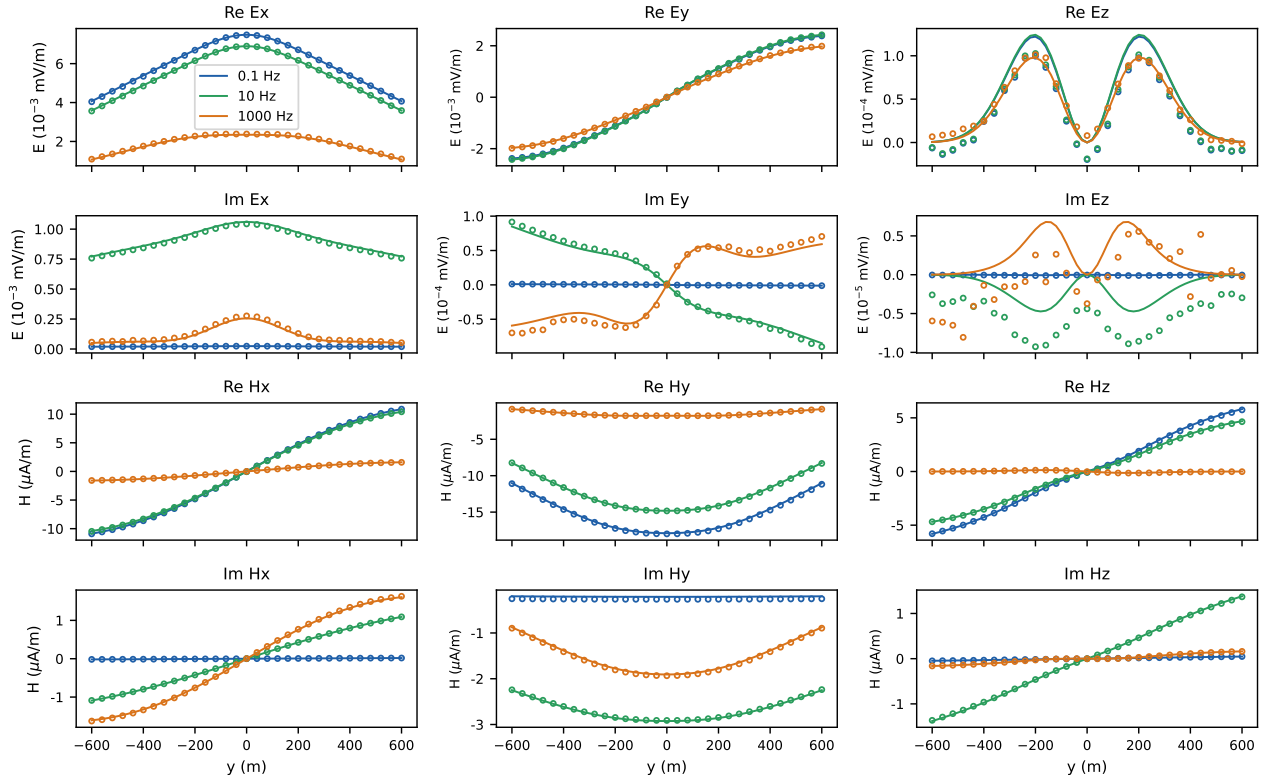


Figure 12: T1_wire_inline, y-profile, depression 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

2.3 hill 90 m

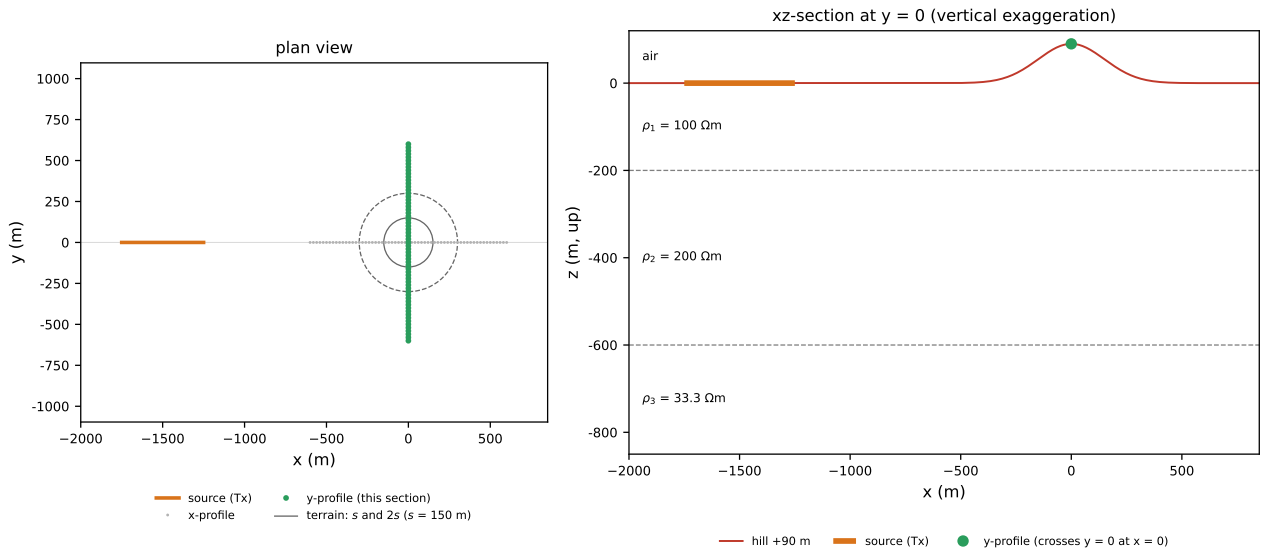


Figure 13: Setup of T1_wire_inline, y-profile, hill 90 m: plan view (left) and xz-section at $y = 0$ (right).

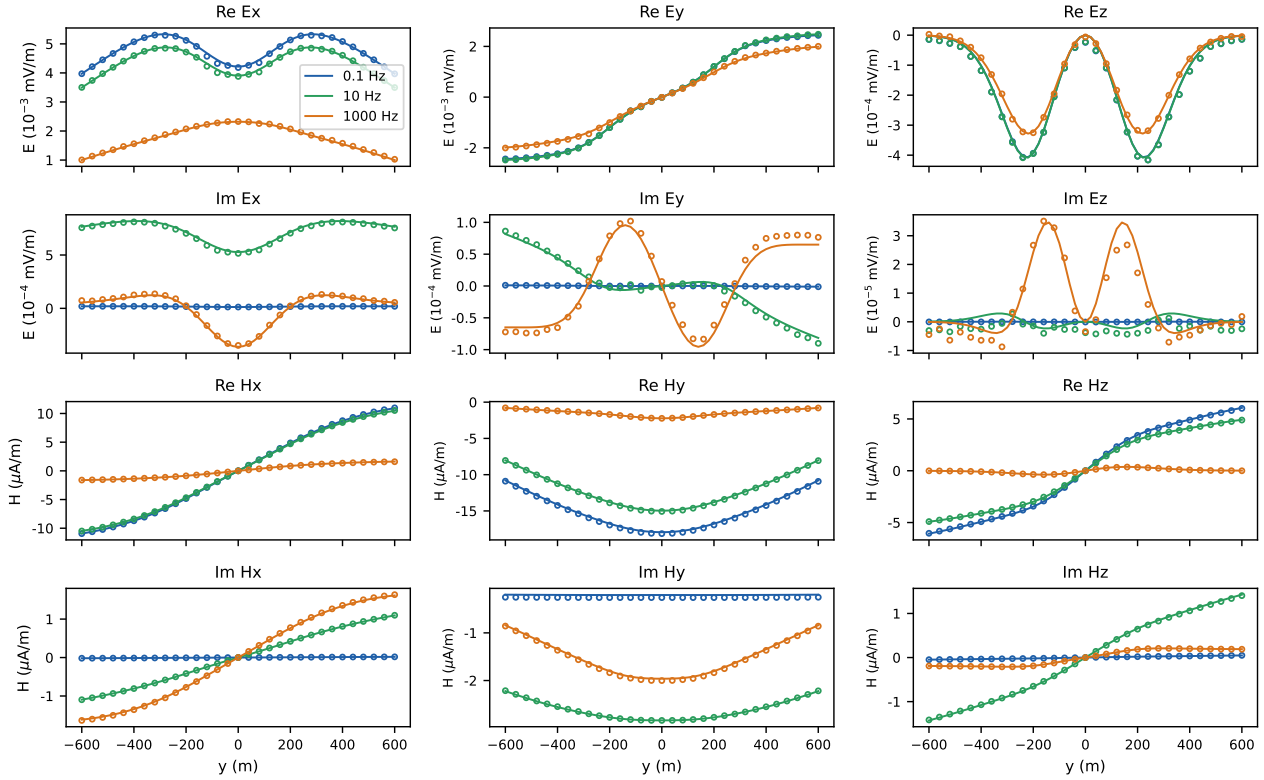


Figure 14: T1_wire_inline, y-profile, hill 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

2.4 depression 90 m

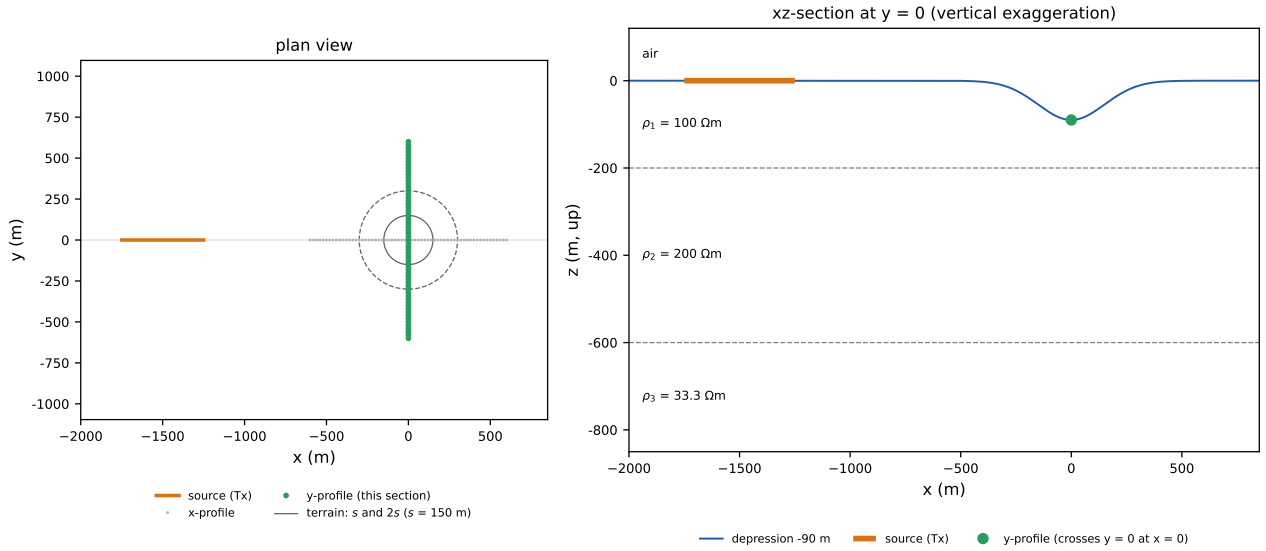


Figure 15: Setup of T1_wire_inline, y-profile, depression 90 m: plan view (left) and xz-section at $y = 0$ (right).

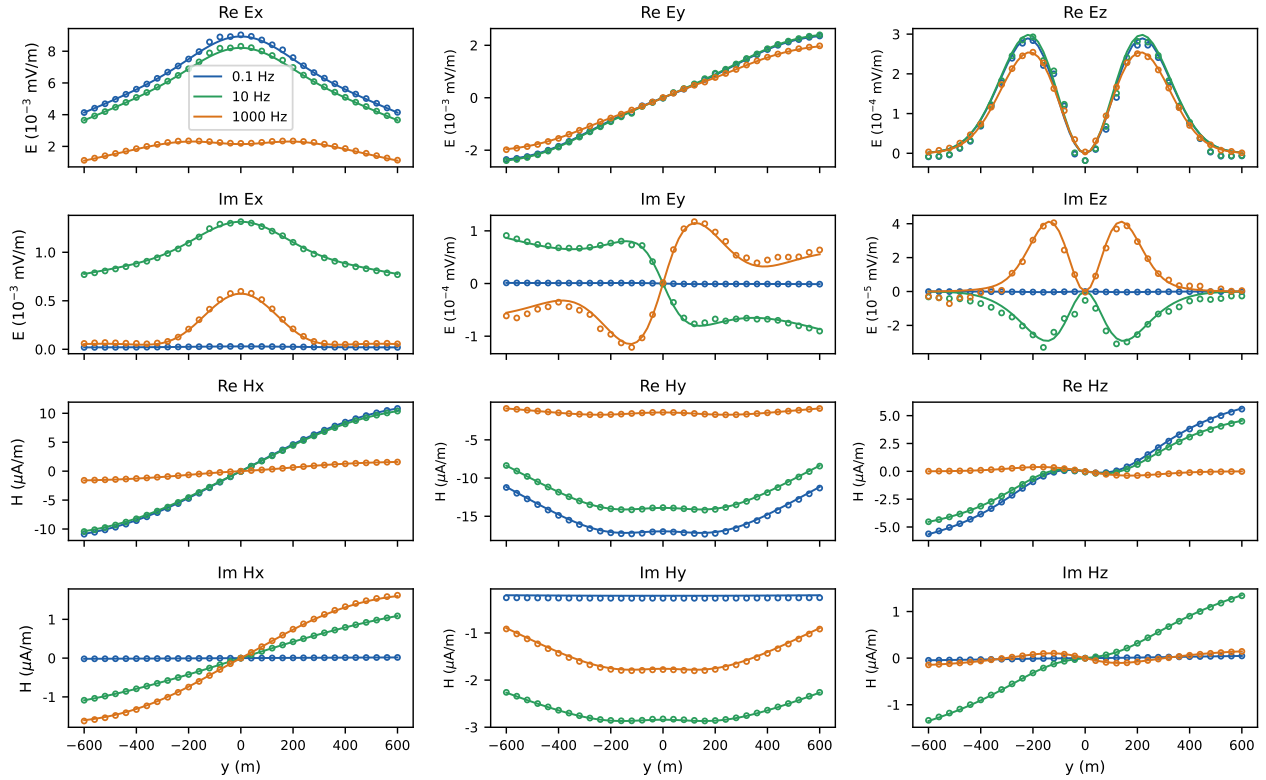


Figure 16: T1_wire_inline, y-profile, depression 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

3 T2_wire_broadside, x-profile

elfe3D comparison T2: grounded wire broadside (along y), 500 m, centre 1500 m left of the hill.

3.1 hill 30 m

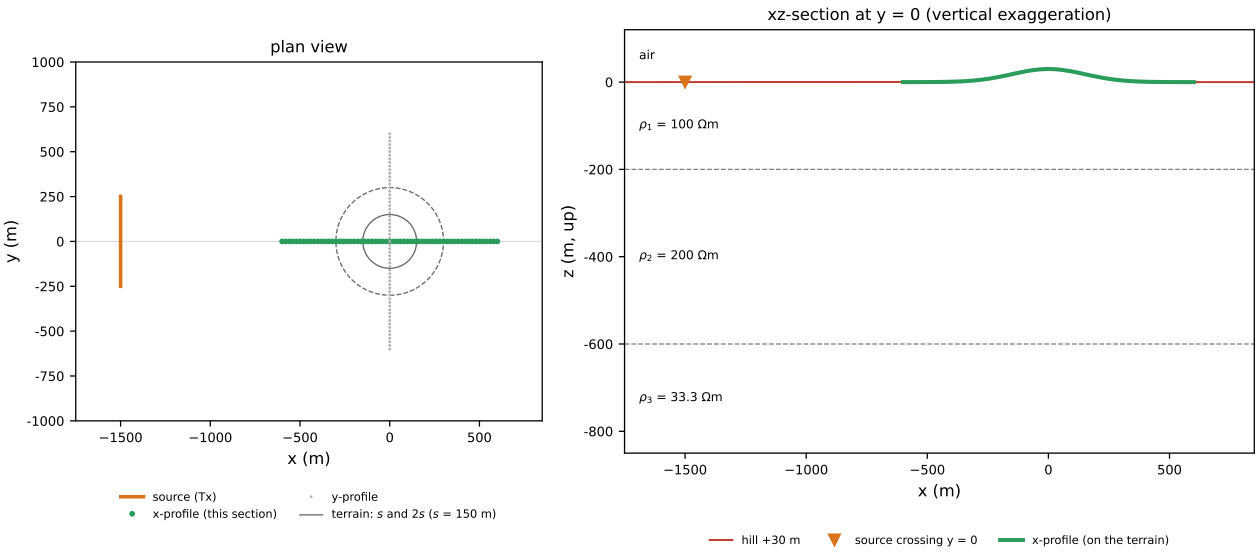


Figure 17: Setup of T2_wire_broadside, x-profile, hill 30 m: plan view (left) and xz-section at $y = 0$ (right).

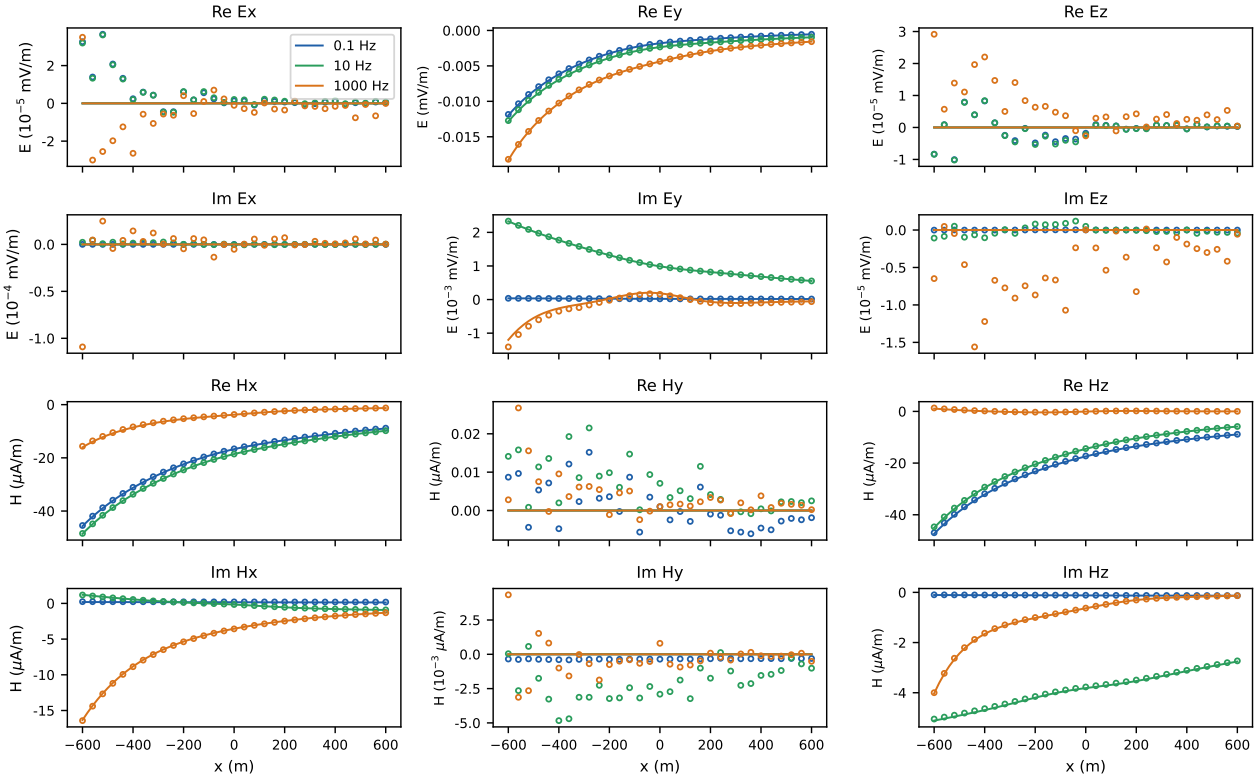


Figure 18: T2_wire_broadside, x-profile, hill 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

3.2 depression 30 m

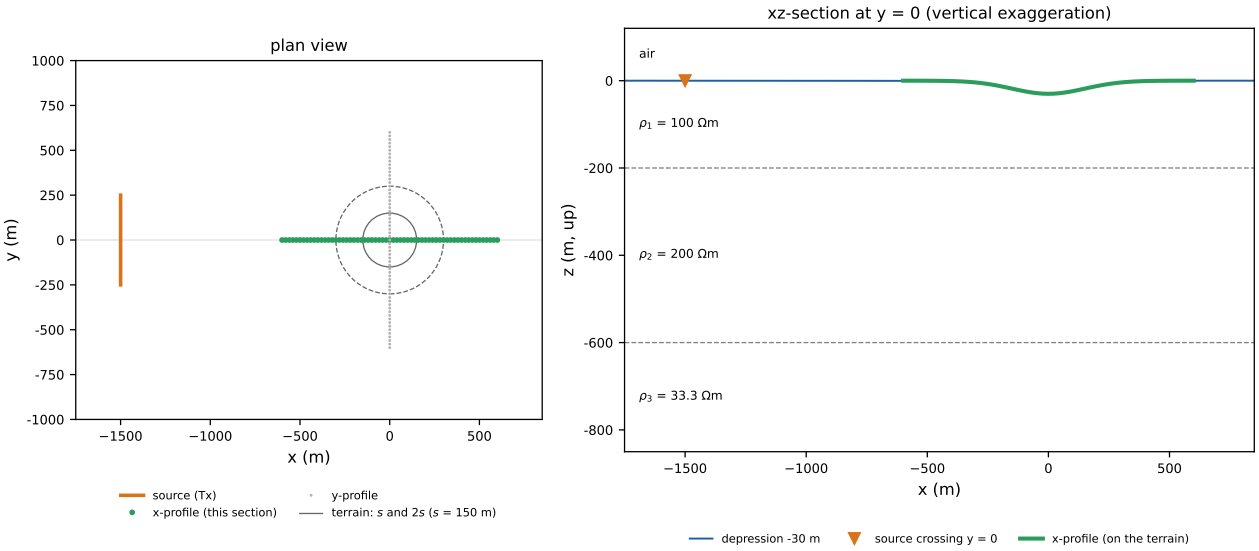


Figure 19: Setup of T2_wire_broadside, x-profile, depression 30 m: plan view (left) and xz-section at $y = 0$ (right).

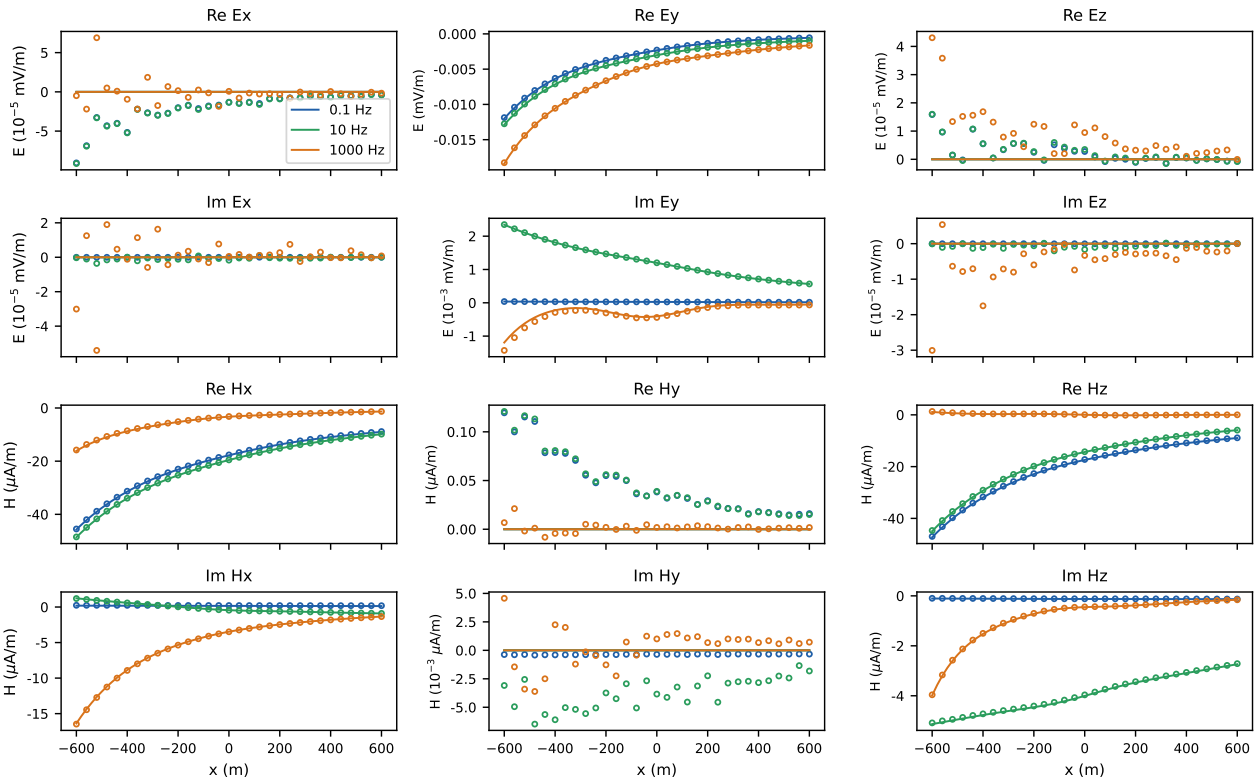


Figure 20: T2_wire_broadside, x-profile, depression 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

3.3 hill 90 m

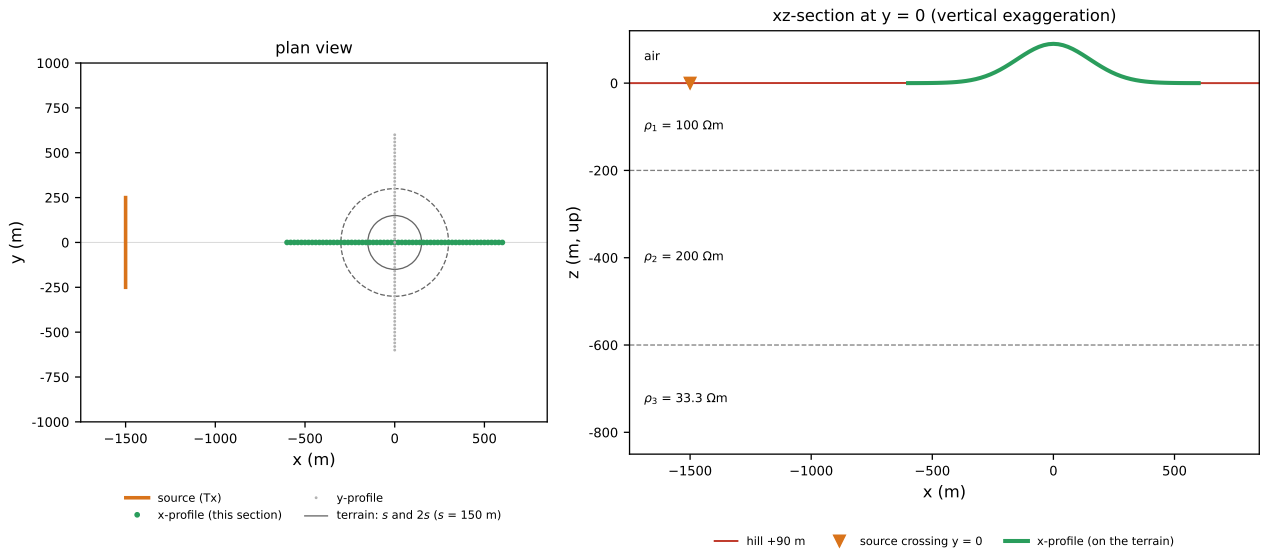


Figure 21: Setup of T2_wire_broadside, x-profile, hill 90 m: plan view (left) and xz-section at $y = 0$ (right).

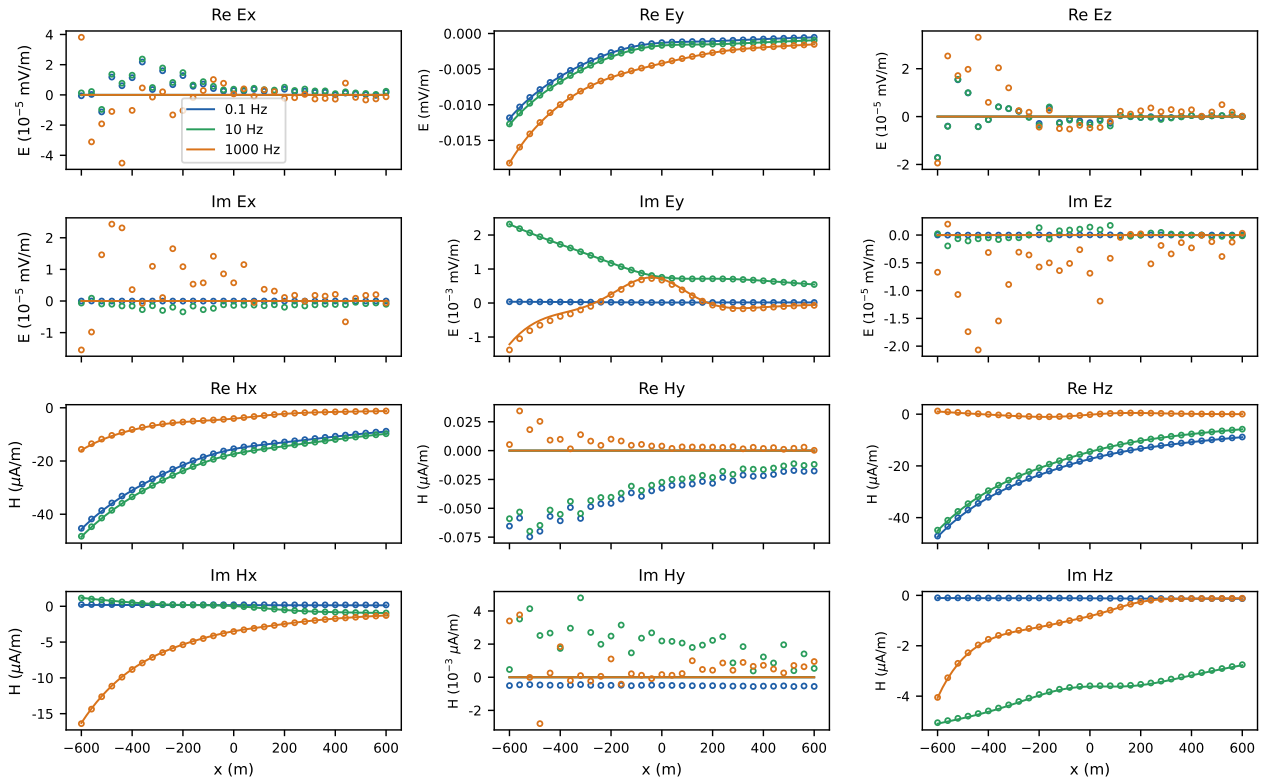


Figure 22: T2_wire_broadside, x-profile, hill 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

3.4 depression 90 m

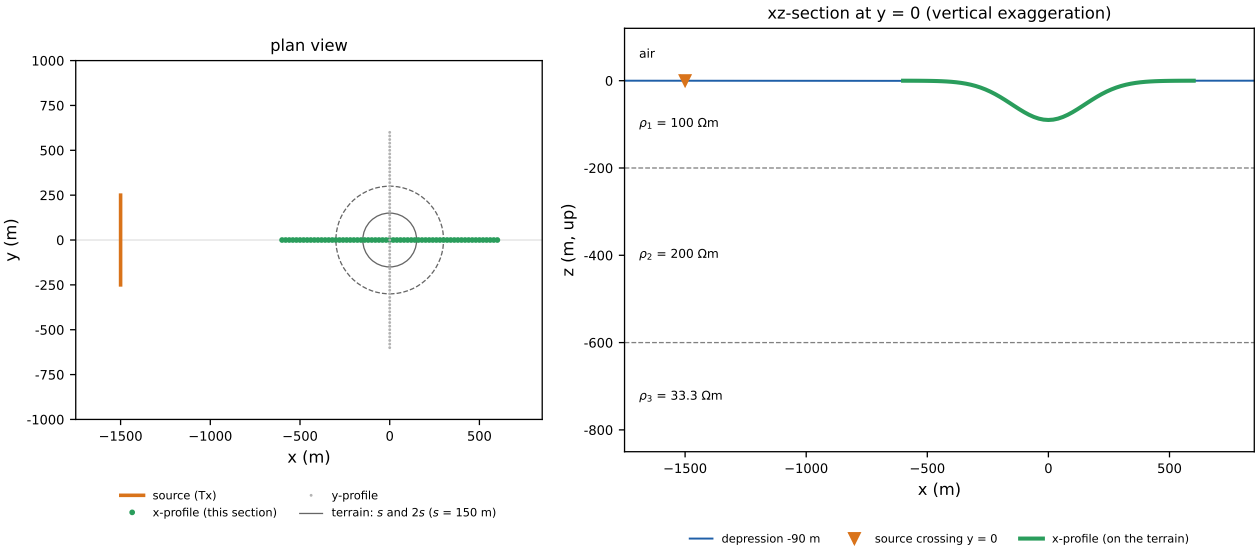


Figure 23: Setup of T2_wire_broadside, x-profile, depression 90 m: plan view (left) and xz-section at $y = 0$ (right).

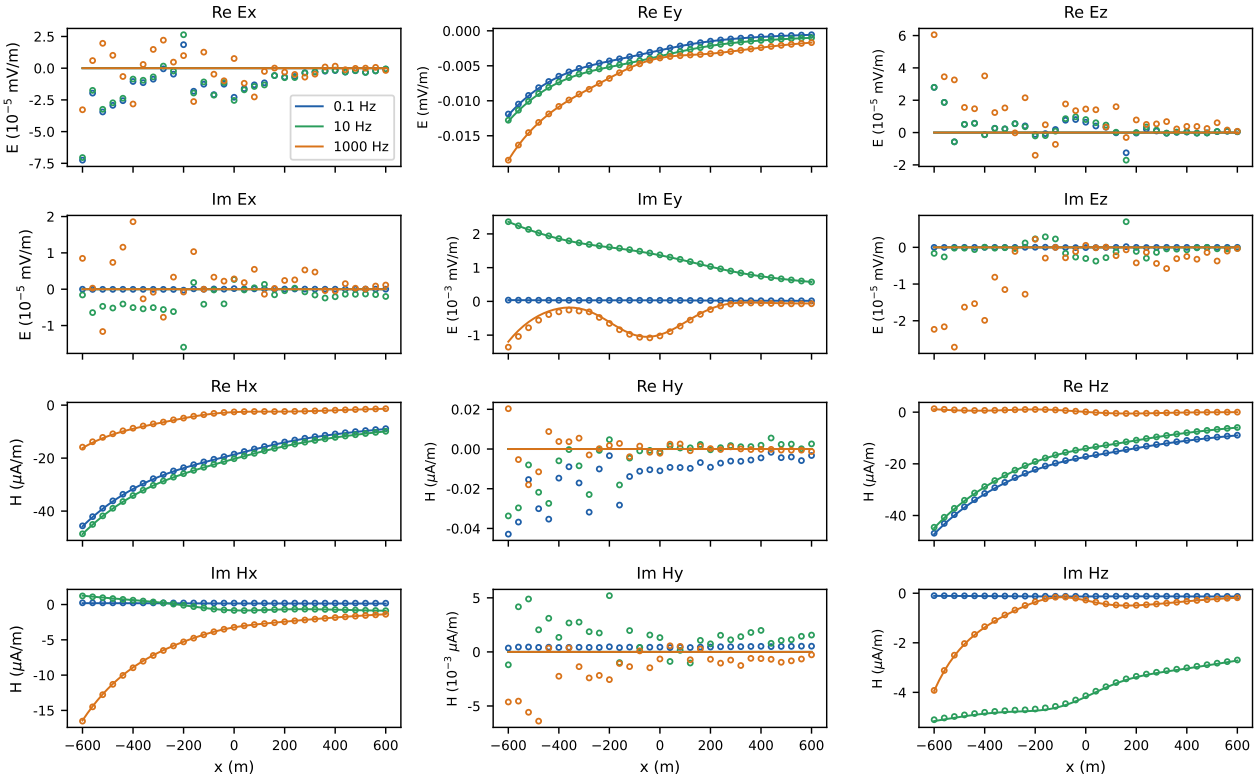


Figure 24: T2_wire_broadside, x-profile, depression 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

4 T2_wire_broadside, y-profile

elfe3D comparison T2: grounded wire broadside (along y), 500 m, centre 1500 m left of the hill.

4.1 hill 30 m

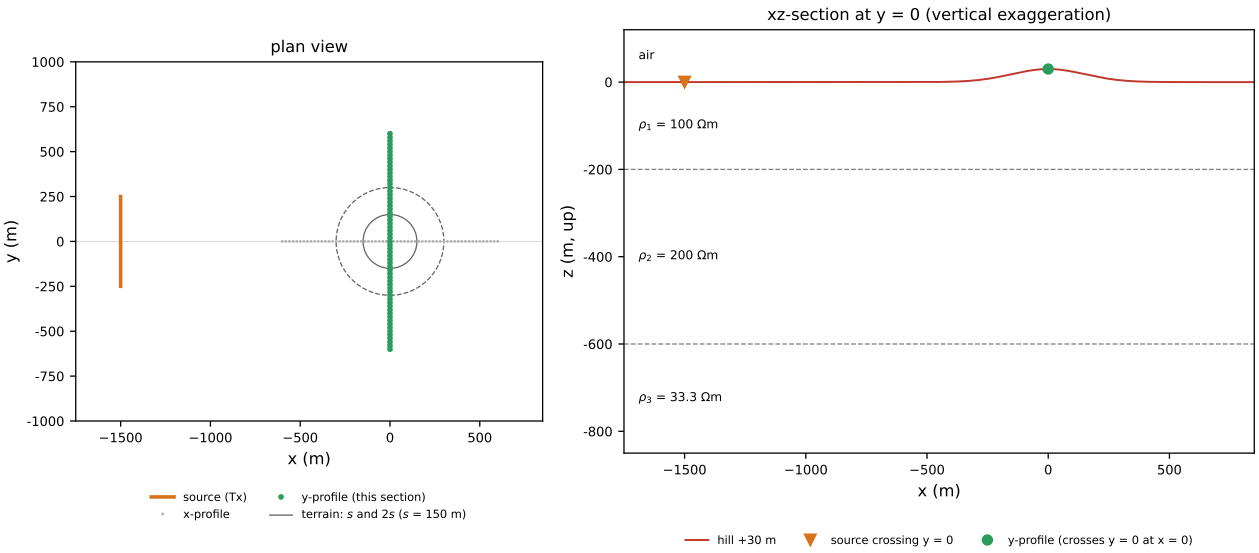


Figure 25: Setup of T2_wire_broadside, y-profile, hill 30 m: plan view (left) and xz-section at $y = 0$ (right).

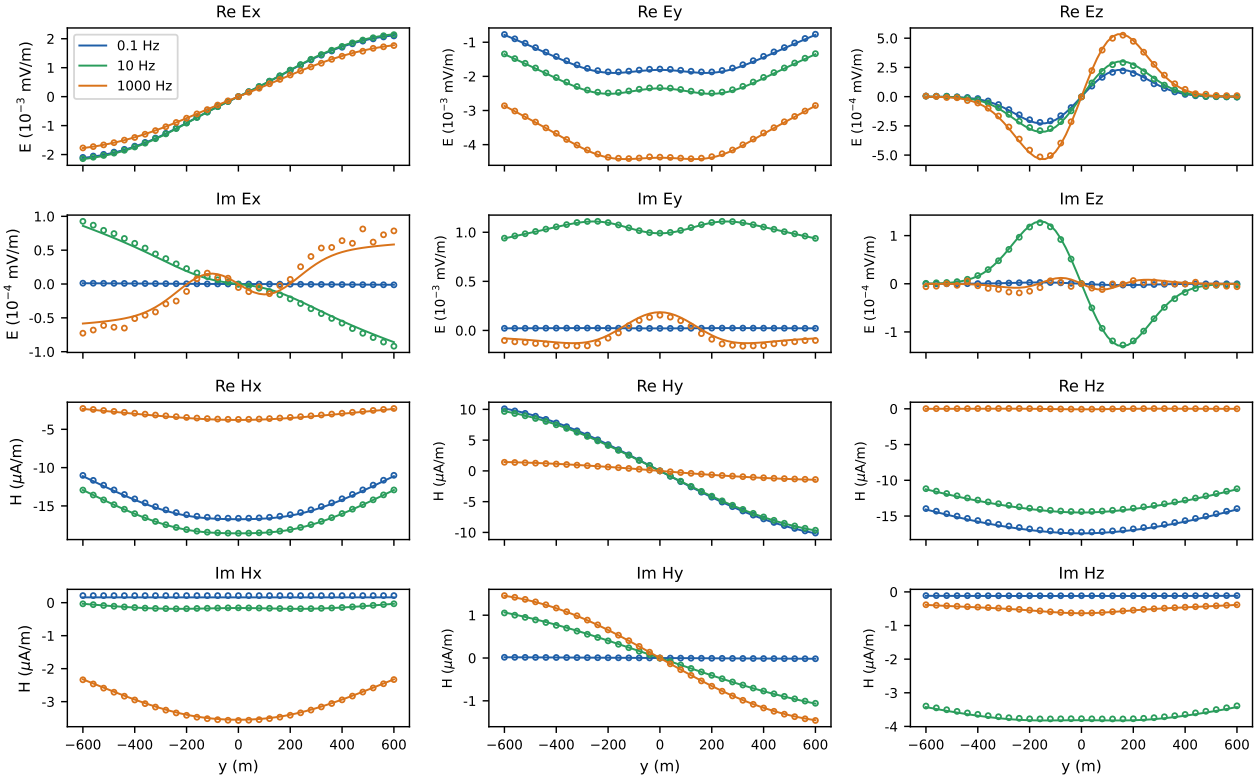


Figure 26: T2_wire_broadside, y-profile, hill 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

4.2 depression 30 m

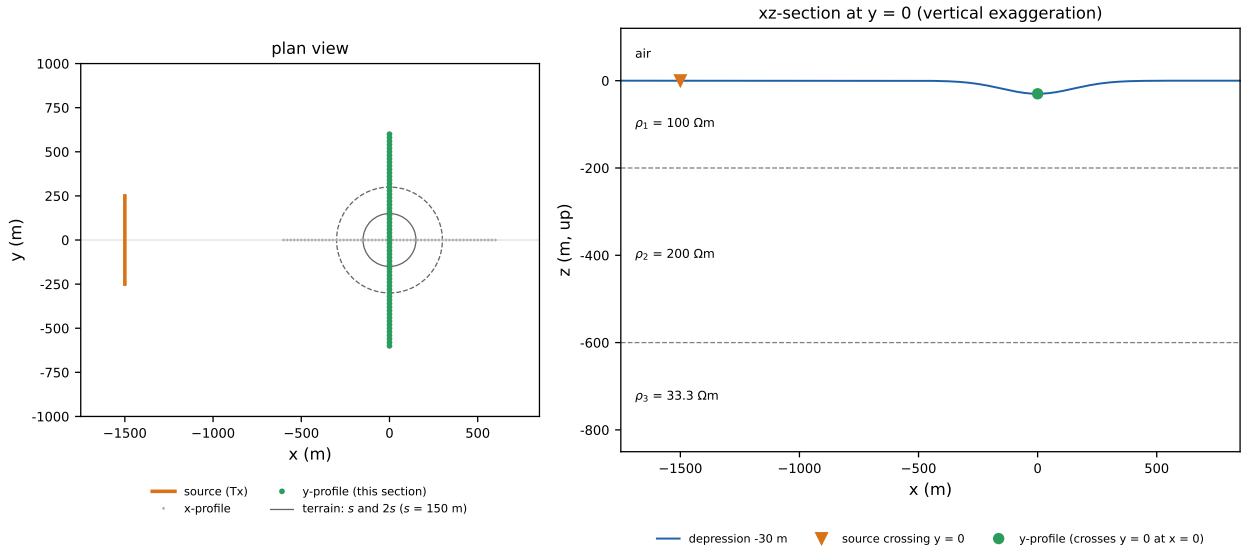


Figure 27: Setup of T2_wire_broadside, y-profile, depression 30 m: plan view (left) and xz-section at $y = 0$ (right).

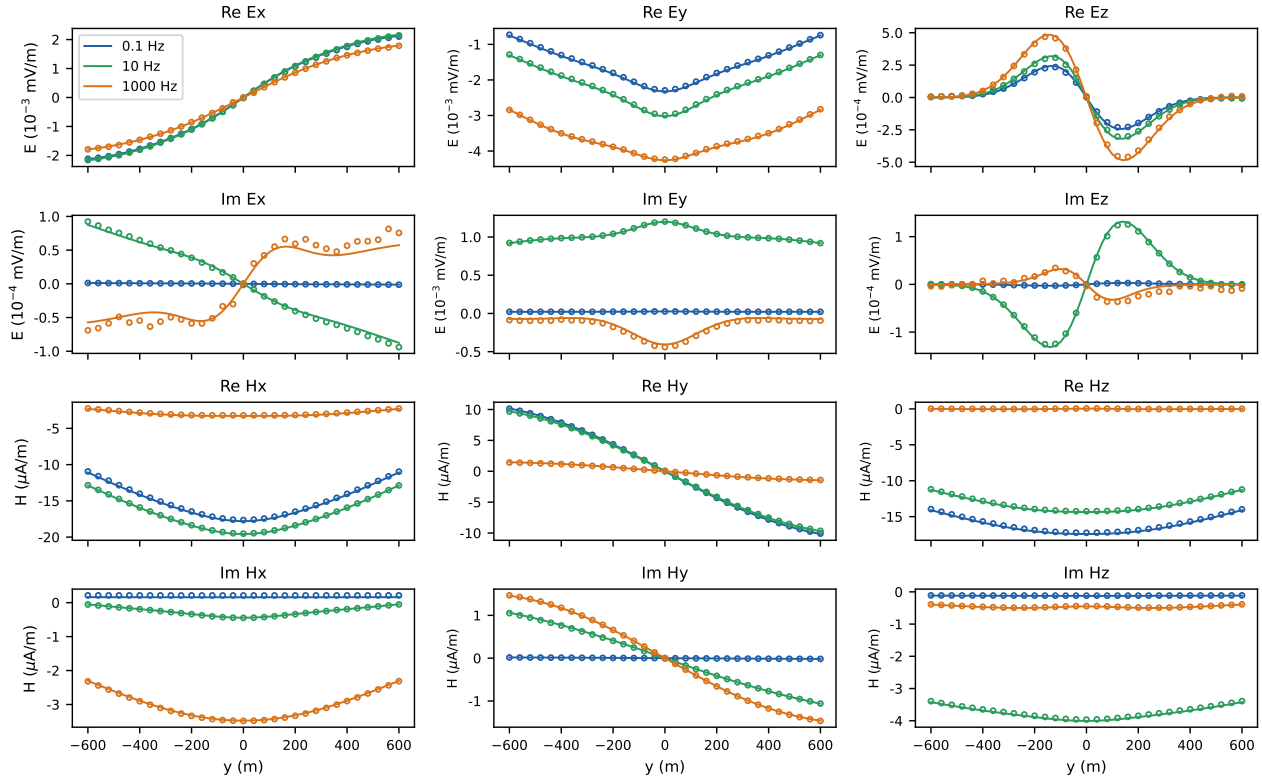


Figure 28: T2_wire_broadside, y-profile, depression 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

4.3 hill 90 m

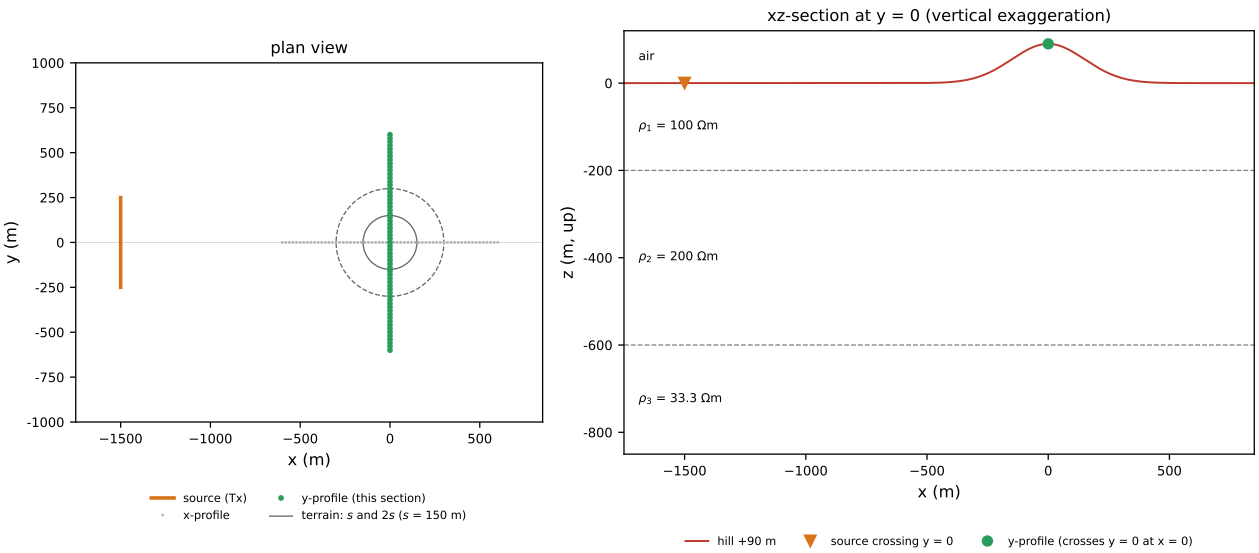


Figure 29: Setup of T2_wire_broadside, y-profile, hill 90 m: plan view (left) and xz-section at $y = 0$ (right).

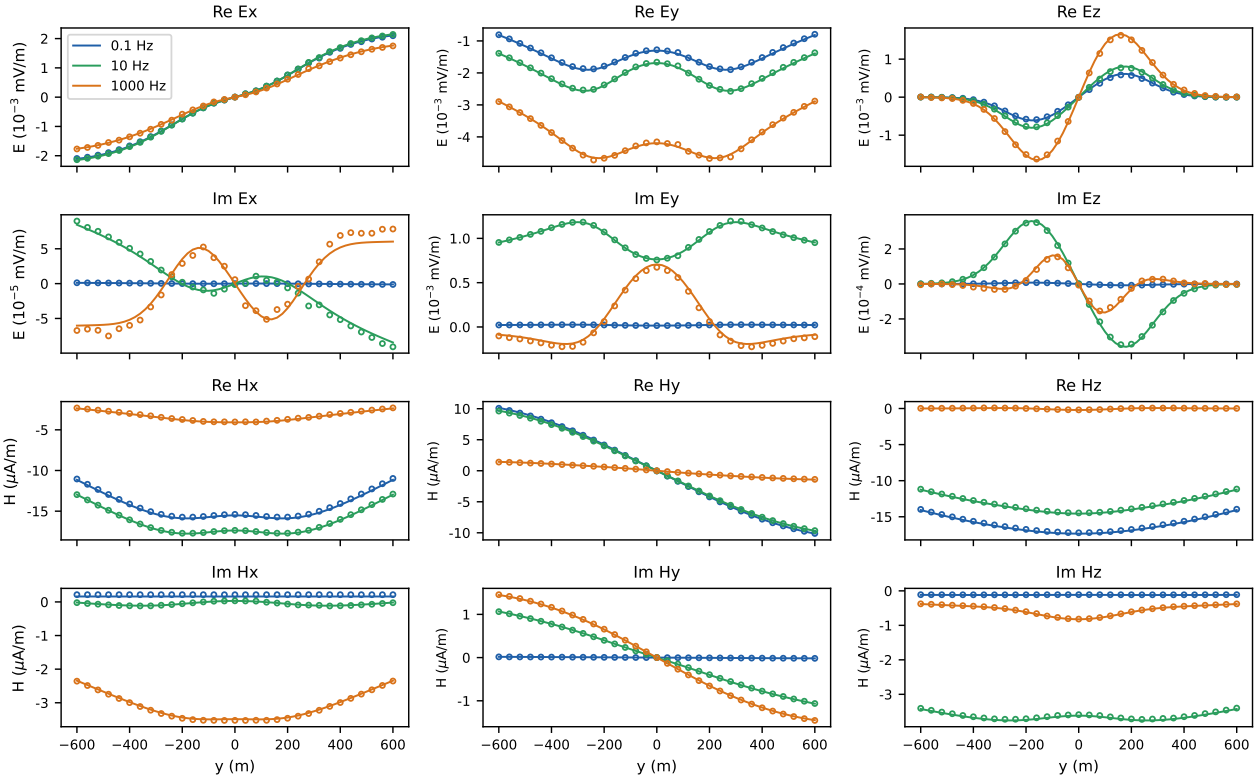


Figure 30: T2_wire_broadside, y-profile, hill 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

4.4 depression 90 m

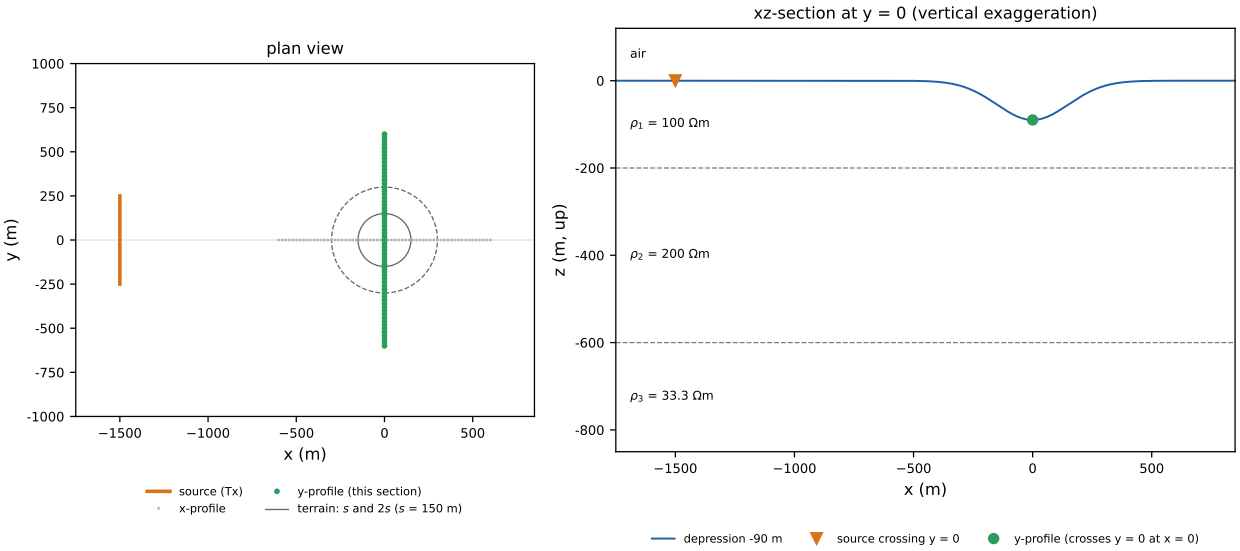


Figure 31: Setup of T2_wire_broadside, y-profile, depression 90 m: plan view (left) and xz-section at $y = 0$ (right).

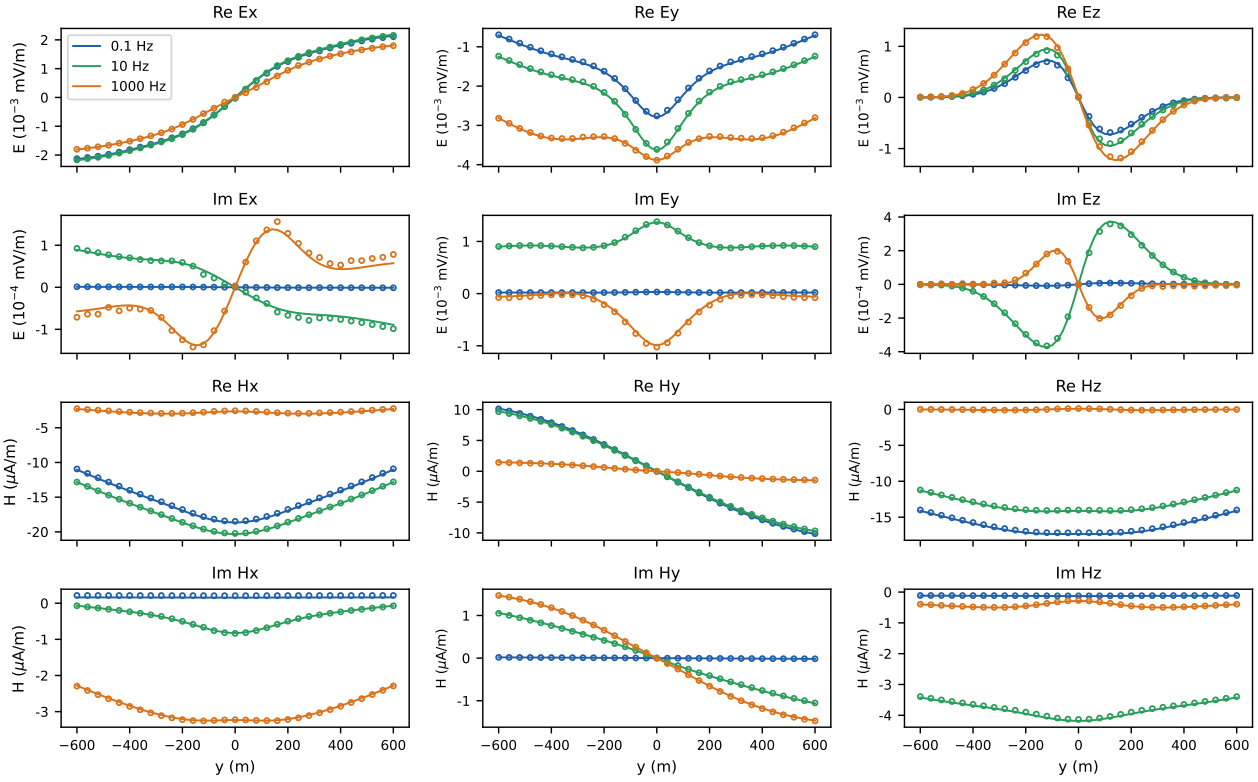


Figure 32: T2_wire_broadside, y-profile, depression 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

5 T3_loop_offset, x-profile

elfe3D comparison T3: square loop 400 x 400 m, centre (-1500, 0), counter-clockwise seen from above.

5.1 hill 30 m

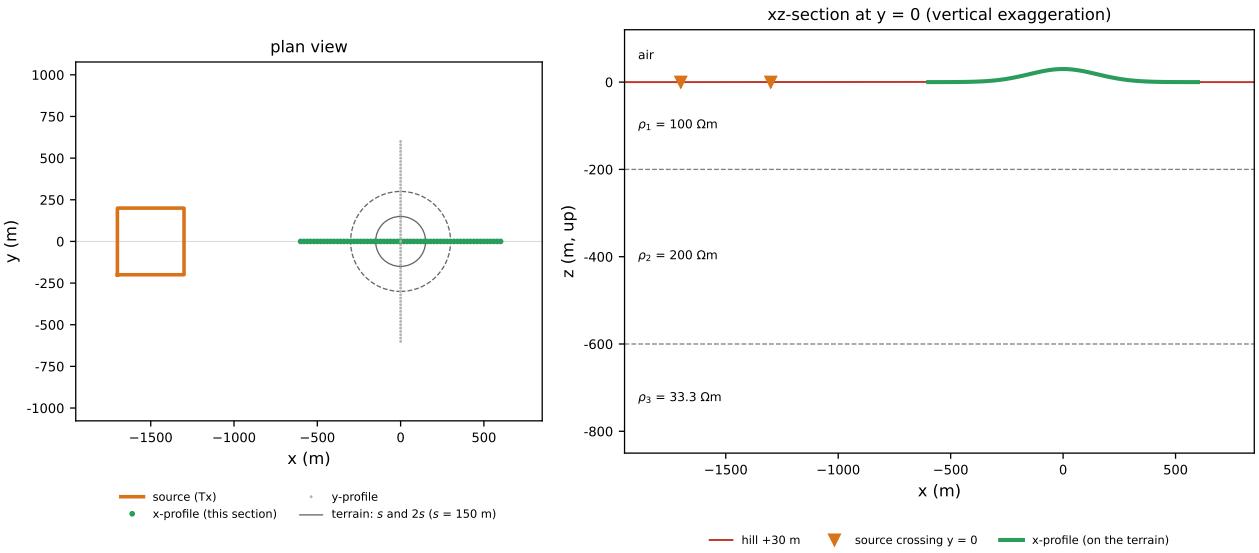


Figure 33: Setup of T3_loop_offset, x-profile, hill 30 m: plan view (left) and xz-section at $y = 0$ (right).

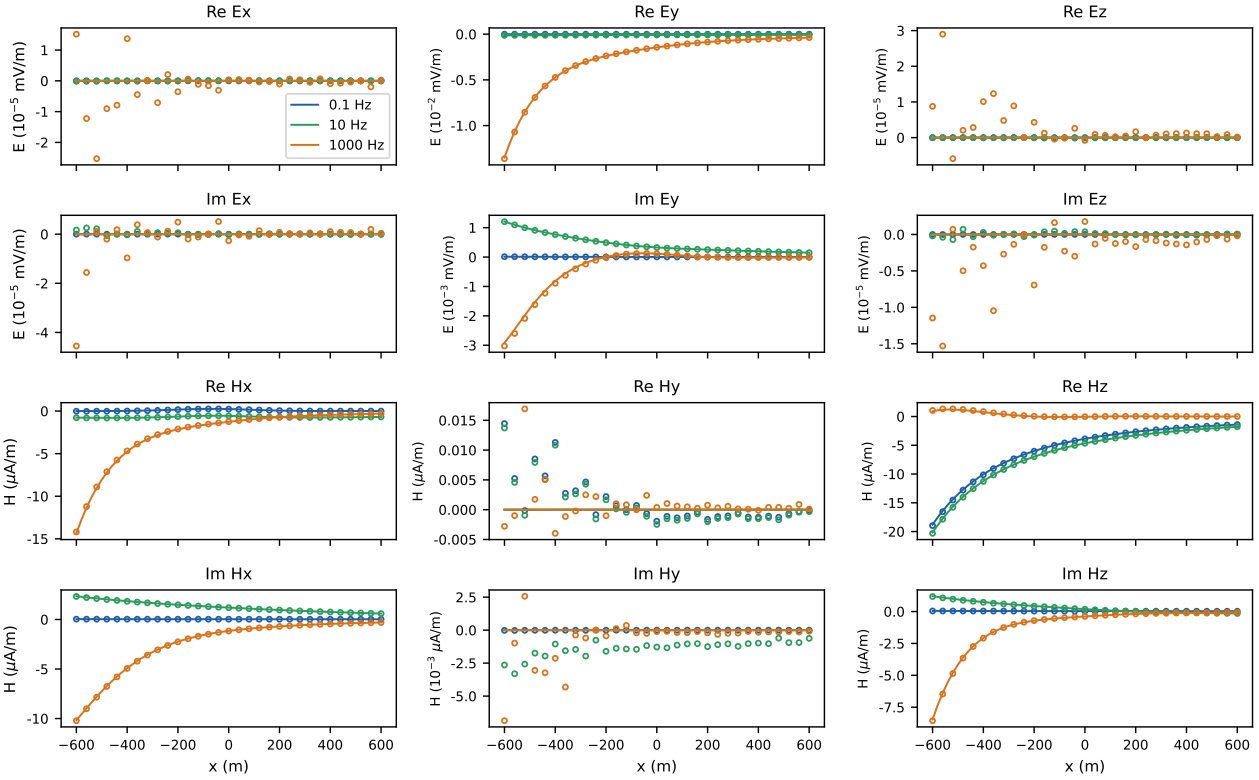


Figure 34: T3_loop_offset, x-profile, hill 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

5.2 depression 30 m

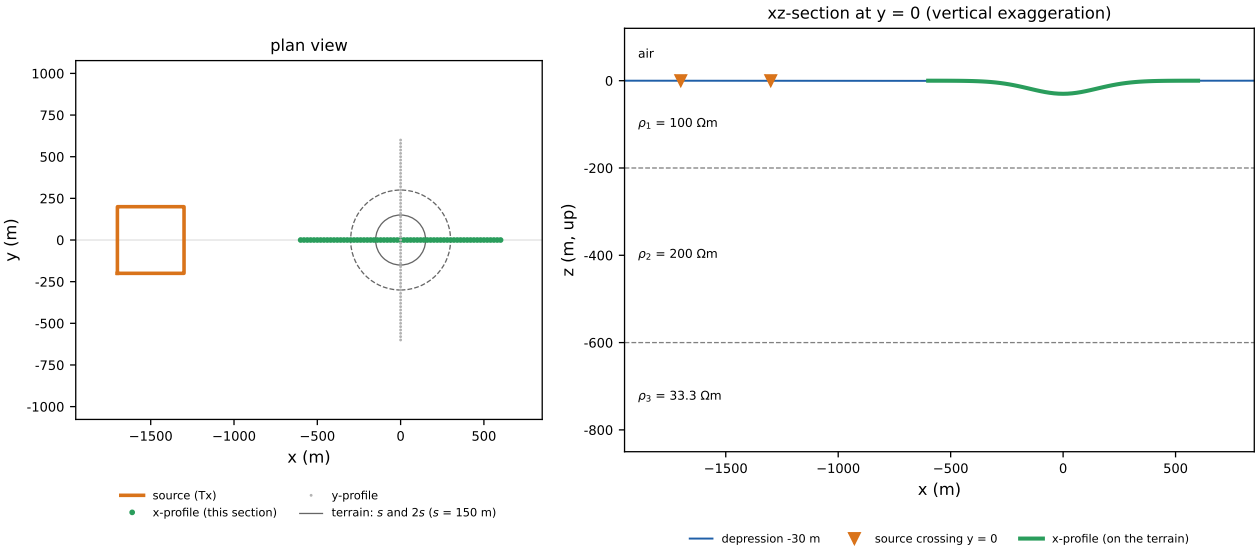


Figure 35: Setup of T3_loop_offset, x-profile, depression 30 m: plan view (left) and xz-section at $y = 0$ (right).

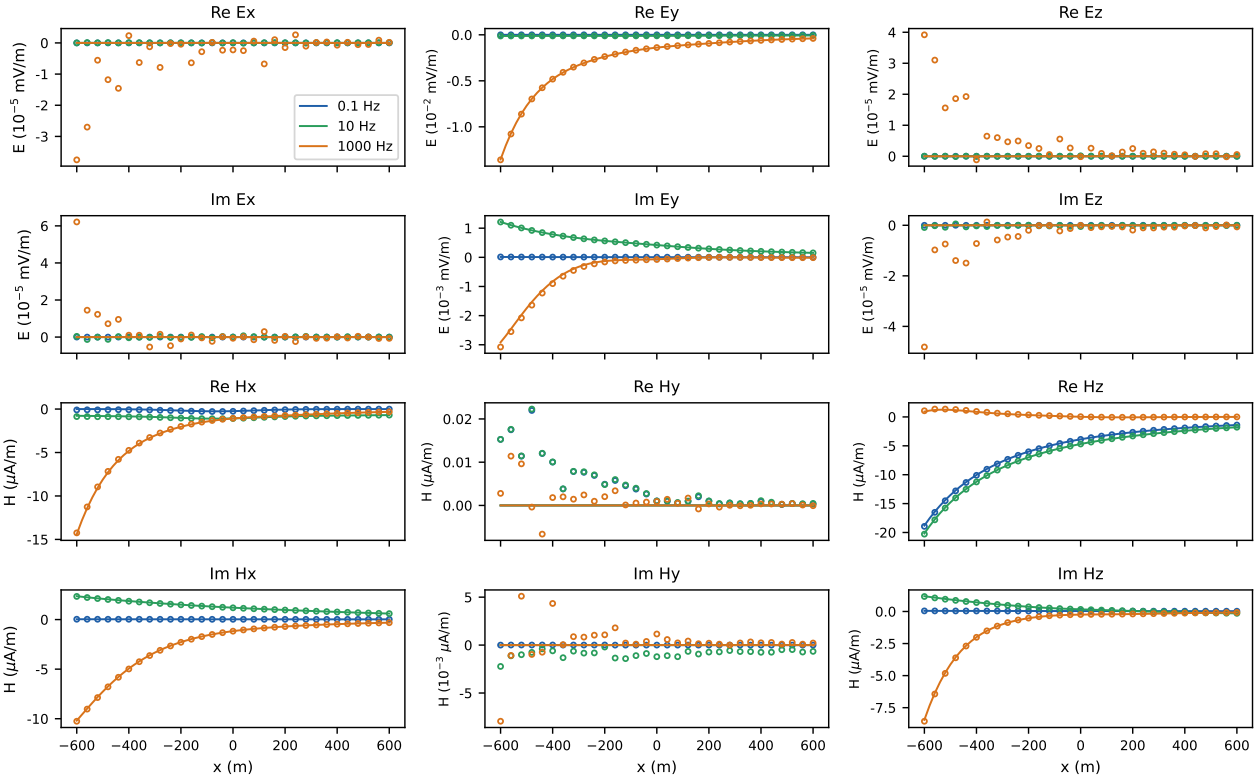


Figure 36: T3_loop_offset, x-profile, depression 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

5.3 hill 90 m

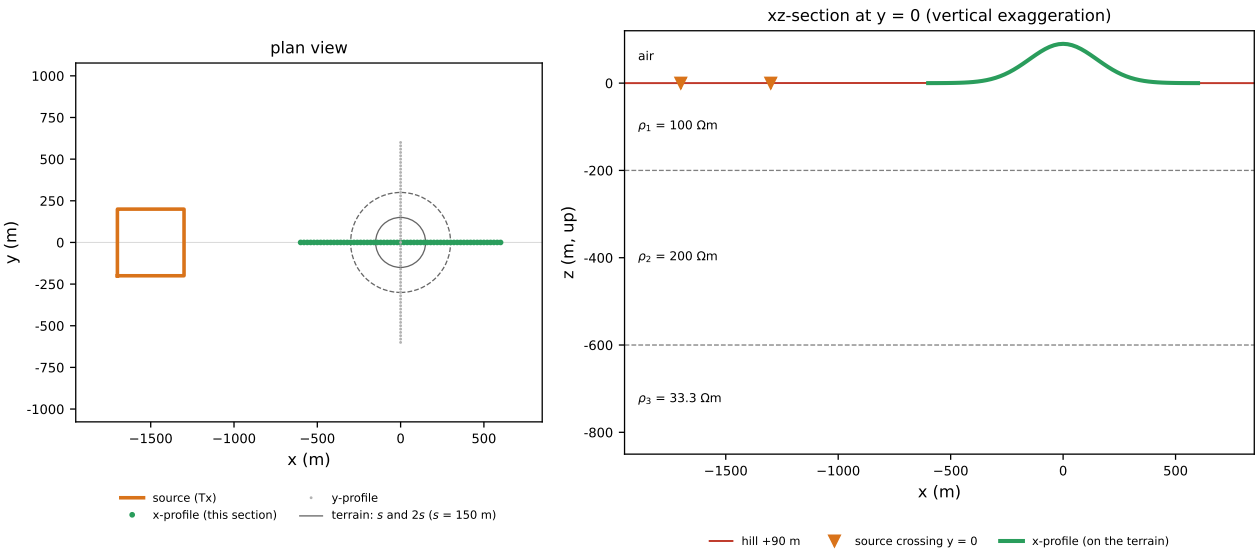


Figure 37: Setup of T3_loop_offset, x-profile, hill 90 m: plan view (left) and xz-section at $y = 0$ (right).

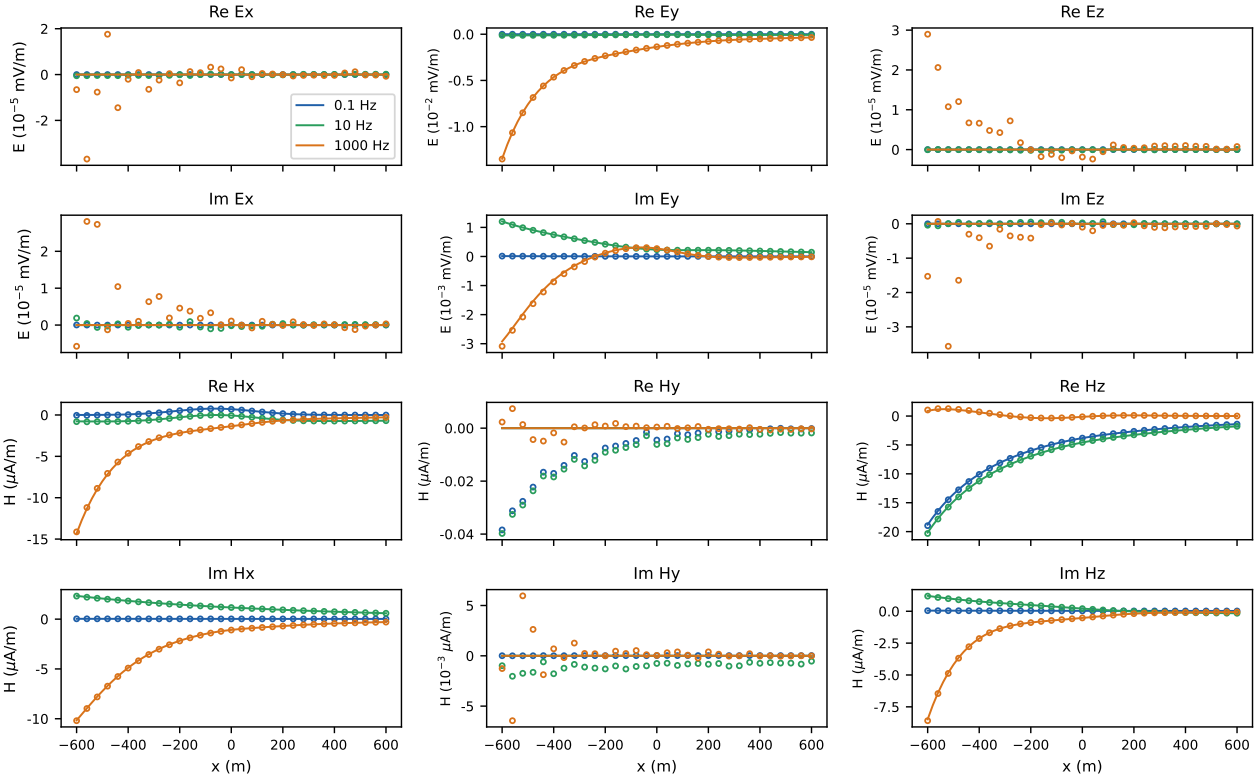


Figure 38: T3_loop_offset, x-profile, hill 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

5.4 depression 90 m

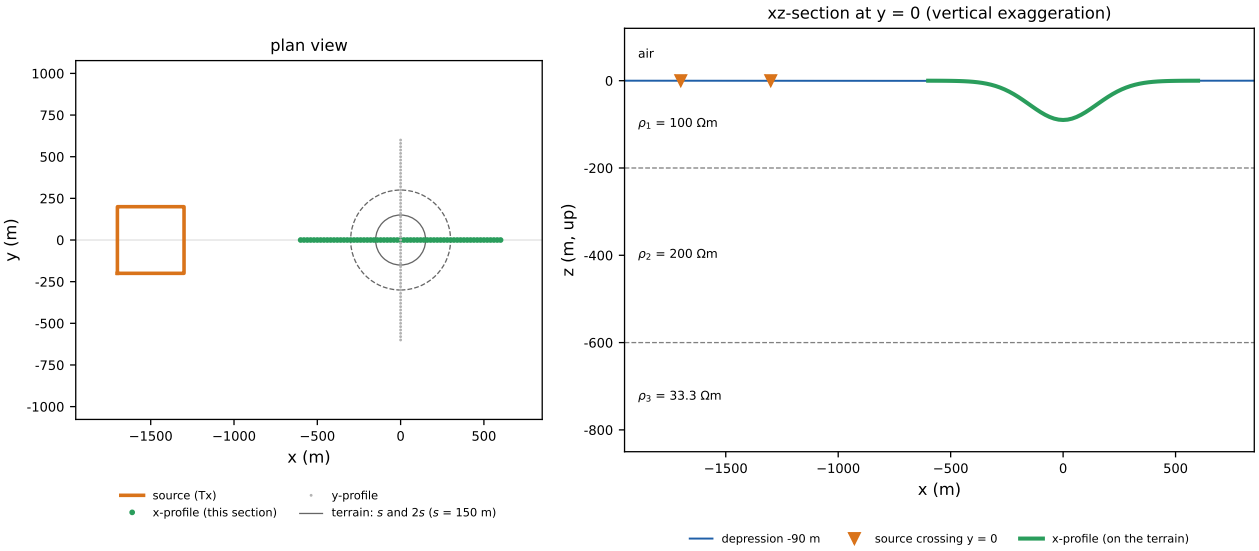


Figure 39: Setup of T3_loop_offset, x-profile, depression 90 m: plan view (left) and xz-section at $y = 0$ (right).

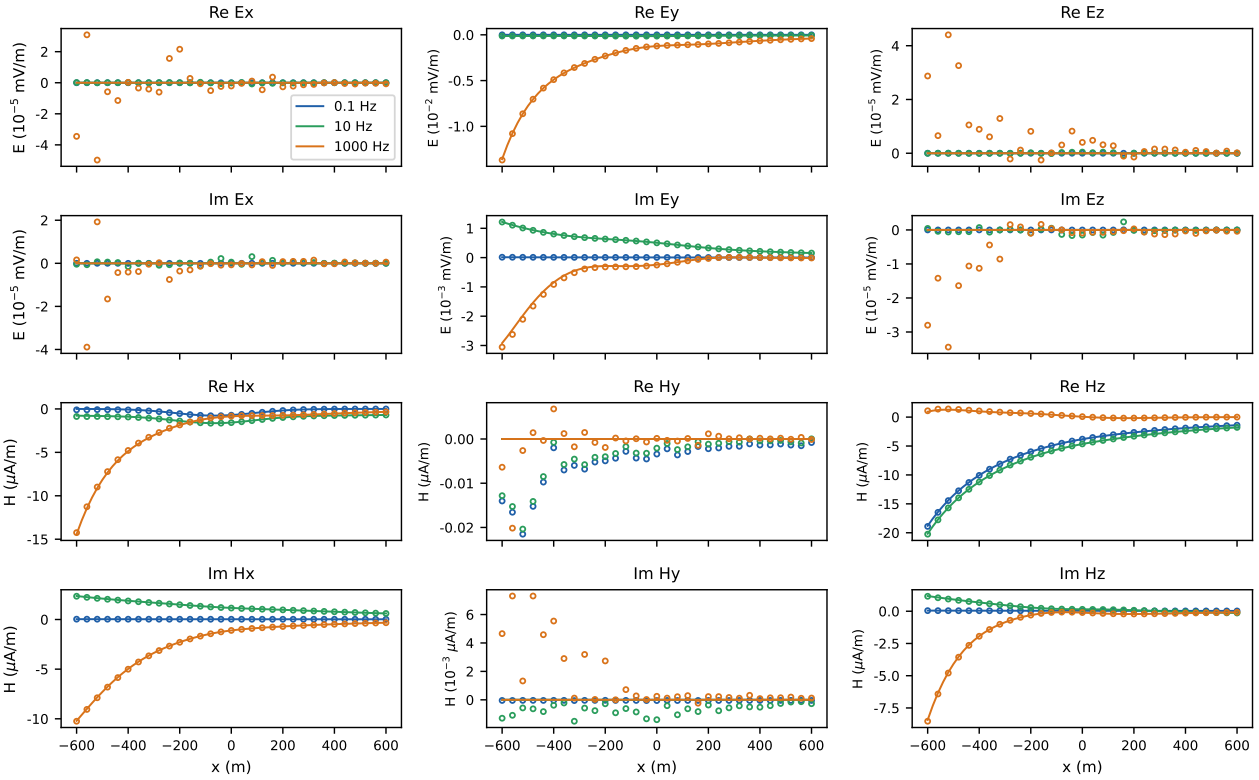


Figure 40: T3_loop_offset, x-profile, depression 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

6 T3_loop_offset, y-profile

elfe3D comparison T3: square loop 400 x 400 m, centre (-1500, 0), counter-clockwise seen from above.

6.1 hill 30 m

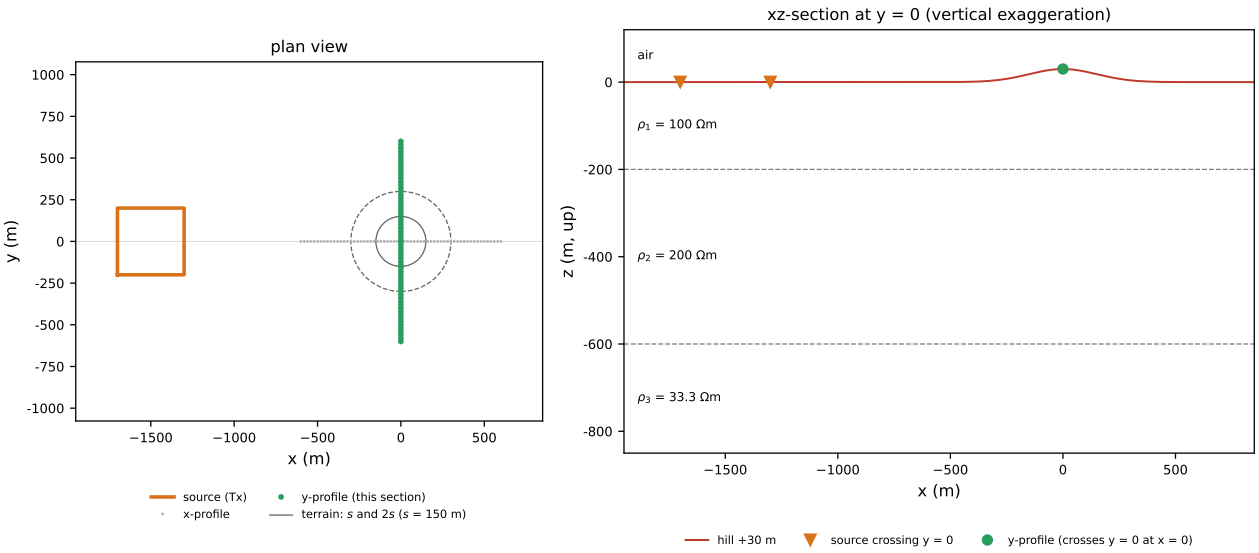


Figure 41: Setup of T3_loop_offset, y-profile, hill 30 m: plan view (left) and xz-section at $y = 0$ (right).

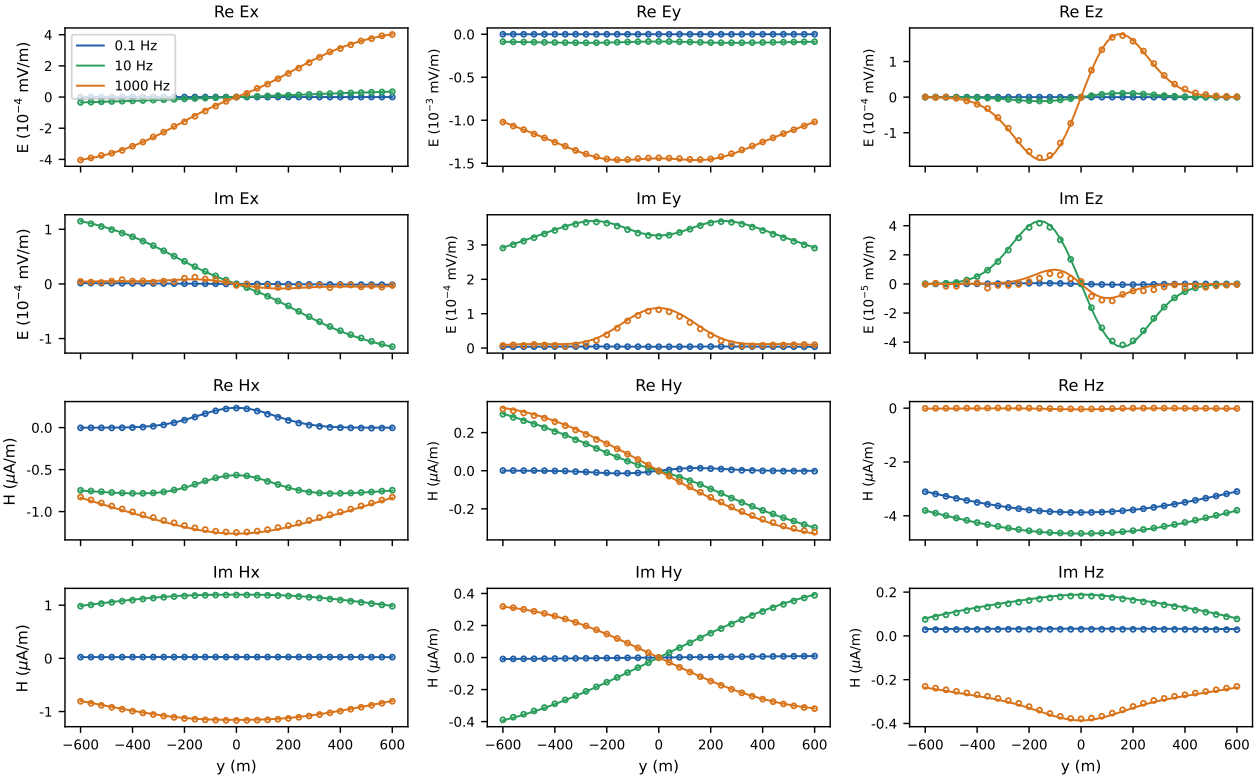


Figure 42: T3_loop_offset, y-profile, hill 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

6.2 depression 30 m

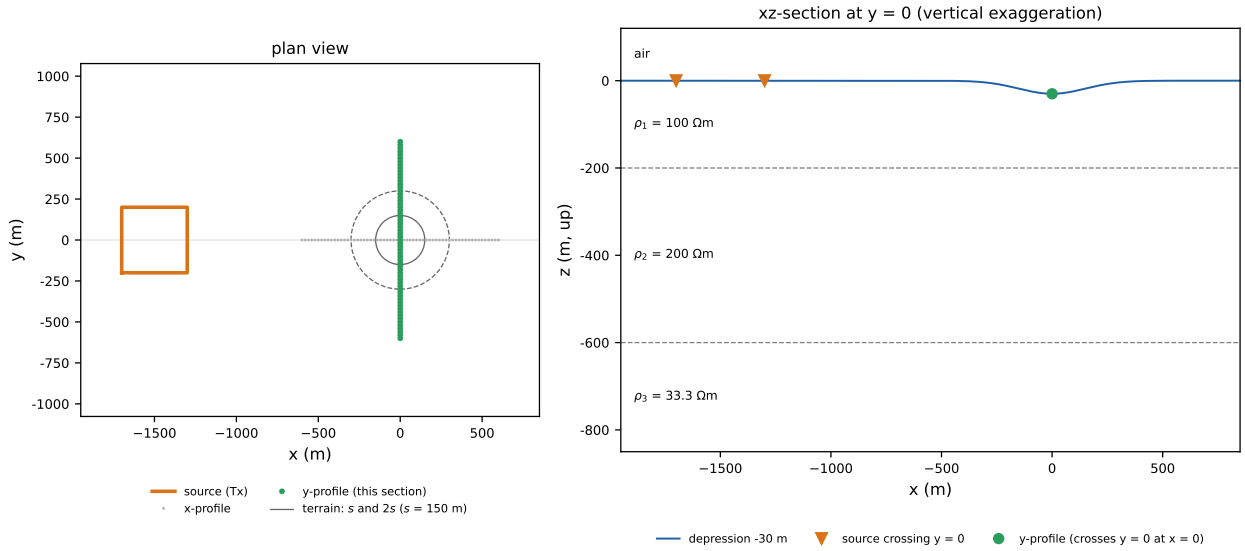


Figure 43: Setup of T3_loop_offset, y-profile, depression 30 m: plan view (left) and xz-section at $y = 0$ (right).

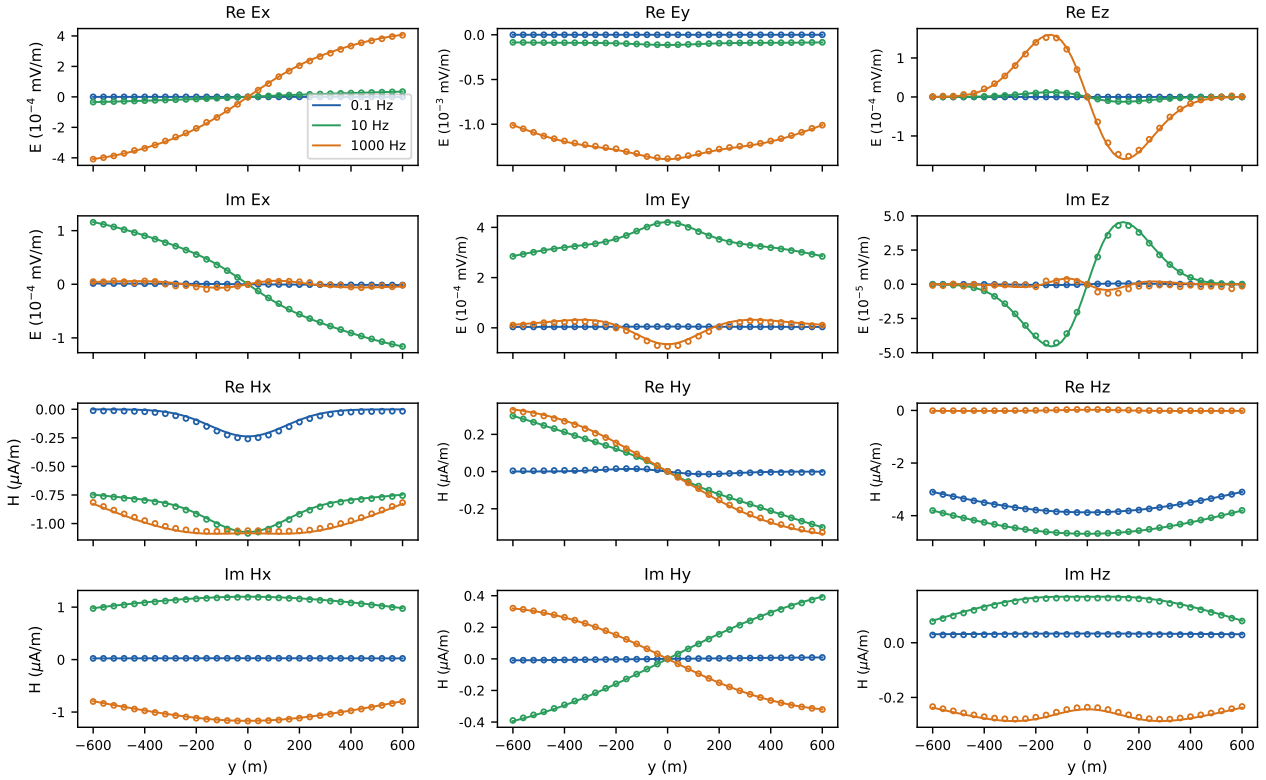


Figure 44: T3_loop_offset, y-profile, depression 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

6.3 hill 90 m

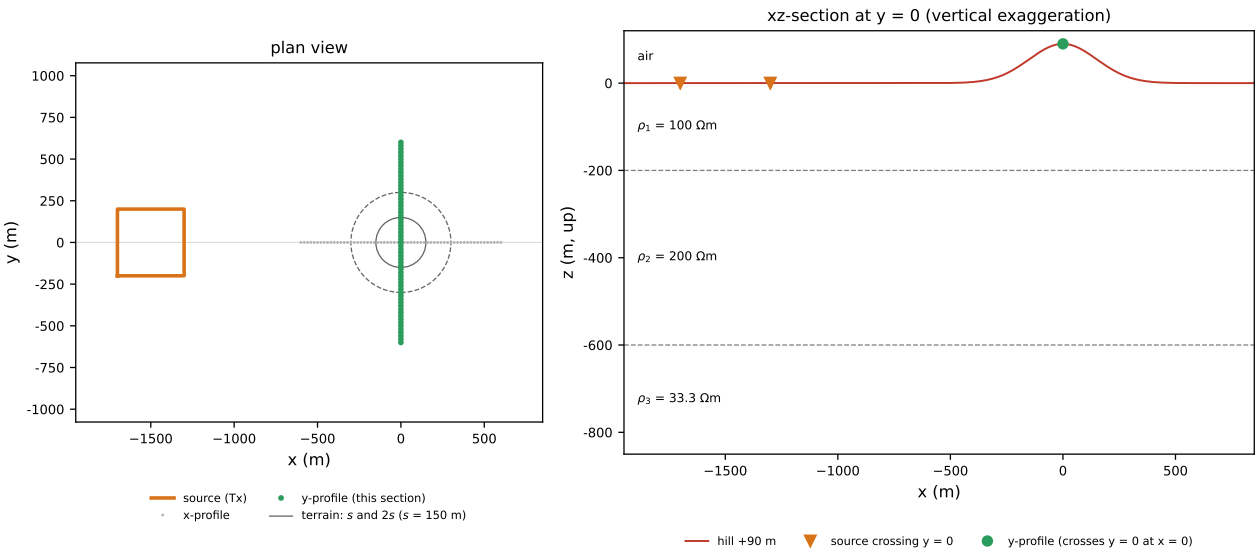


Figure 45: Setup of T3_loop_offset, y-profile, hill 90 m: plan view (left) and xz-section at $y = 0$ (right).

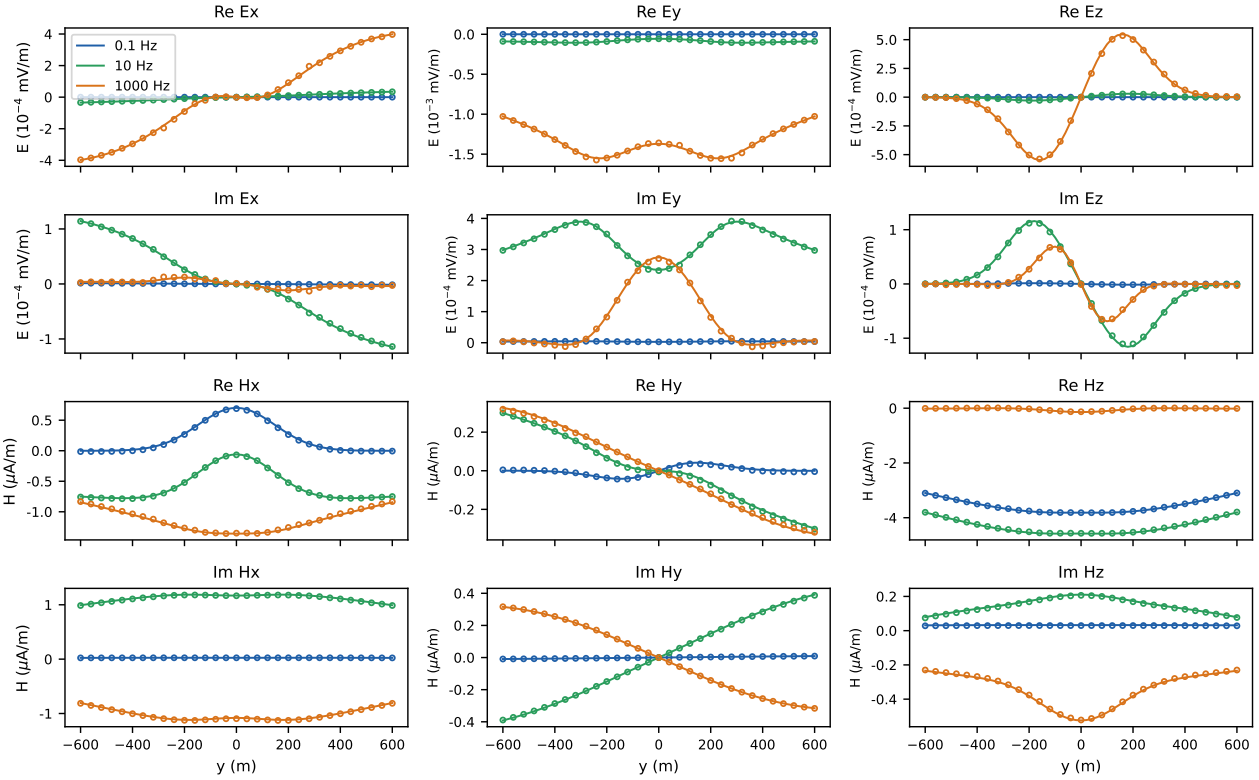


Figure 46: T3_loop_offset, y-profile, hill 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

6.4 depression 90 m

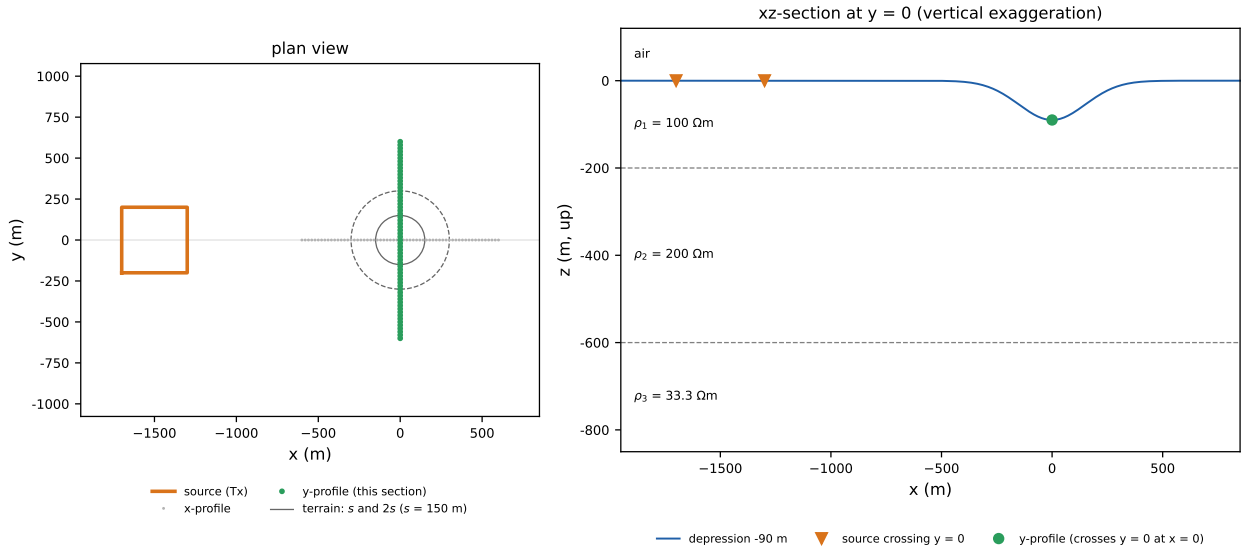


Figure 47: Setup of T3_loop_offset, y-profile, depression 90 m: plan view (left) and xz-section at $y = 0$ (right).

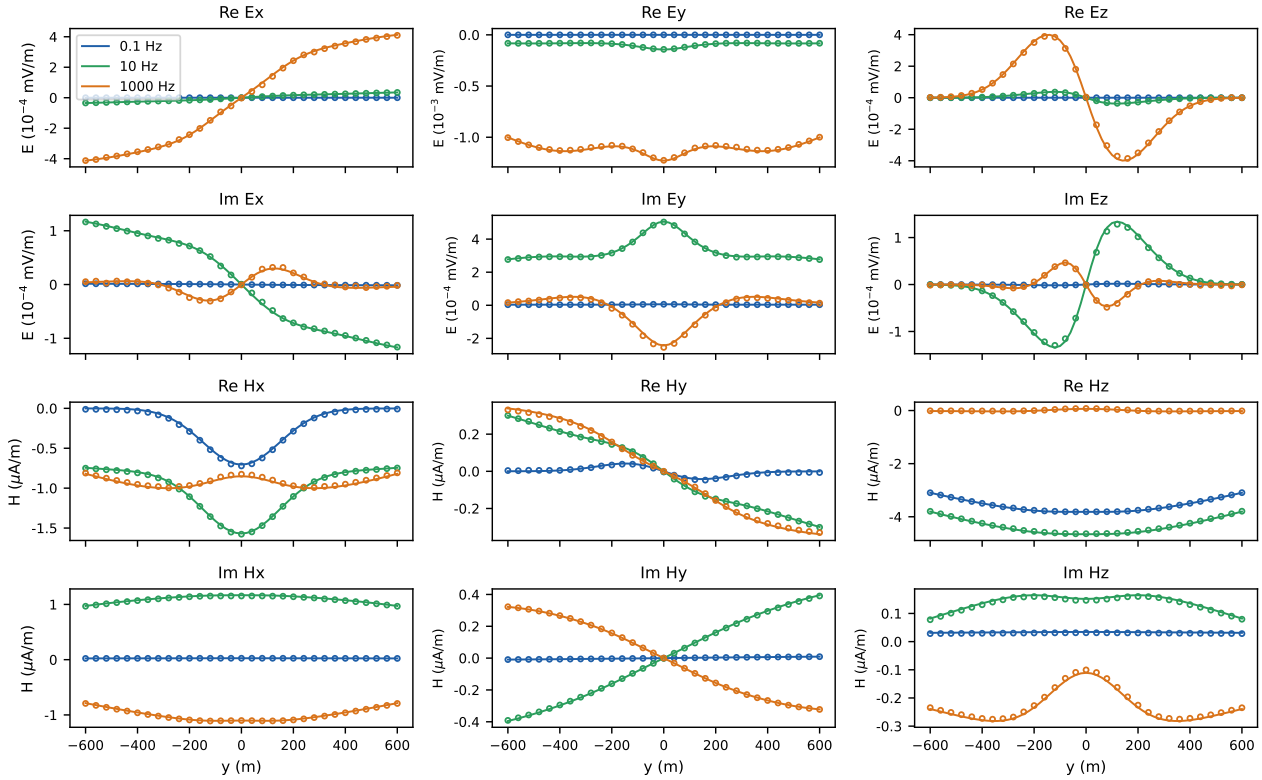


Figure 48: T3_loop_offset, y-profile, depression 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

7 T4_loop_large, x-profile

elfe3D comparison T4: square loop 2.6 x 2.6 km around the hill, counter-clockwise seen from above.

7.1 hill 30 m

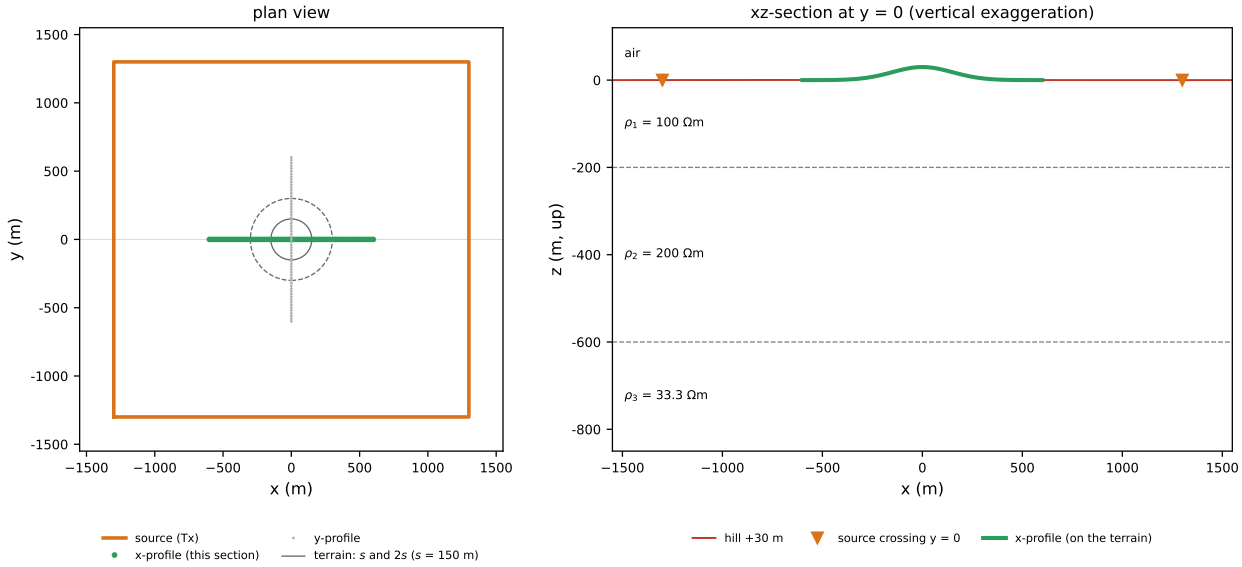


Figure 49: Setup of T4_loop_large, x-profile, hill 30 m: plan view (left) and xz-section at $y = 0$ (right).

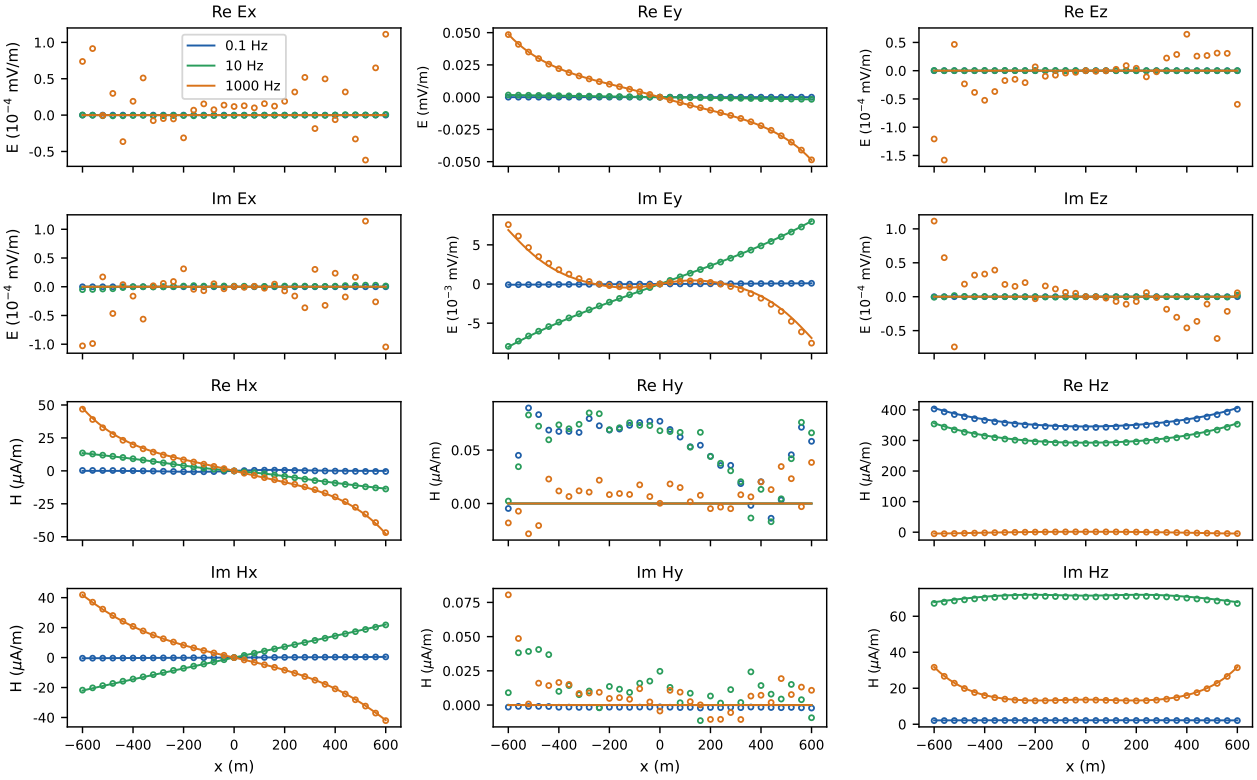


Figure 50: T4_loop_large, x-profile, hill 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

7.2 depression 30 m

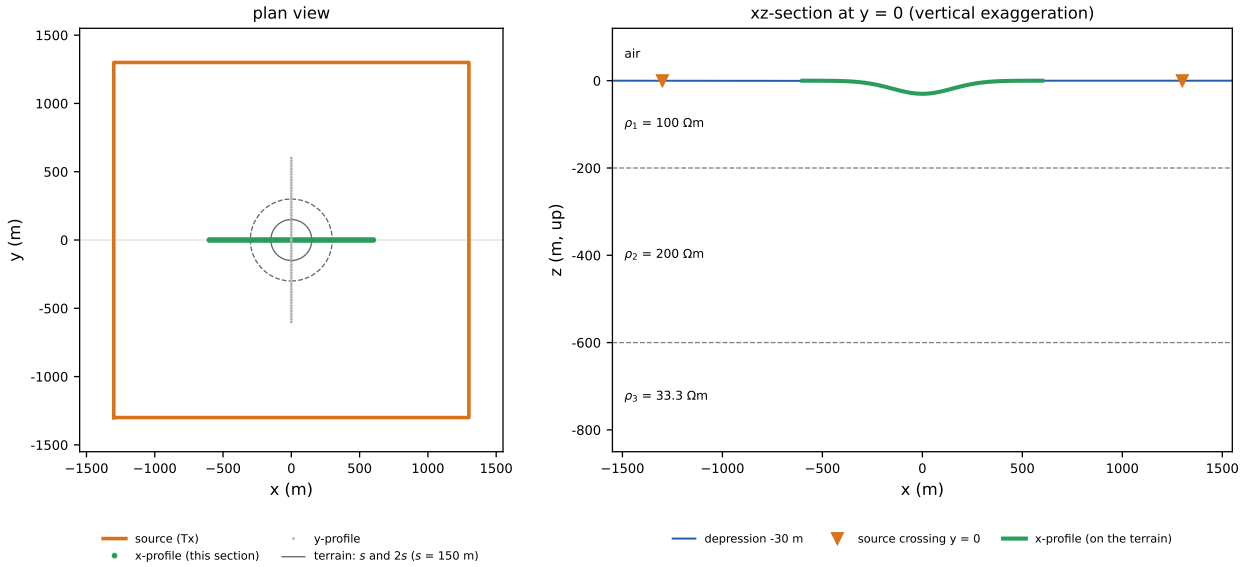


Figure 51: Setup of T4_loop_large, x-profile, depression 30 m: plan view (left) and xz-section at $y = 0$ (right).

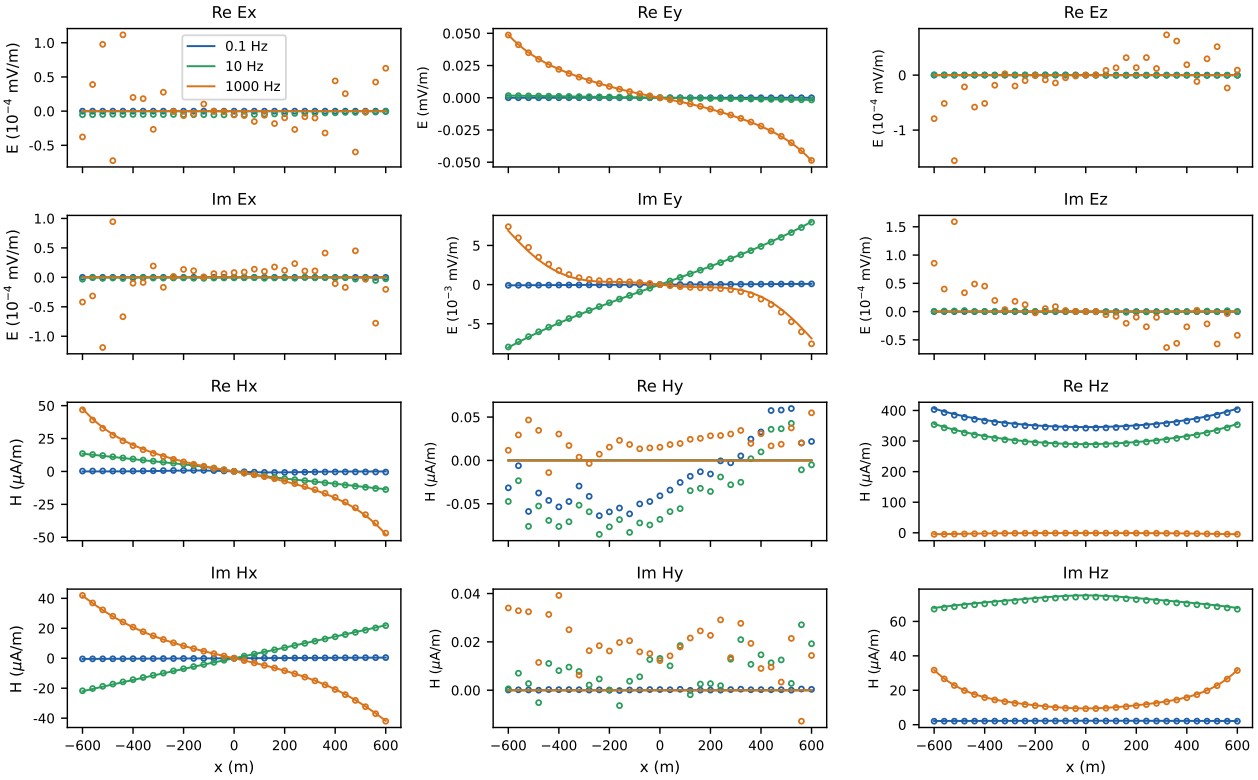


Figure 52: T4_loop_large, x-profile, depression 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

7.3 hill 90 m

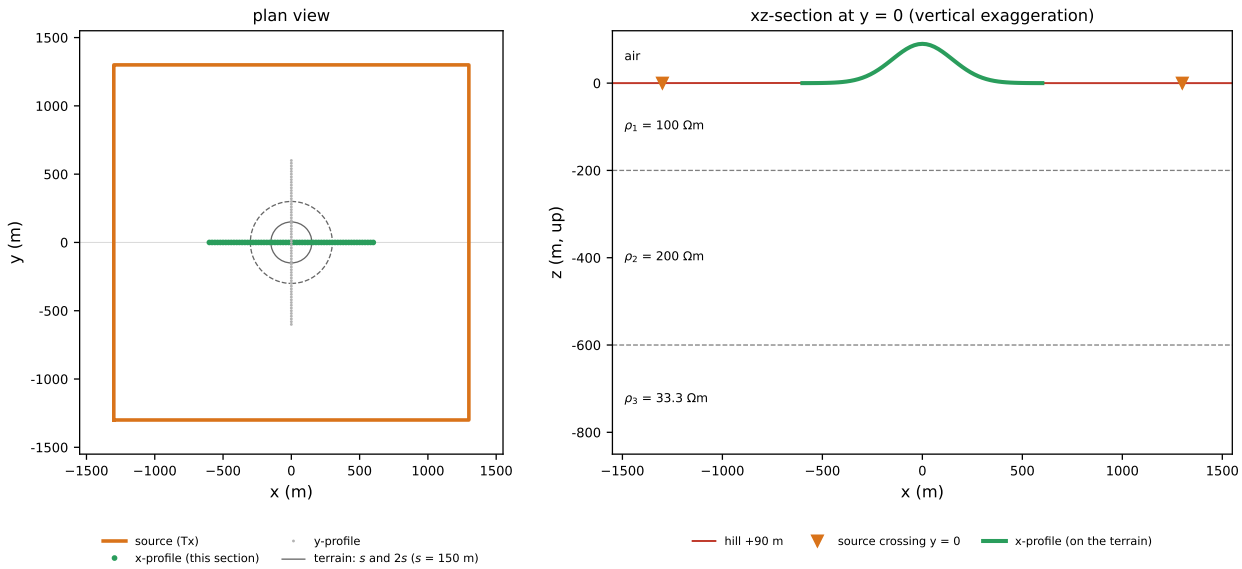


Figure 53: Setup of T4_loop_large, x-profile, hill 90 m: plan view (left) and xz-section at $y = 0$ (right).

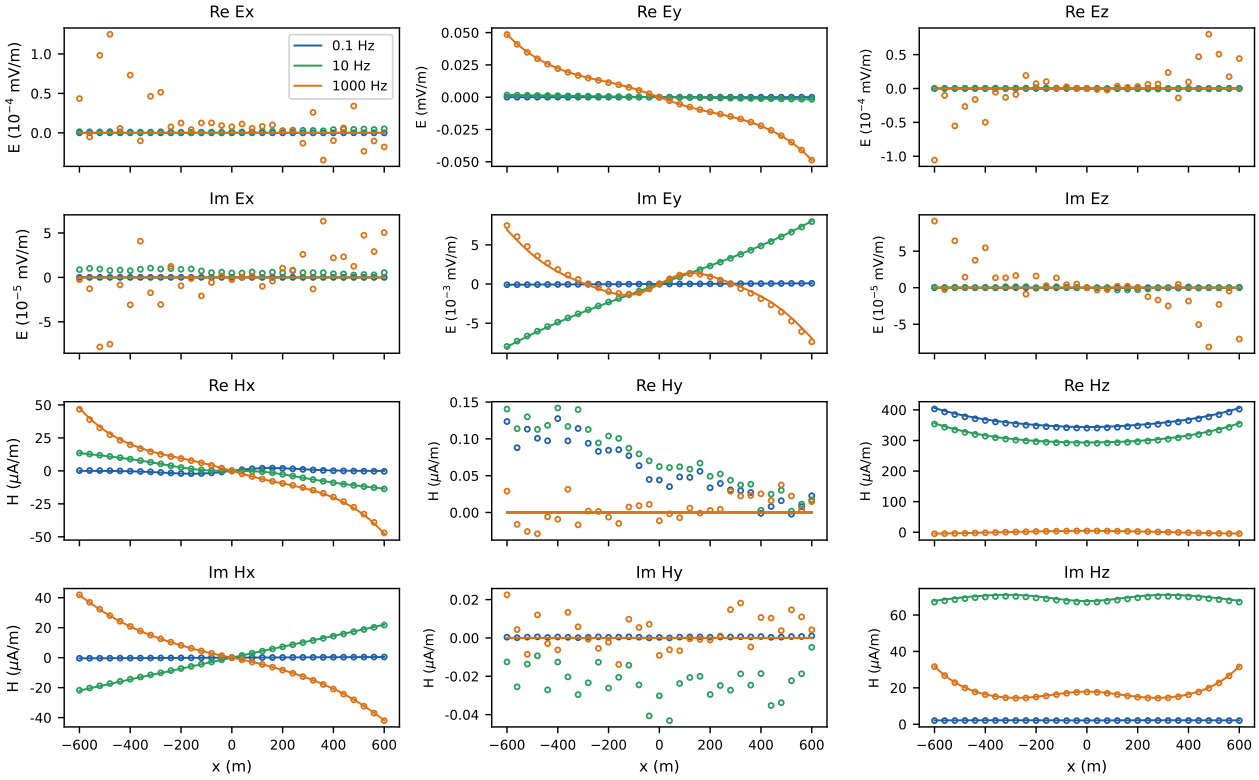


Figure 54: T4_loop_large, x-profile, hill 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

7.4 depression 90 m

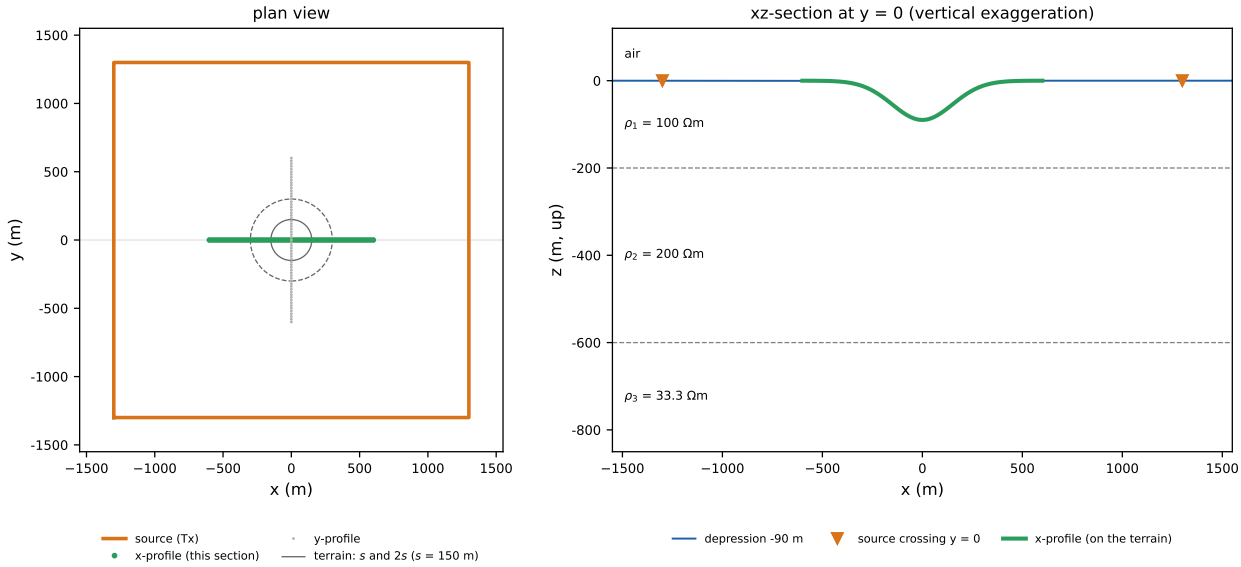


Figure 55: Setup of T4_loop_large, x-profile, depression 90 m: plan view (left) and xz-section at $y = 0$ (right).

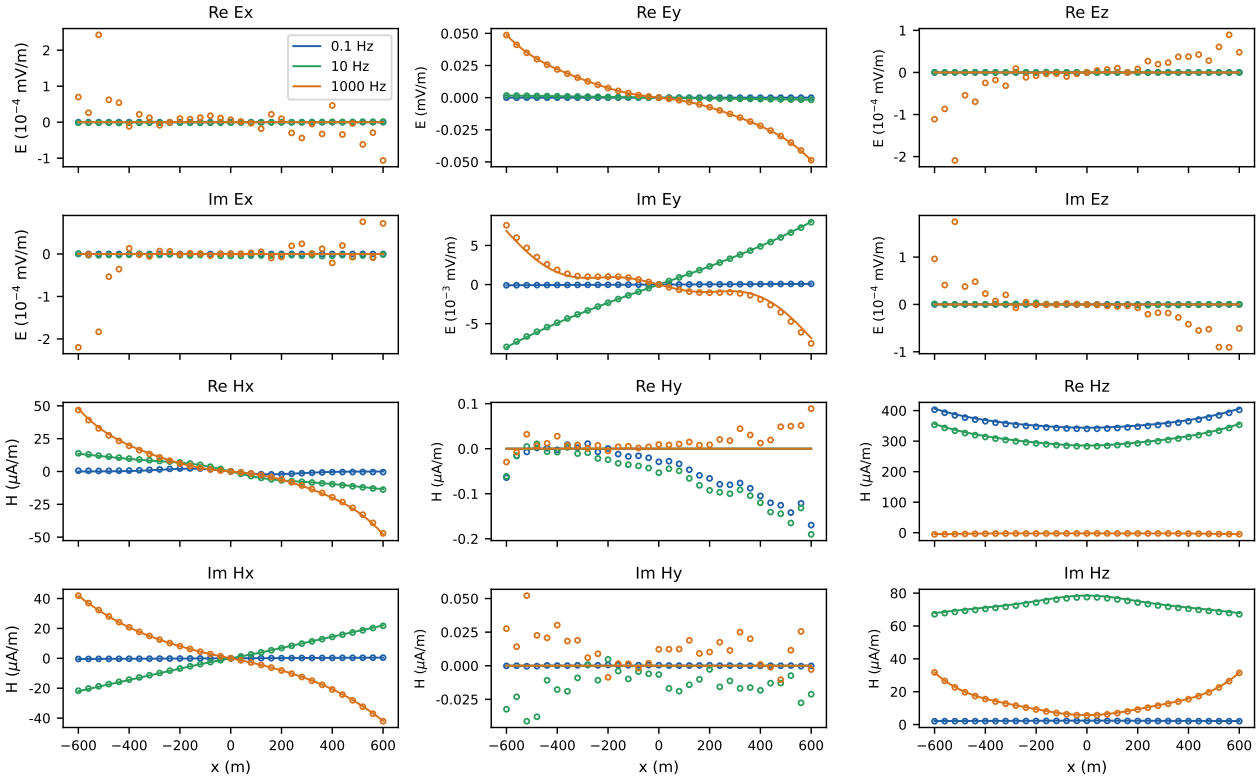


Figure 56: T4_loop_large, x-profile, depression 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

8 T4_loop_large, y-profile

elfe3D comparison T4: square loop 2.6 x 2.6 km around the hill, counter-clockwise seen from above.

8.1 hill 30 m

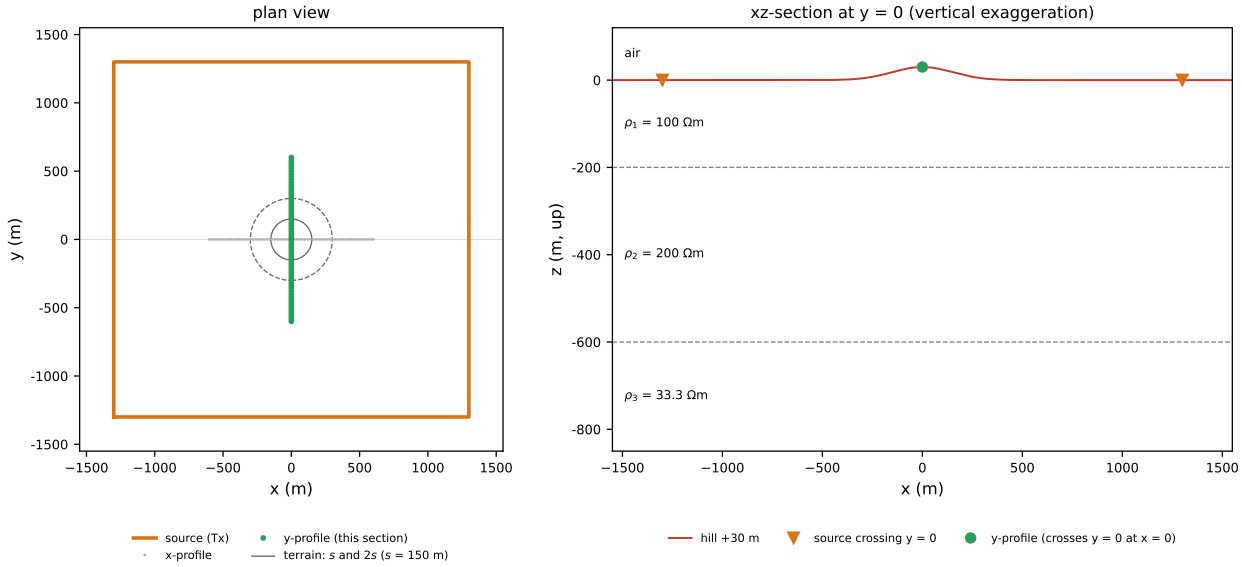


Figure 57: Setup of T4_loop_large, y-profile, hill 30 m: plan view (left) and xz-section at $y = 0$ (right).

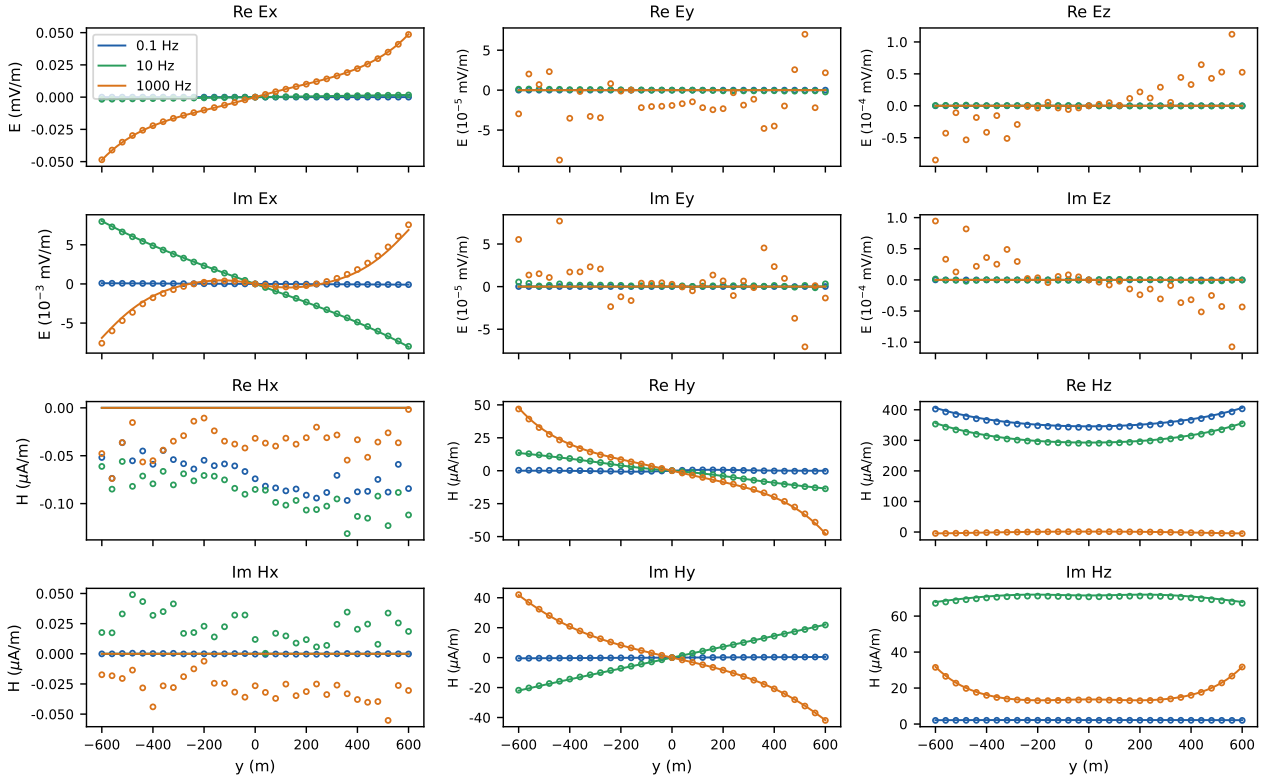


Figure 58: T4_loop_large, y-profile, hill 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

8.2 depression 30 m

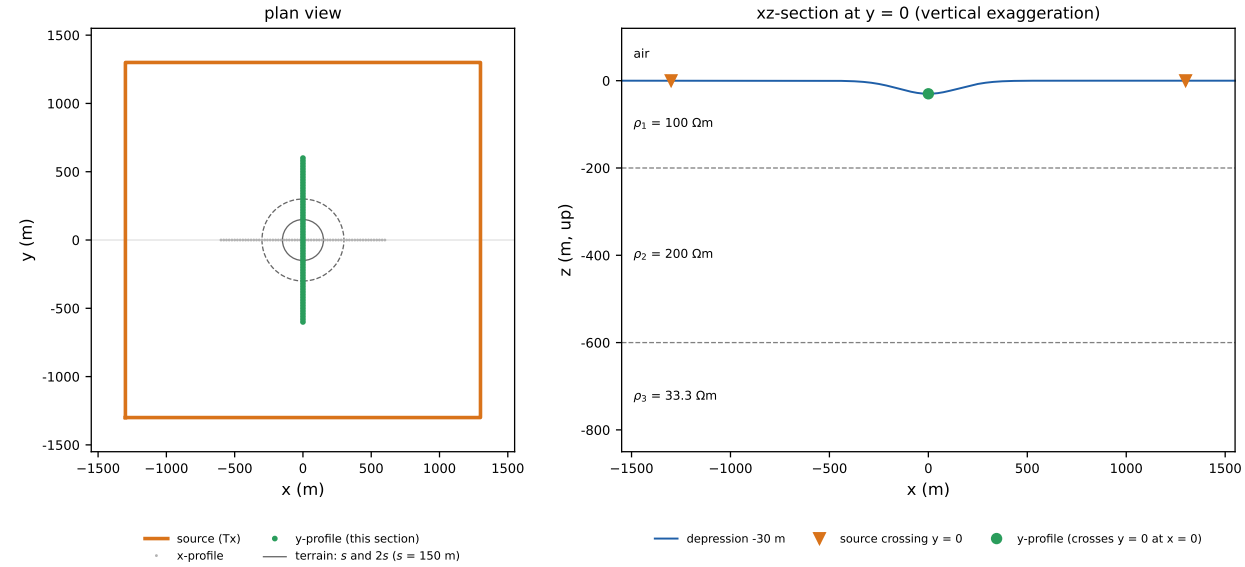


Figure 59: Setup of T4_loop_large, y-profile, depression 30 m: plan view (left) and xz-section at $y = 0$ (right).

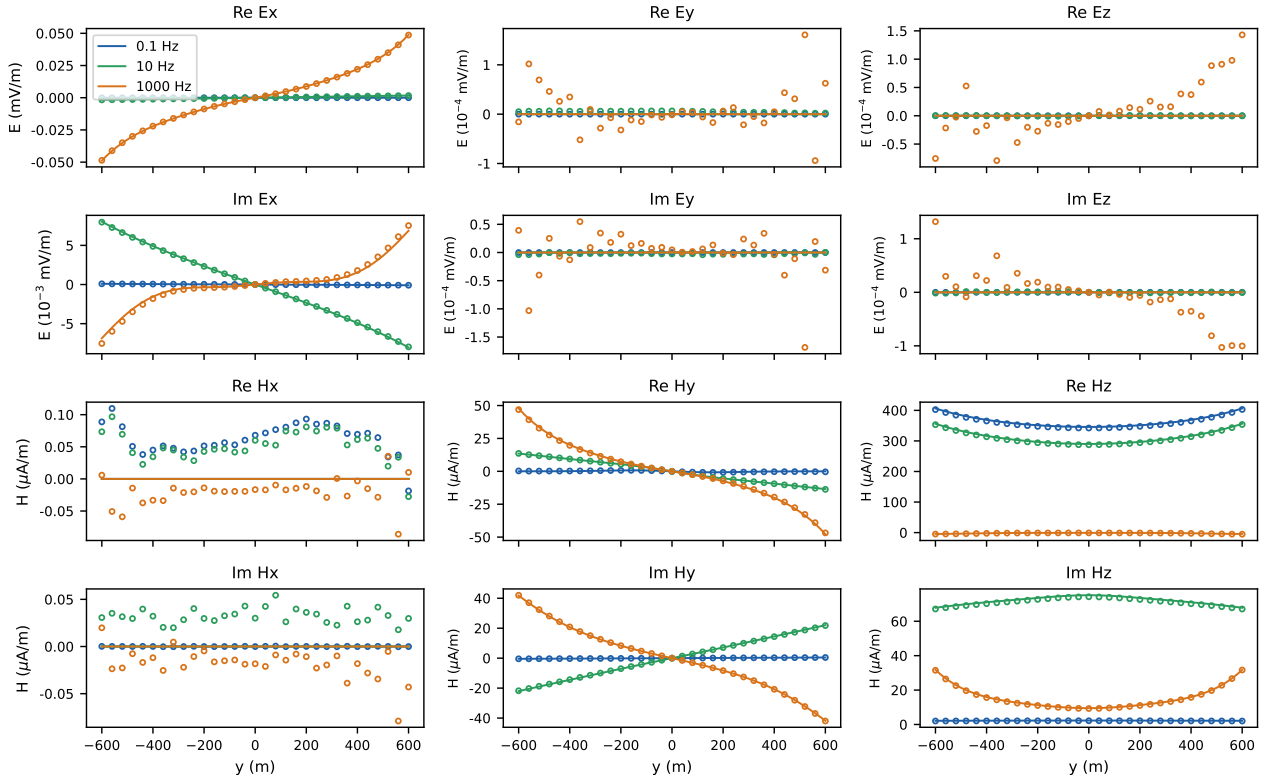


Figure 60: T4_loop_large, y-profile, depression 30 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

8.3 hill 90 m

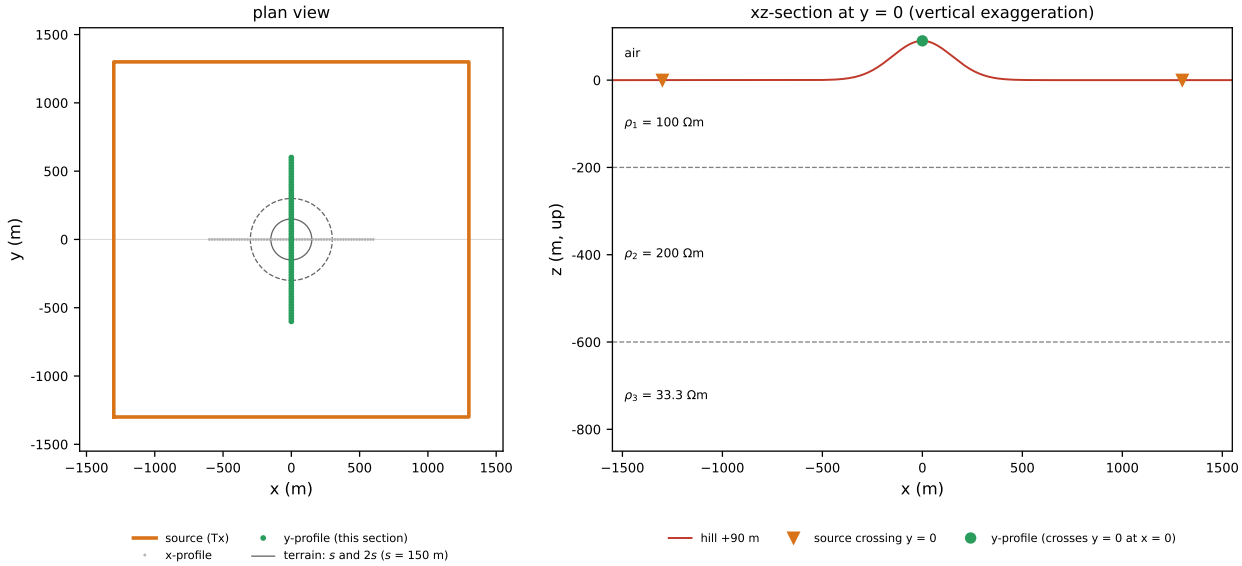


Figure 61: Setup of T4_loop_large, y-profile, hill 90 m: plan view (left) and xz-section at $y = 0$ (right).

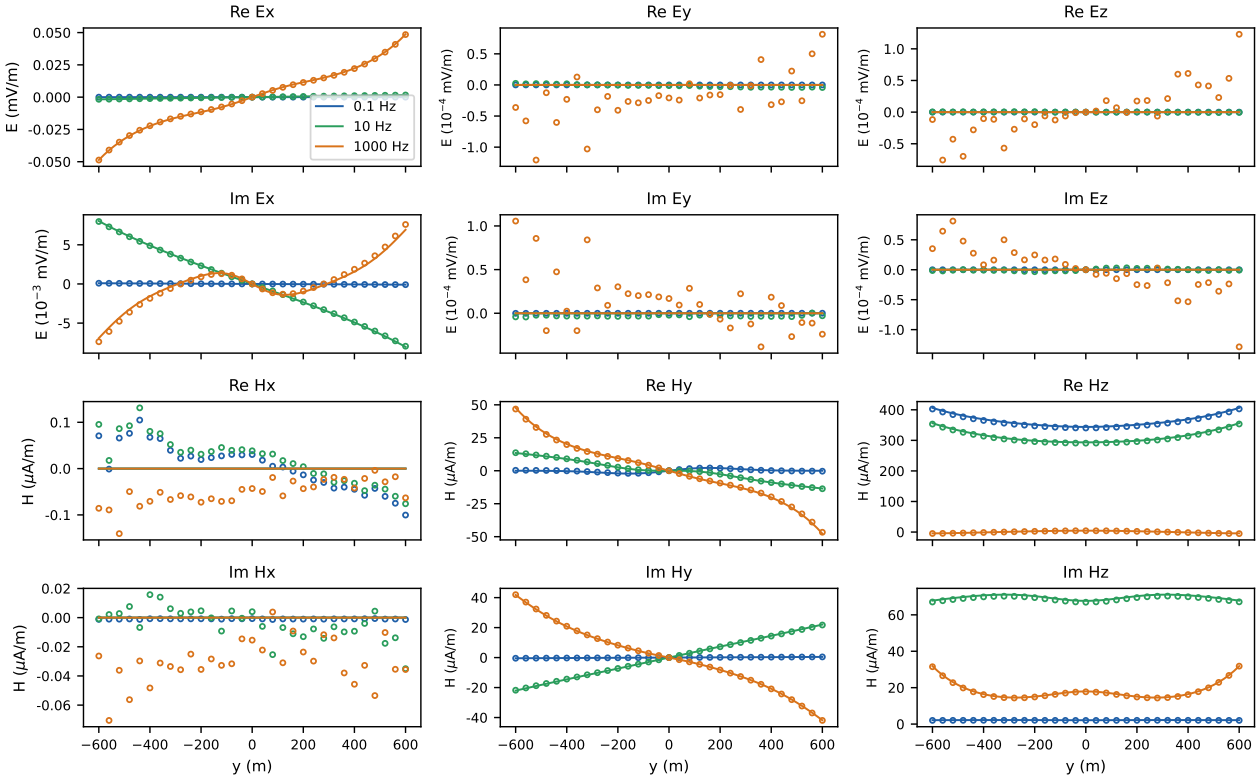


Figure 62: T4_loop_large, y-profile, hill 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

8.4 depression 90 m

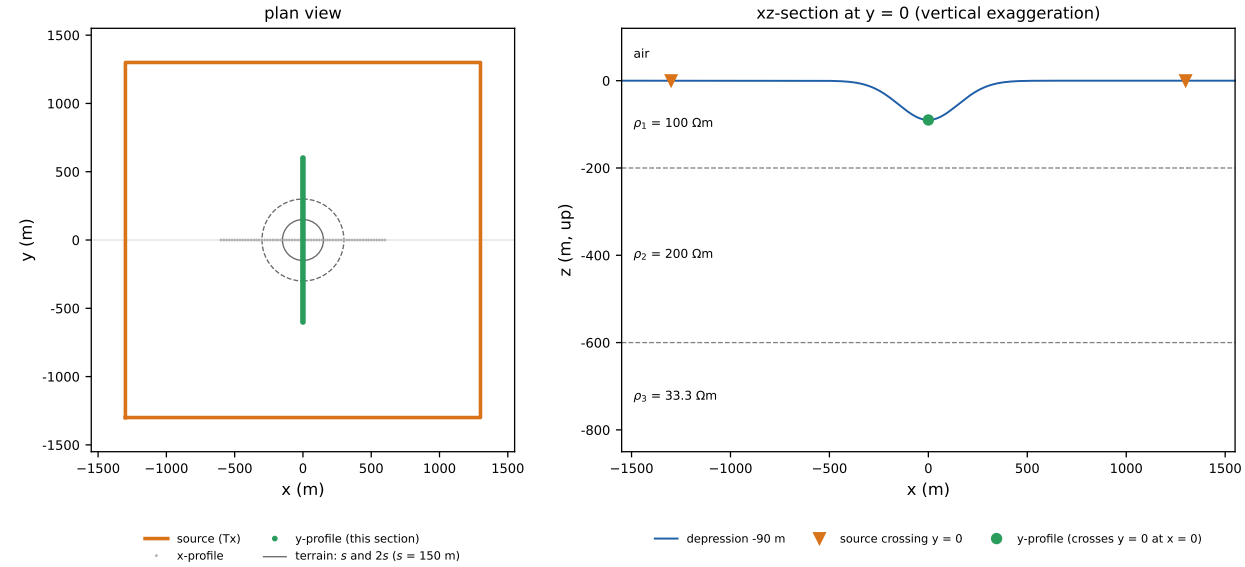


Figure 63: Setup of T4_loop_large, y-profile, depression 90 m: plan view (left) and xz-section at $y = 0$ (right).

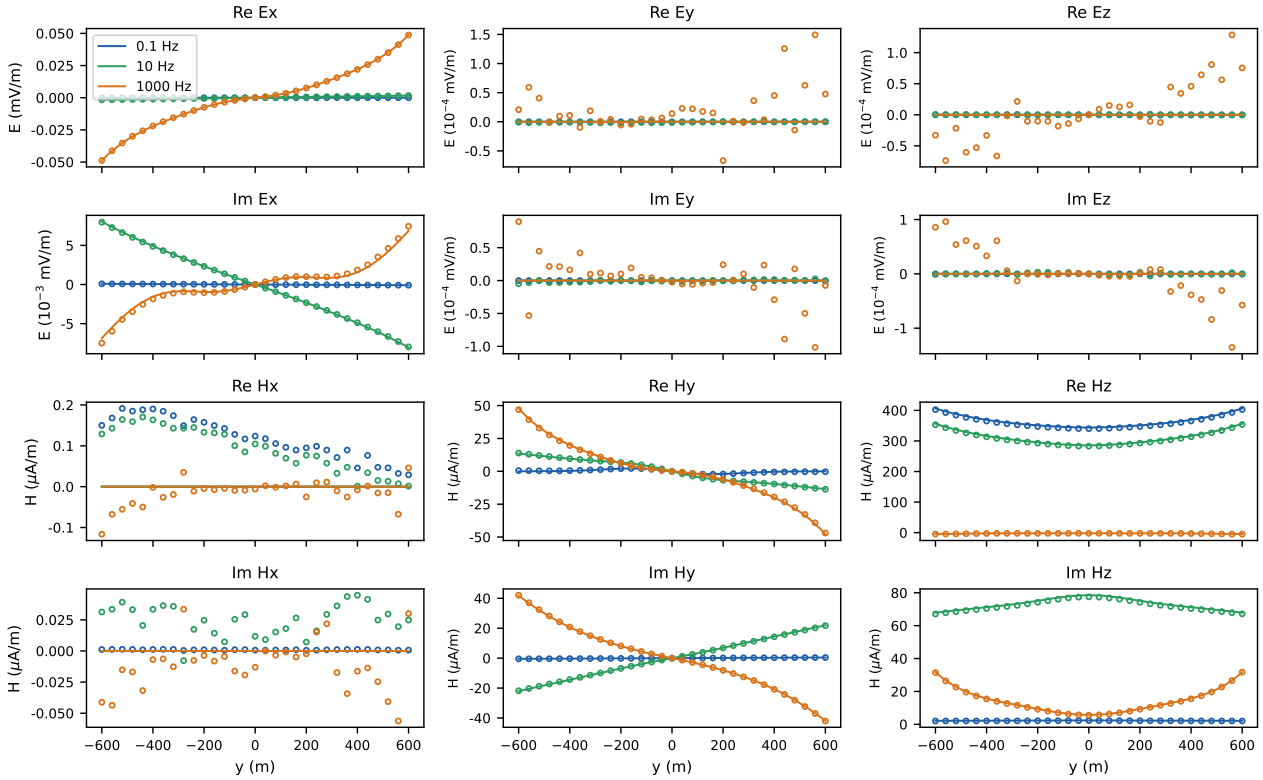


Figure 64: T4_loop_large, y-profile, depression 90 m: series (lines) and elfe3D (circles, every second receiver). elfe3D runs: 0.1 Hz: nf/out; 10 Hz: nf/out; 1000 Hz: hf1k/out.

terrasphere_csem: sphere under a hill against elfe3D

October 14, 2026

Series solution of this package (lines) against the 3D finite-element code elfe3D (circles) for the four sources of the elfe3D comparison with a sphere: Gaussian hill of height 90 m and width 150 m, three layers (interfaces at 200 m and 600 m depth), sphere of 10 Ωm, radius 180 m, centre (0, 0) m at 400 m depth (in layer 2, under the hill). For each source two figures: the total field with the sphere, and the anomaly of the sphere. The elfe3D anomaly is the difference of the run with the sphere and a run on the identical mesh with the host resistivity in the sphere, so that the discretisation error of the background largely cancels; the series anomaly is the series with the sphere minus the series without it. Receivers 1 cm below the surface on the x- and y-profile from −600 m to 600 m every 20 m. Fields per 1 A source current, E in mV/m, H in μA/m, time dependence $e^{-i\omega t}$, in the convention of this package: the series exactly as it writes them, elfe3D converted ($H = -H_{\mathrm{elfe3D}}$).

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1 T1_wire_inline, x-profile

elfe3D sphere comparison T1: grounded wire in line with the x-profile, 500 m, centre 1500 m left of the hill.

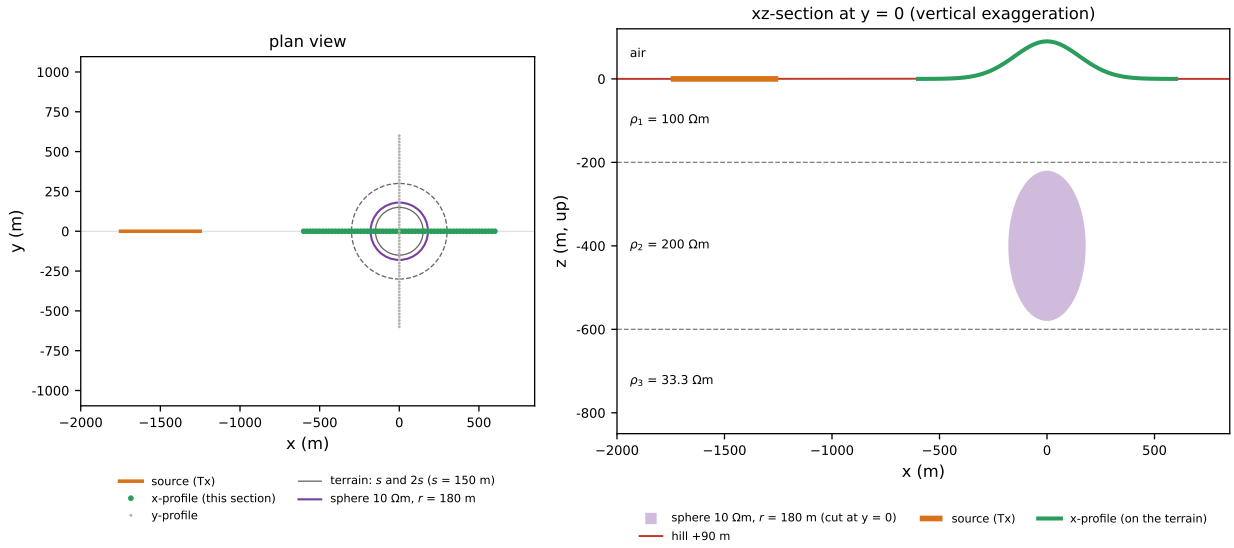


Figure 1: Setup of T1_wire_inline, x-profile, hill 90 m with sphere: plan view (left) and xz-section at $y = 0$ (right).

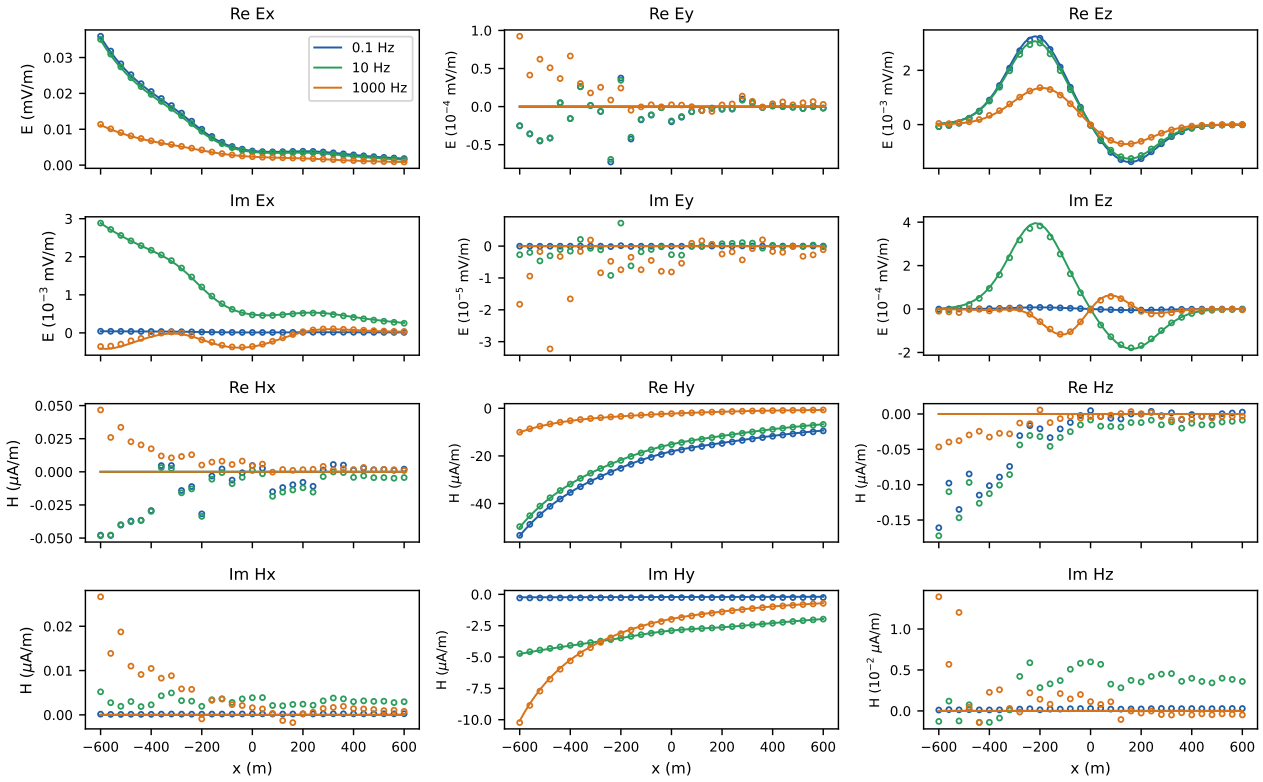


Figure 2: T1_wire_inline, x-profile, total field with sphere: series (lines) and elfe3D (circles, every second receiver).

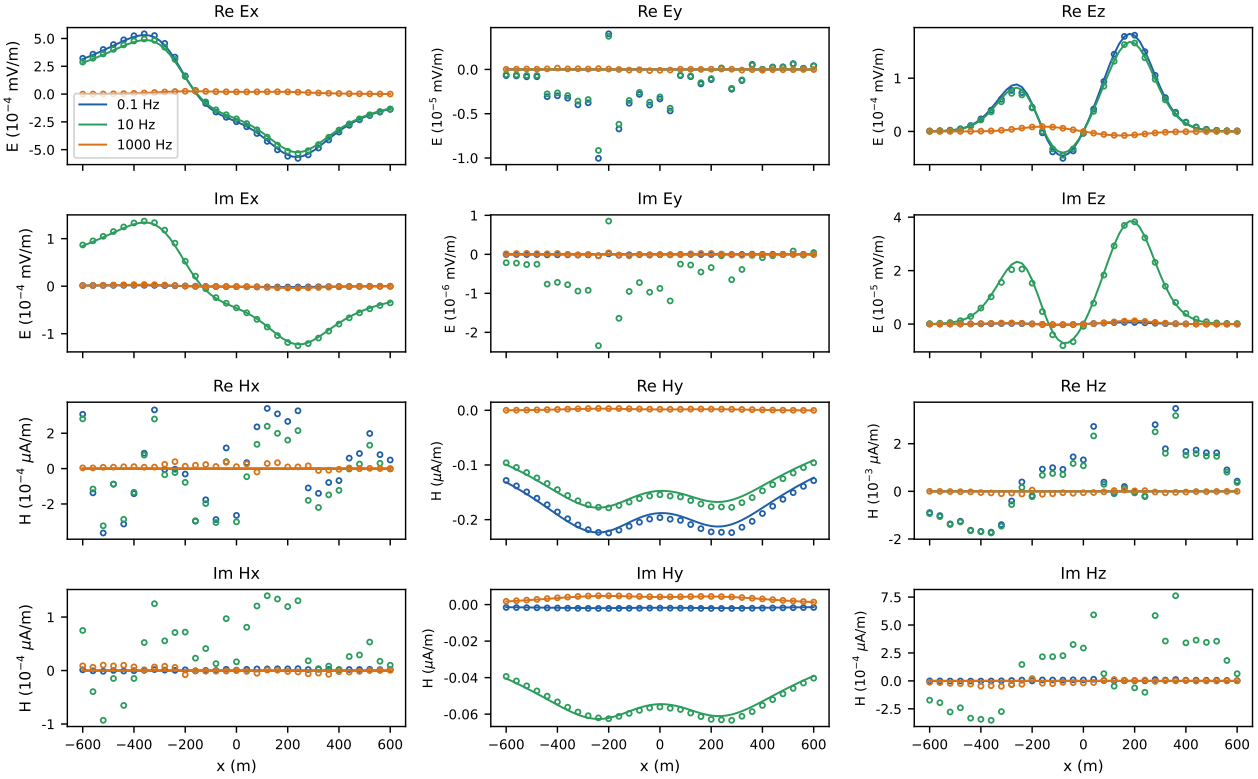


Figure 3: T1_wire_inline, x-profile, sphere anomaly $V(\text{sphere}) - V(\text{no sphere})$: series (lines) and elfe3D run with sphere minus reference run on the same mesh (circles).

2 T1_wire_inline, y-profile

elfe3D sphere comparison T1: grounded wire in line with the x-profile, 500 m, centre 1500 m left of the hill.

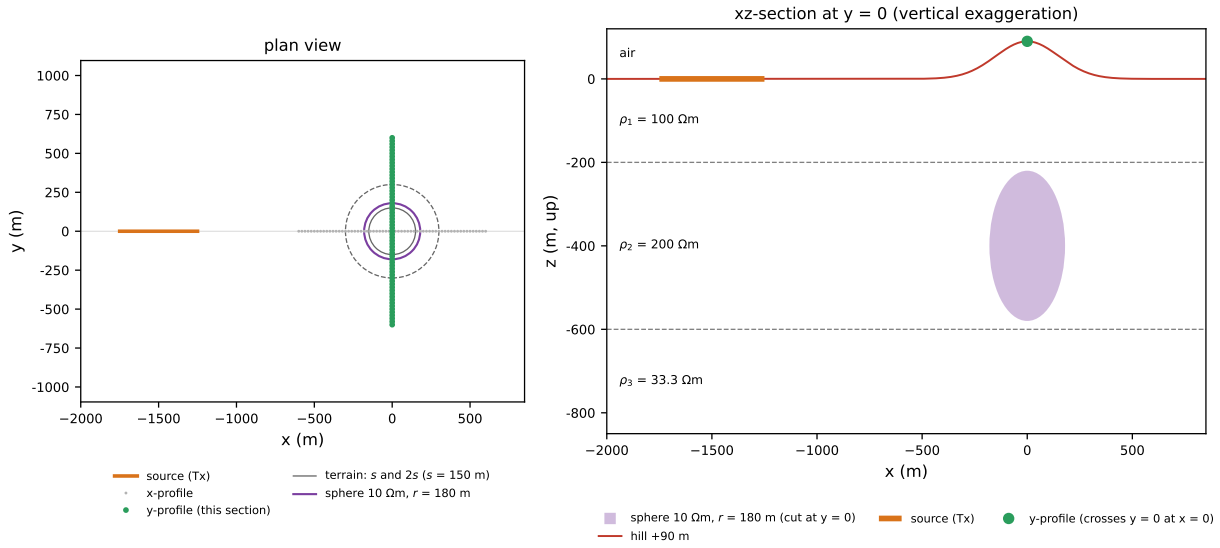


Figure 4: Setup of T1_wire_inline, y-profile, hill 90 m with sphere: plan view (left) and xz-section at $y = 0$ (right).

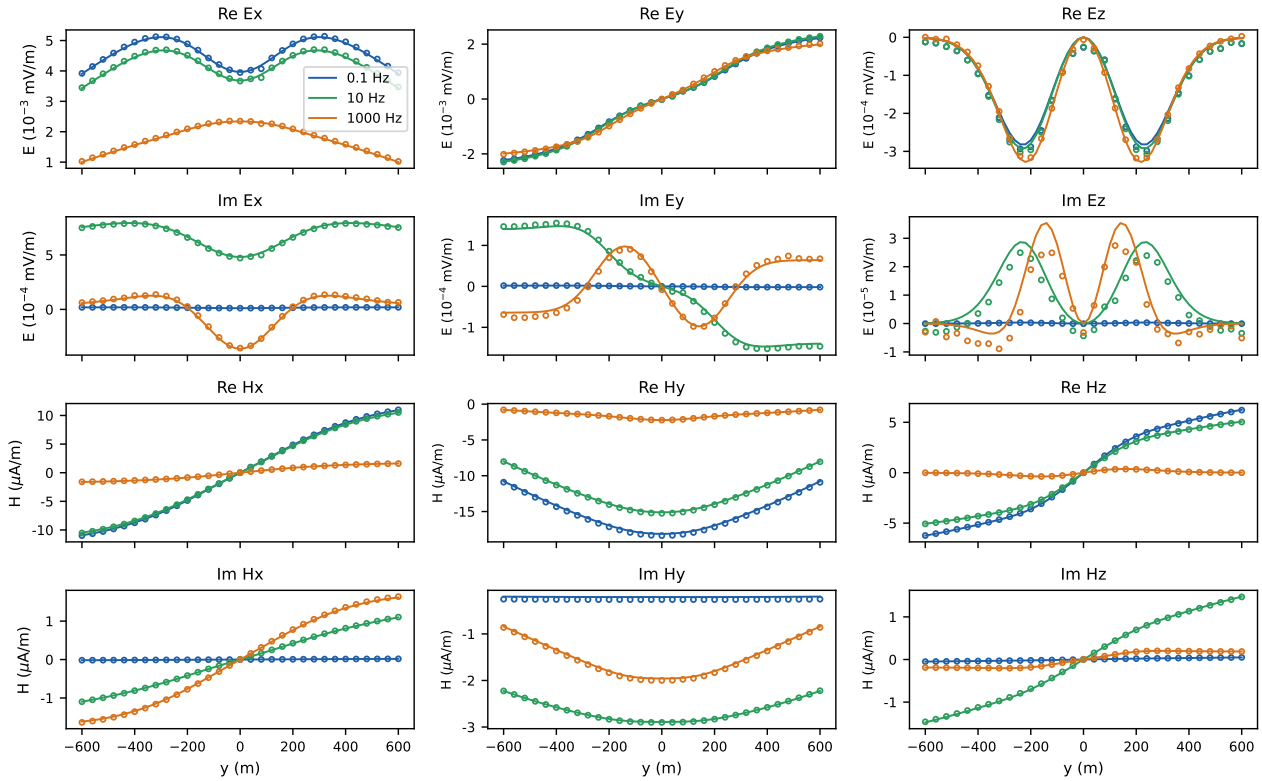


Figure 5: T1_wire_inline, y-profile, total field with sphere: series (lines) and elfe3D (circles, every second receiver).

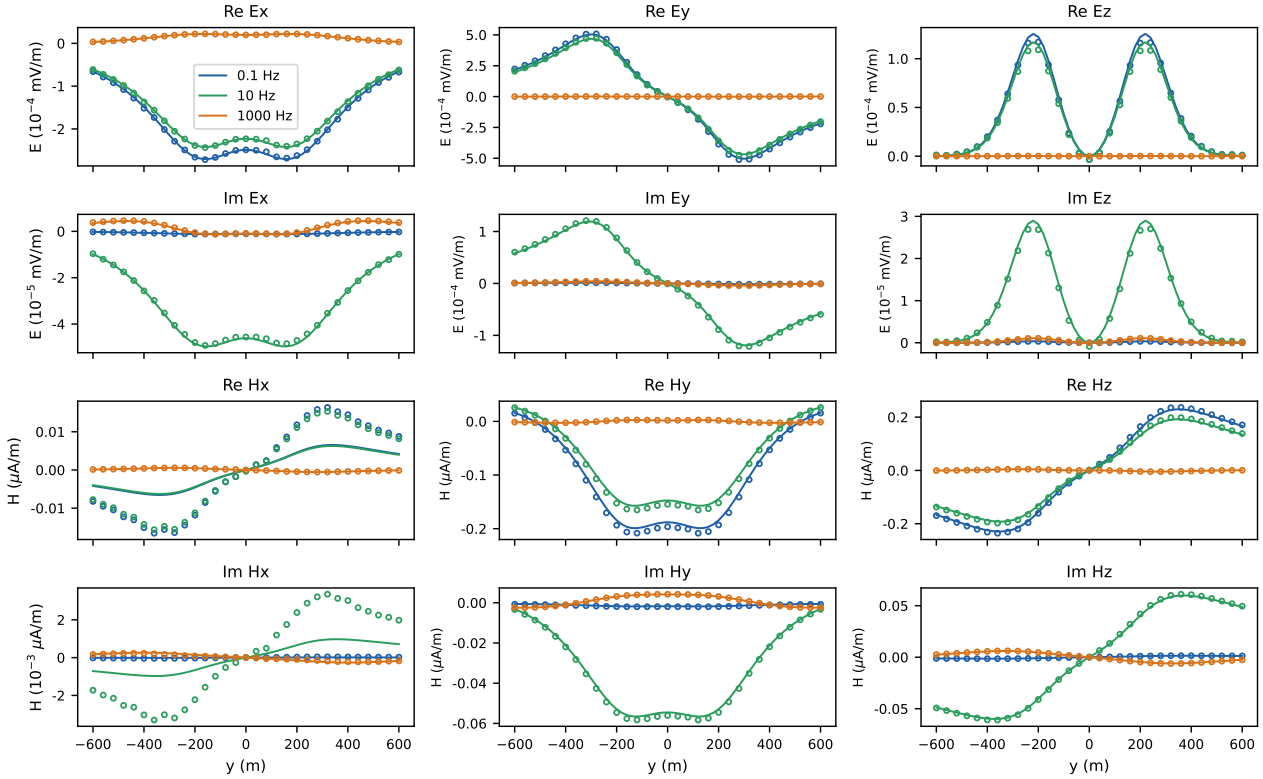


Figure 6: T1_wire_inline, y-profile, sphere anomaly $V(\text{sphere}) - V(\text{no sphere})$: series (lines) and elfe3D run with sphere minus reference run on the same mesh (circles).

3 T2_wire_broadside, x-profile

elfe3D sphere comparison T2: grounded wire broadside (along y), 500 m, centre 1500 m left of the hill.

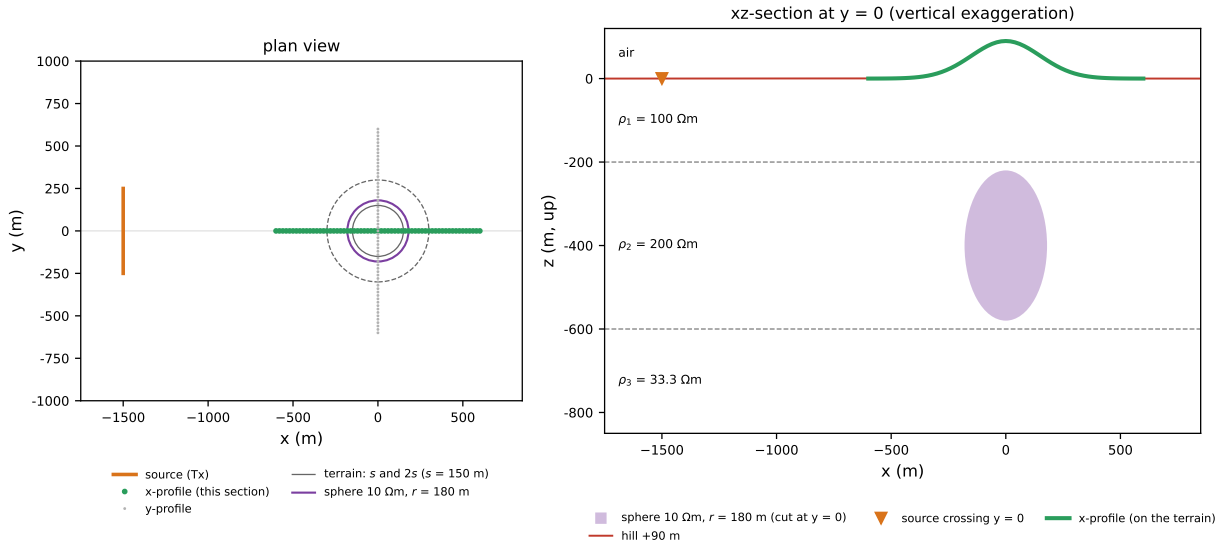


Figure 7: Setup of T2_wire_broadside, x-profile, hill 90 m with sphere: plan view (left) and xz-section at $y = 0$ (right).

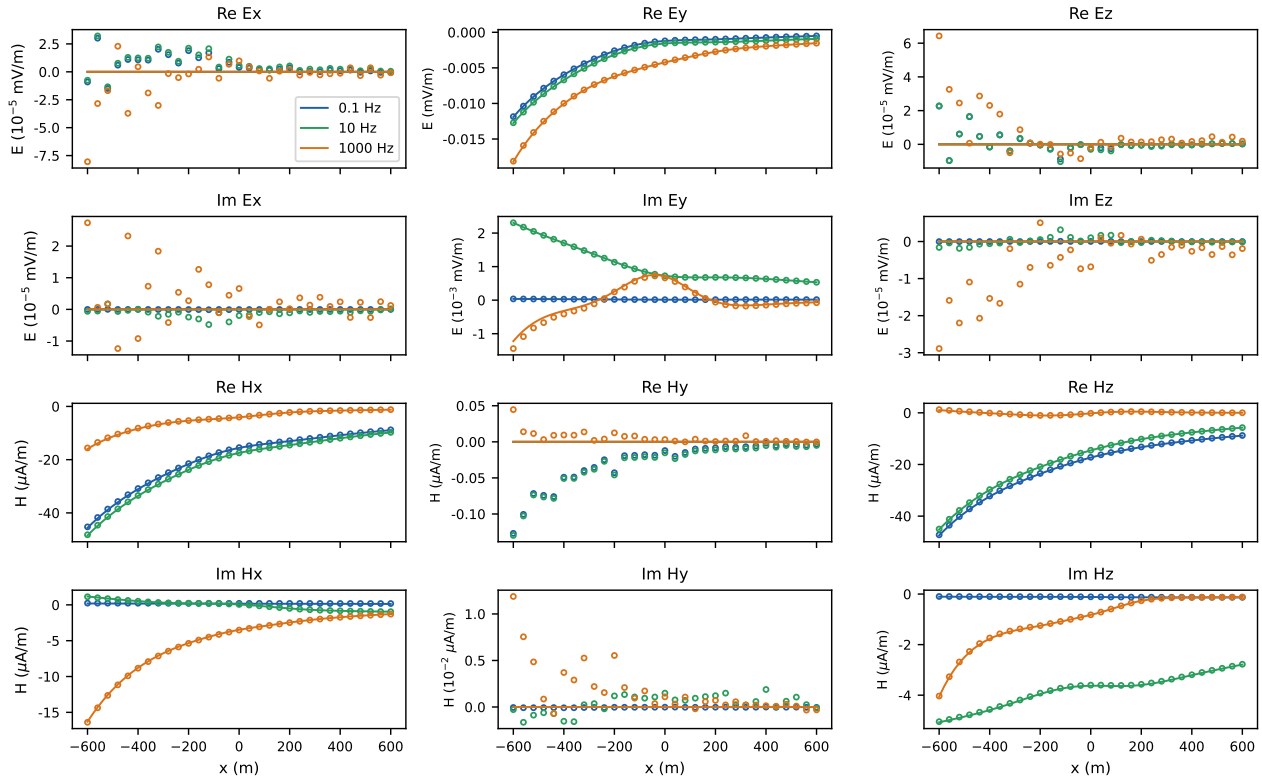


Figure 8: T2_wire_broadside, x-profile, total field with sphere: series (lines) and elfe3D (circles, every second receiver).

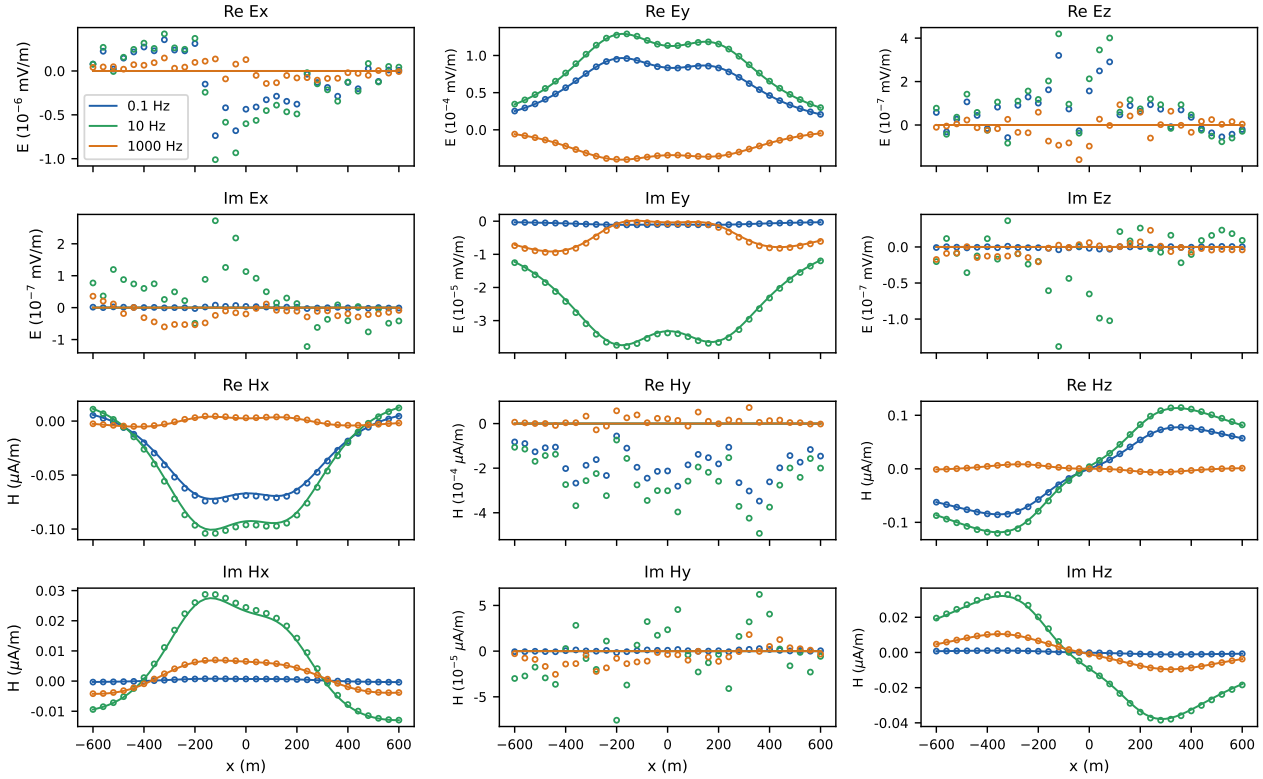


Figure 9: T2_wire_broadside, x-profile, sphere anomaly $V(\text{sphere}) - V(\text{no sphere})$: series (lines) and elfe3D run with sphere minus reference run on the same mesh (circles).

4 T2_wire_broadside, y-profile

elfe3D sphere comparison T2: grounded wire broadside (along y), 500 m, centre 1500 m left of the hill.

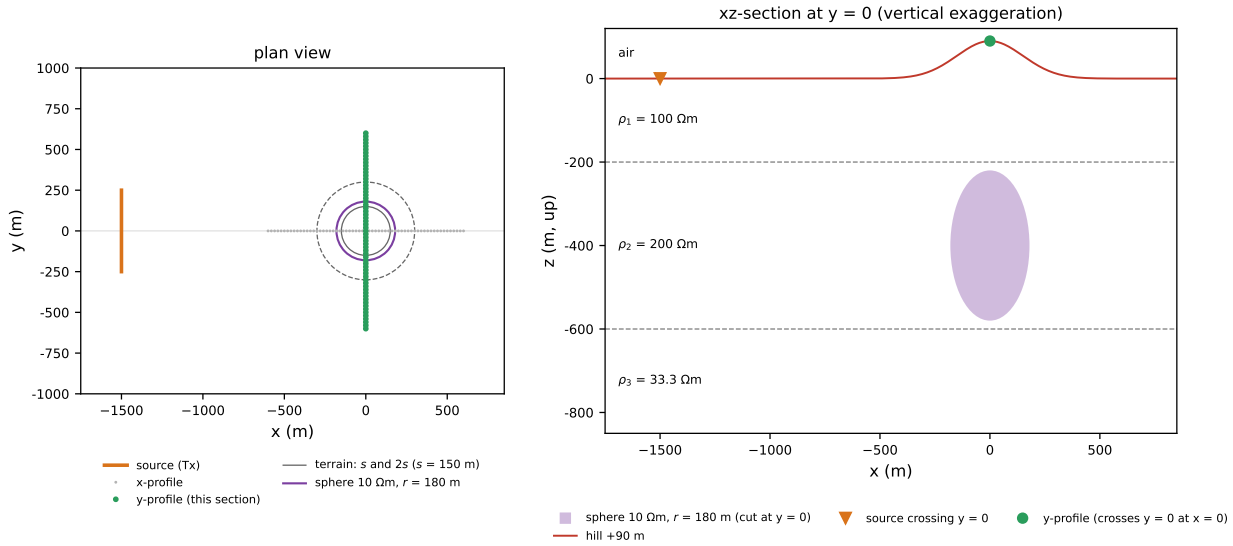


Figure 10: Setup of T2_wire_broadside, y-profile, hill 90 m with sphere: plan view (left) and xz-section at $y = 0$ (right).

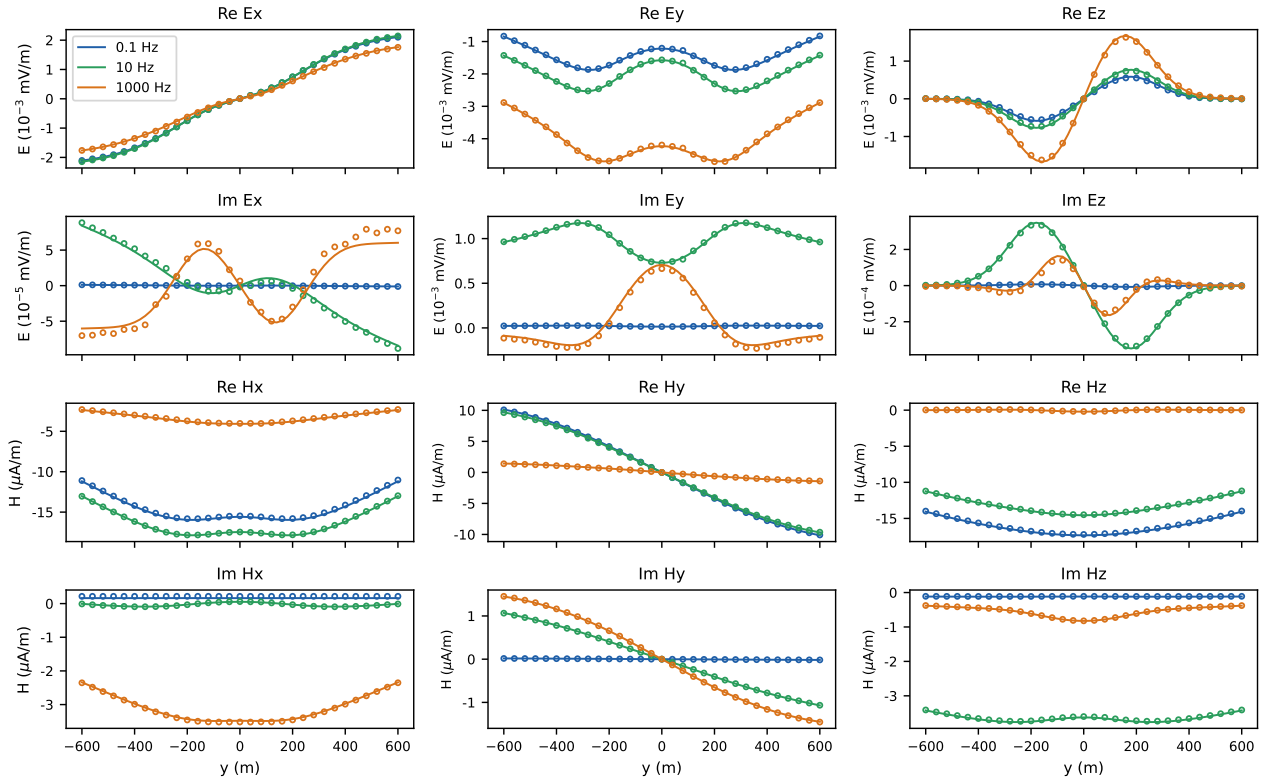


Figure 11: T2_wire_broadside, y-profile, total field with sphere: series (lines) and elfe3D (circles, every second receiver).

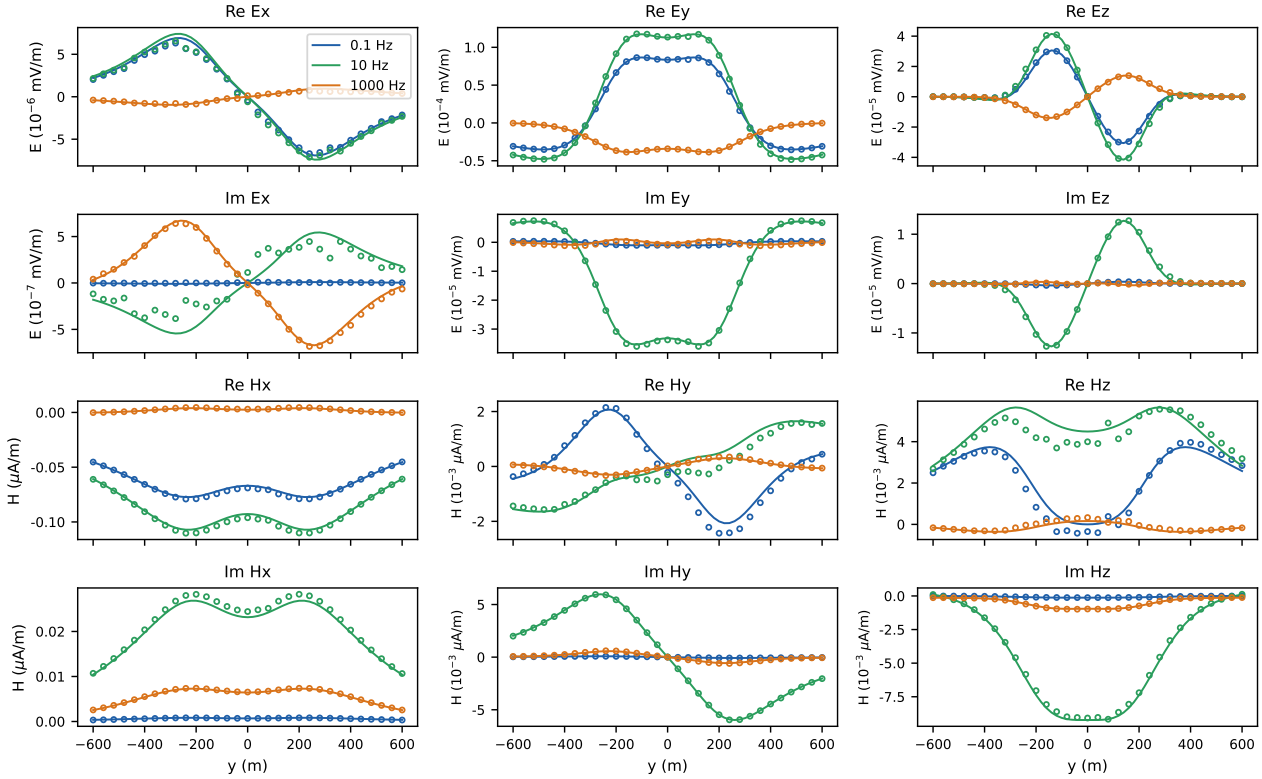


Figure 12: T2_wire_broadside, y-profile, sphere anomaly $V(\text{sphere}) - V(\text{no sphere})$: series (lines) and elfe3D run with sphere minus reference run on the same mesh (circles).

5 T3_loop_offset, x-profile

elfe3D sphere comparison T3: square loop 400 x 400 m, centre (-1500, 0), counter-clockwise seen from above.

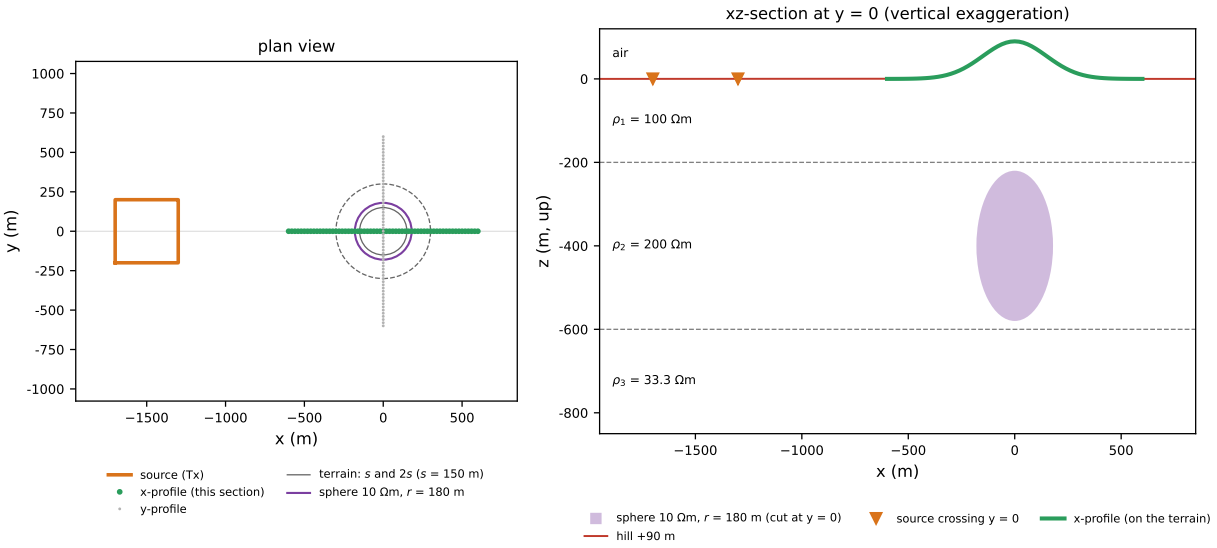


Figure 13: Setup of T3_loop_offset, x-profile, hill 90 m with sphere: plan view (left) and xz-section at $y = 0$ (right).

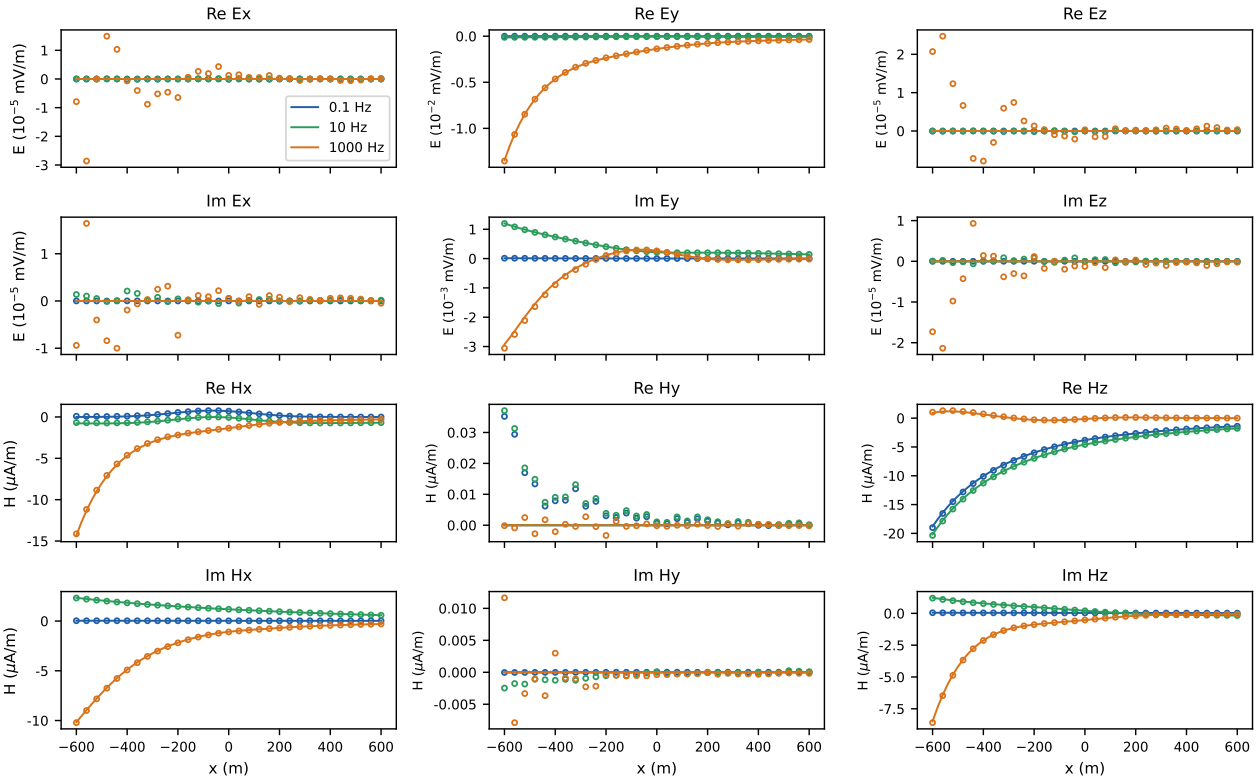


Figure 14: T3_loop_offset, x-profile, total field with sphere: series (lines) and elfe3D (circles, every second receiver).

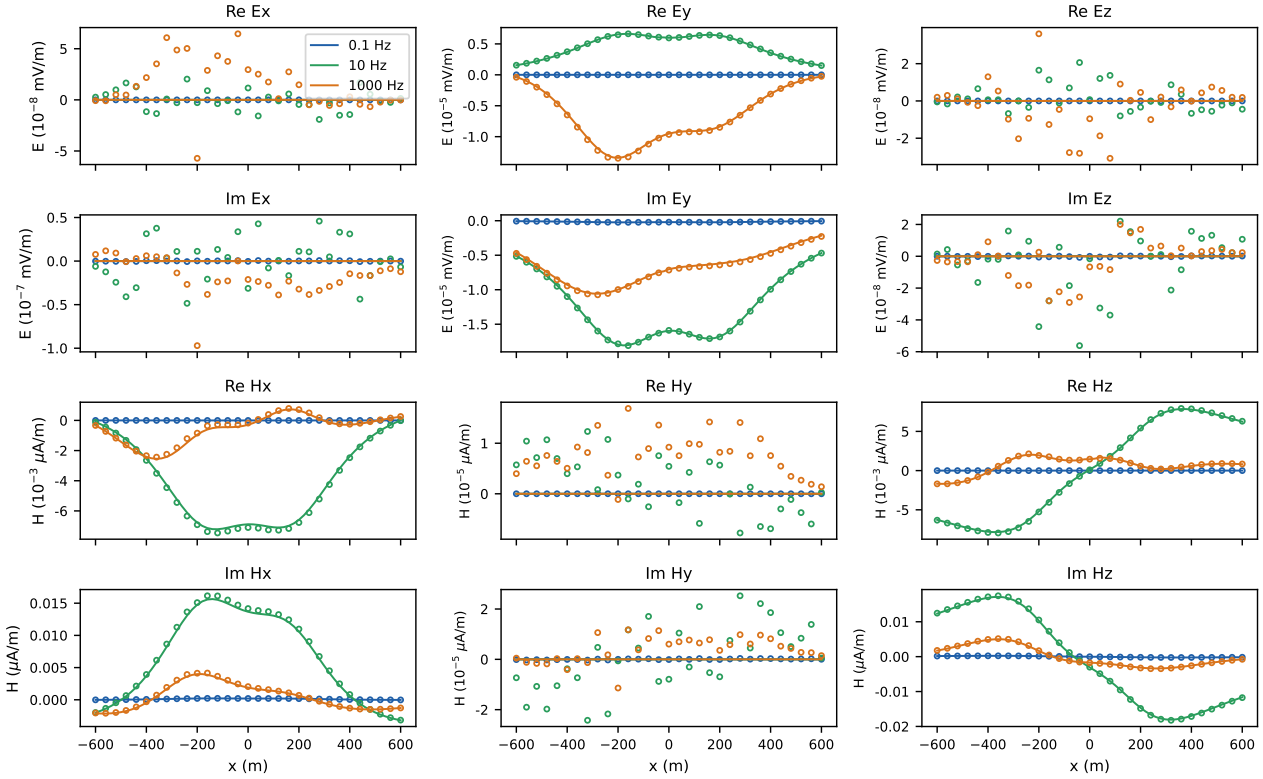


Figure 15: T3_loop_offset, x-profile, sphere anomaly $V(\text{sphere}) - V(\text{no sphere})$: series (lines) and elfe3D run with sphere minus reference run on the same mesh (circles).

6 T3_loop_offset, y-profile

elfe3D sphere comparison T3: square loop 400 x 400 m, centre (-1500, 0), counter-clockwise seen from above.

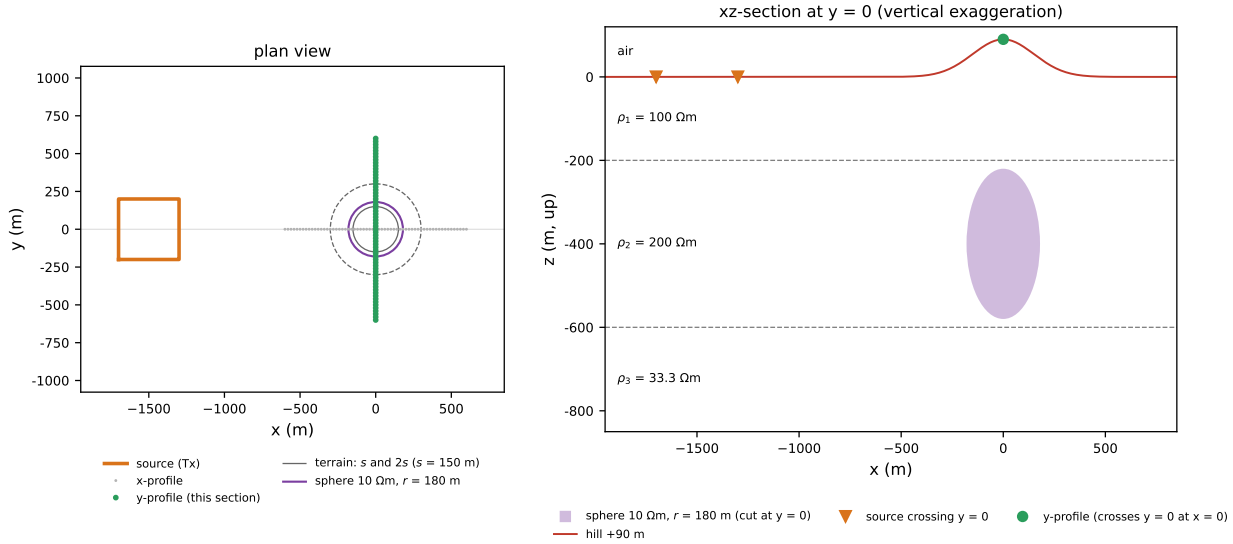


Figure 16: Setup of T3_loop_offset, y-profile, hill 90 m with sphere: plan view (left) and xz-section at $y = 0$ (right).

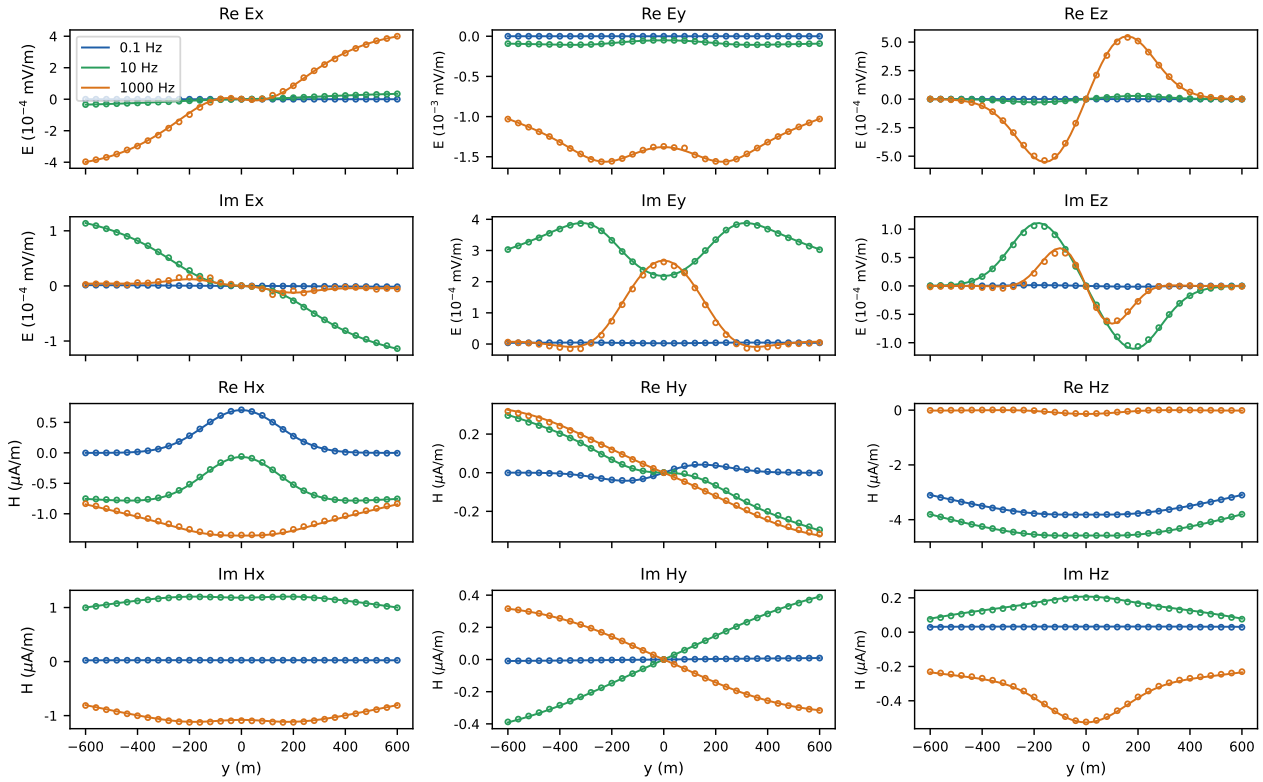


Figure 17: T3_loop_offset, y-profile, total field with sphere: series (lines) and elfe3D (circles, every second receiver).

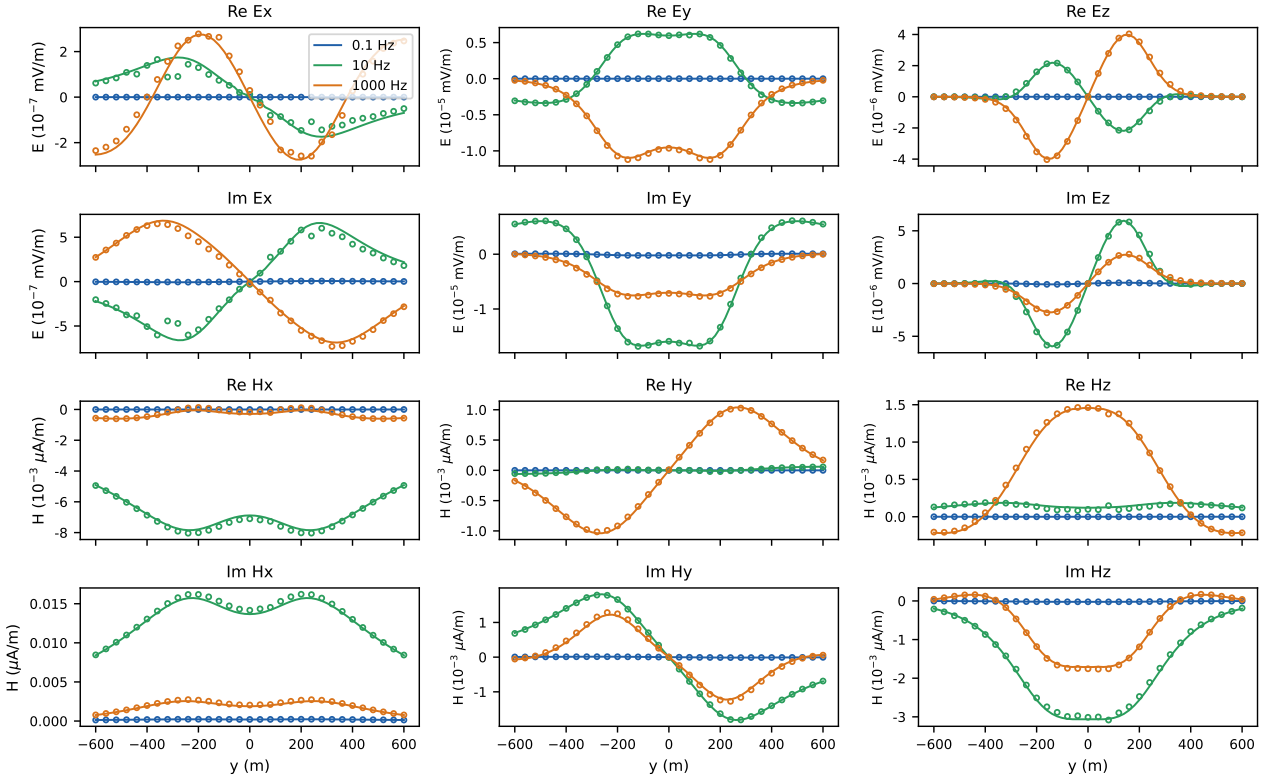


Figure 18: T3_loop_offset, y-profile, sphere anomaly $V(\text{sphere}) - V(\text{no sphere})$: series (lines) and elfe3D run with sphere minus reference run on the same mesh (circles).

7 T4_loop_large, x-profile

elfe3D sphere comparison T4: square loop 2.6 x 2.6 km around the hill, counter-clockwise seen from above.

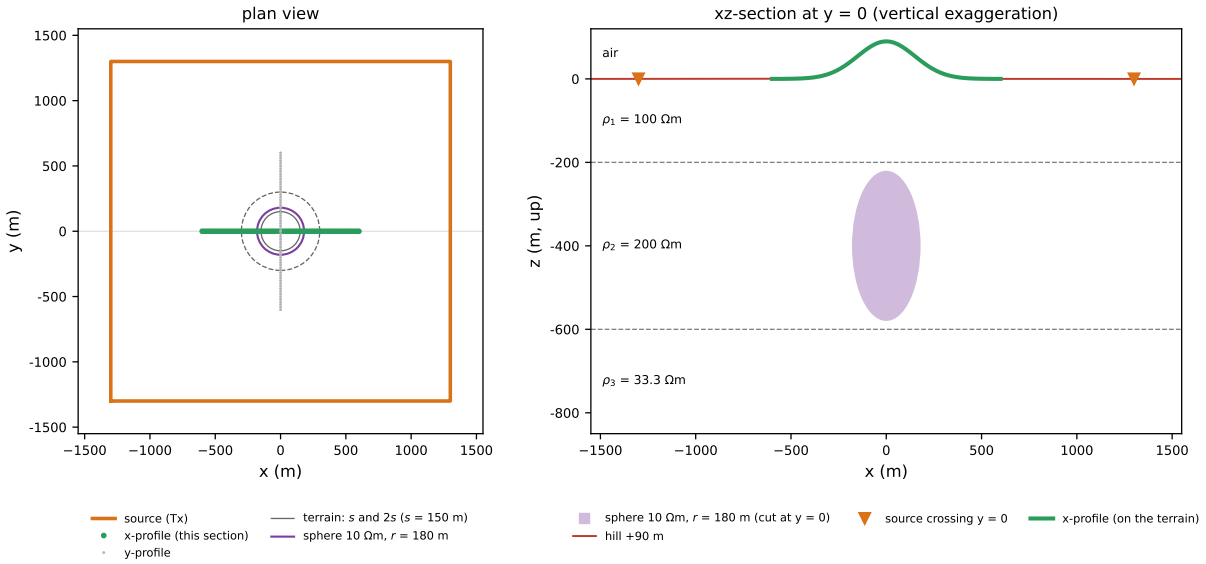


Figure 19: Setup of T4_loop_large, x-profile, hill 90 m with sphere: plan view (left) and xz-section at $y = 0$ (right).

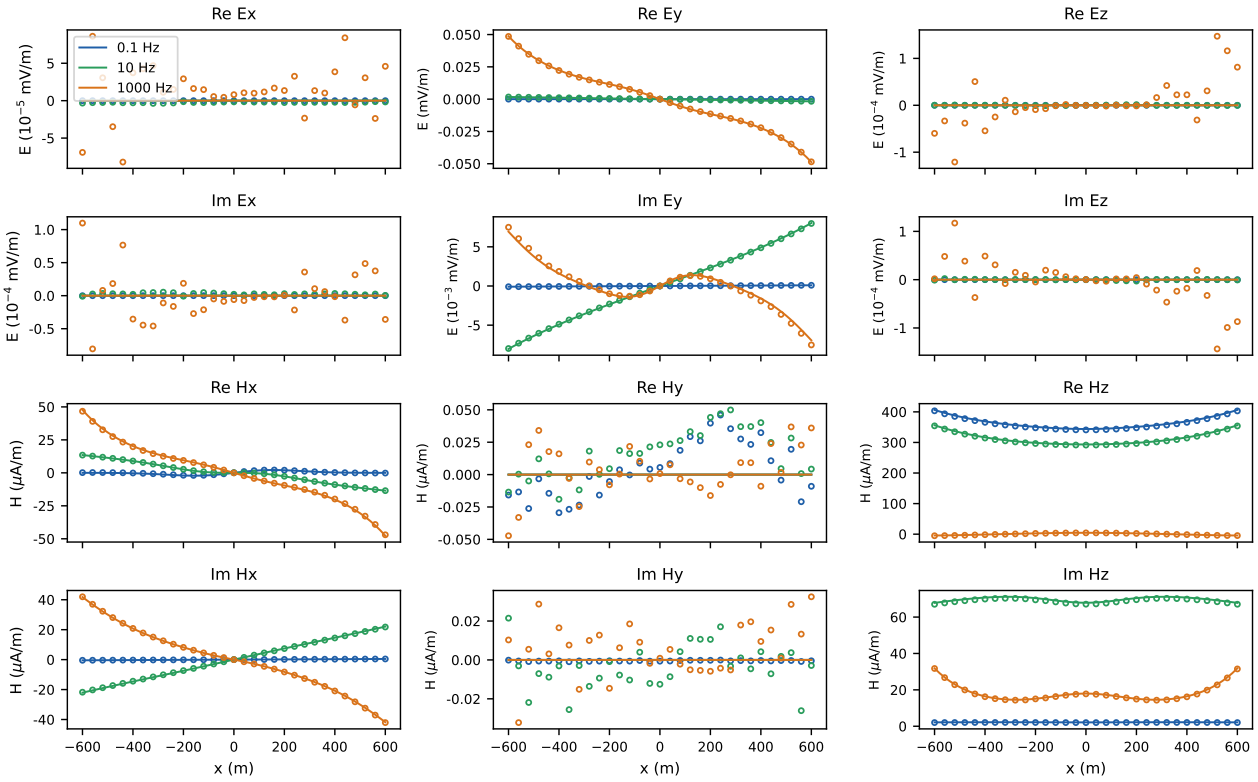


Figure 20: T4_loop_large, x-profile, total field with sphere: series (lines) and elfe3D (circles, every second receiver).

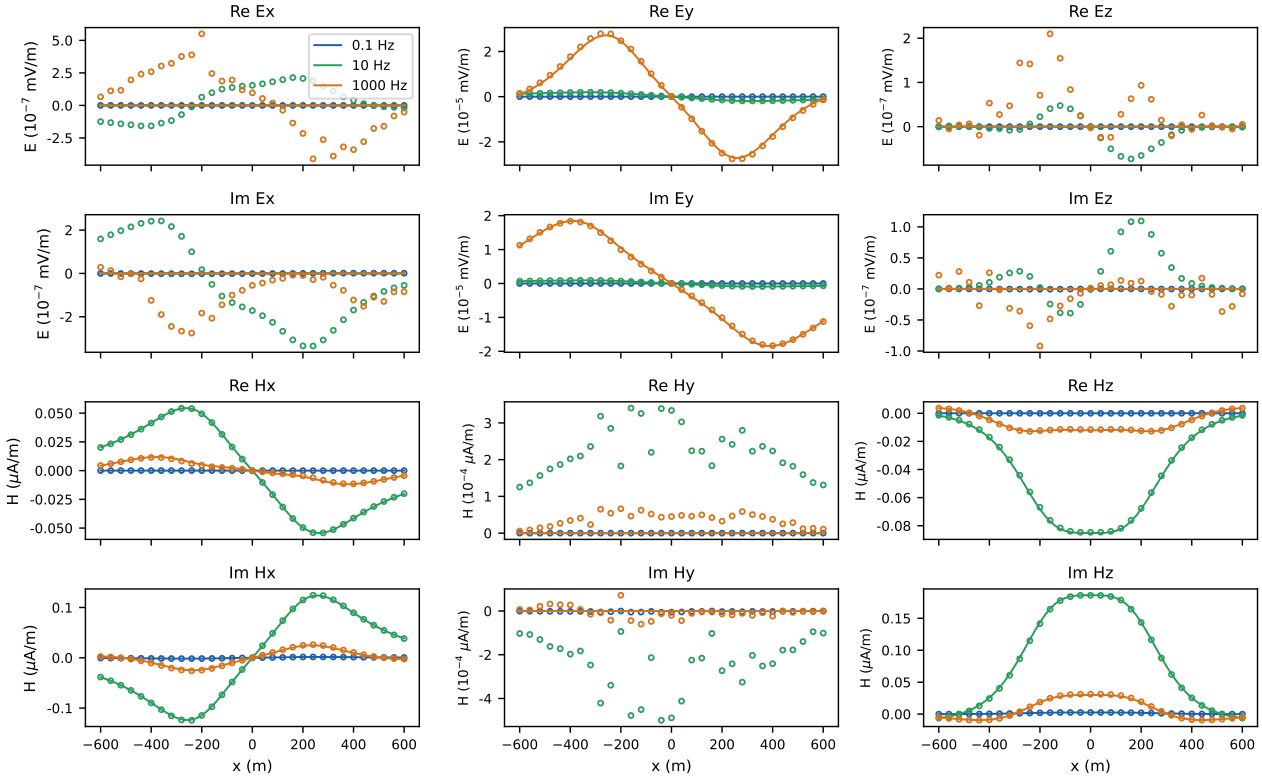


Figure 21: T4_loop_large, x-profile, sphere anomaly $V(\text{sphere}) - V(\text{no sphere})$: series (lines) and elfe3D run with sphere minus reference run on the same mesh (circles).

8 T4_loop_large, y-profile

elfe3D sphere comparison T4: square loop 2.6 x 2.6 km around the hill, counter-clockwise seen from above.

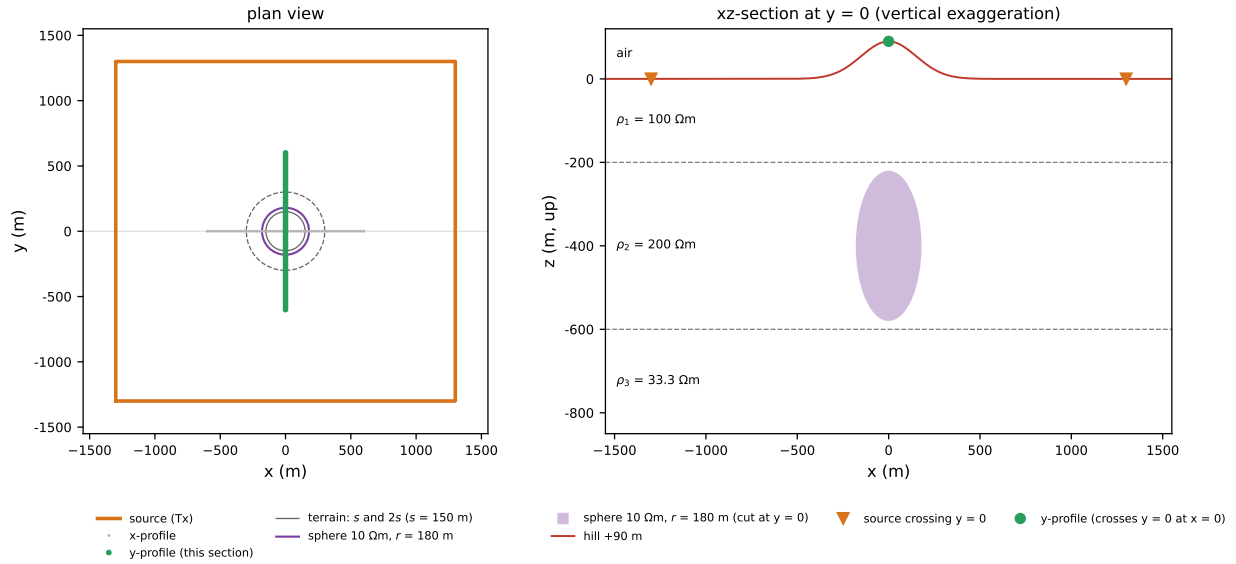


Figure 22: Setup of T4_loop_large, y-profile, hill 90 m with sphere: plan view (left) and xz-section at $y = 0$ (right).

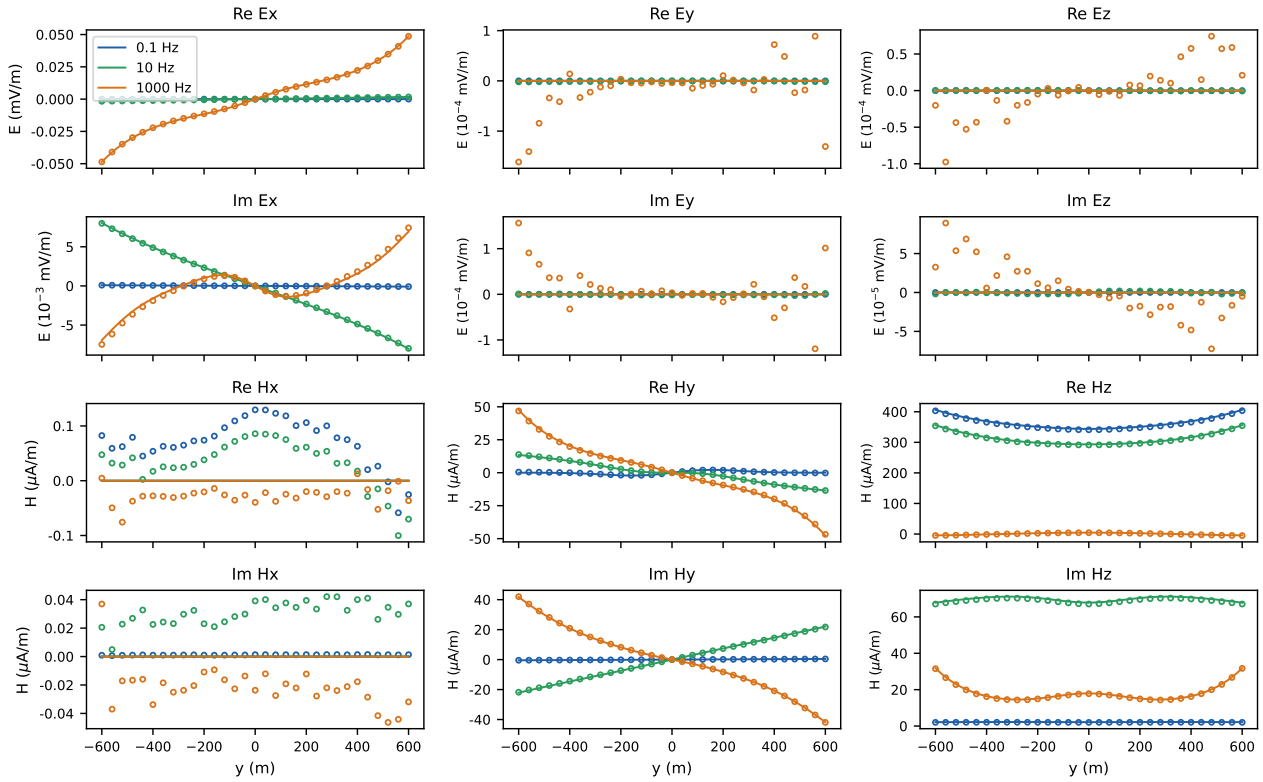


Figure 23: T4_loop_large, y-profile, total field with sphere: series (lines) and elfe3D (circles, every second receiver).

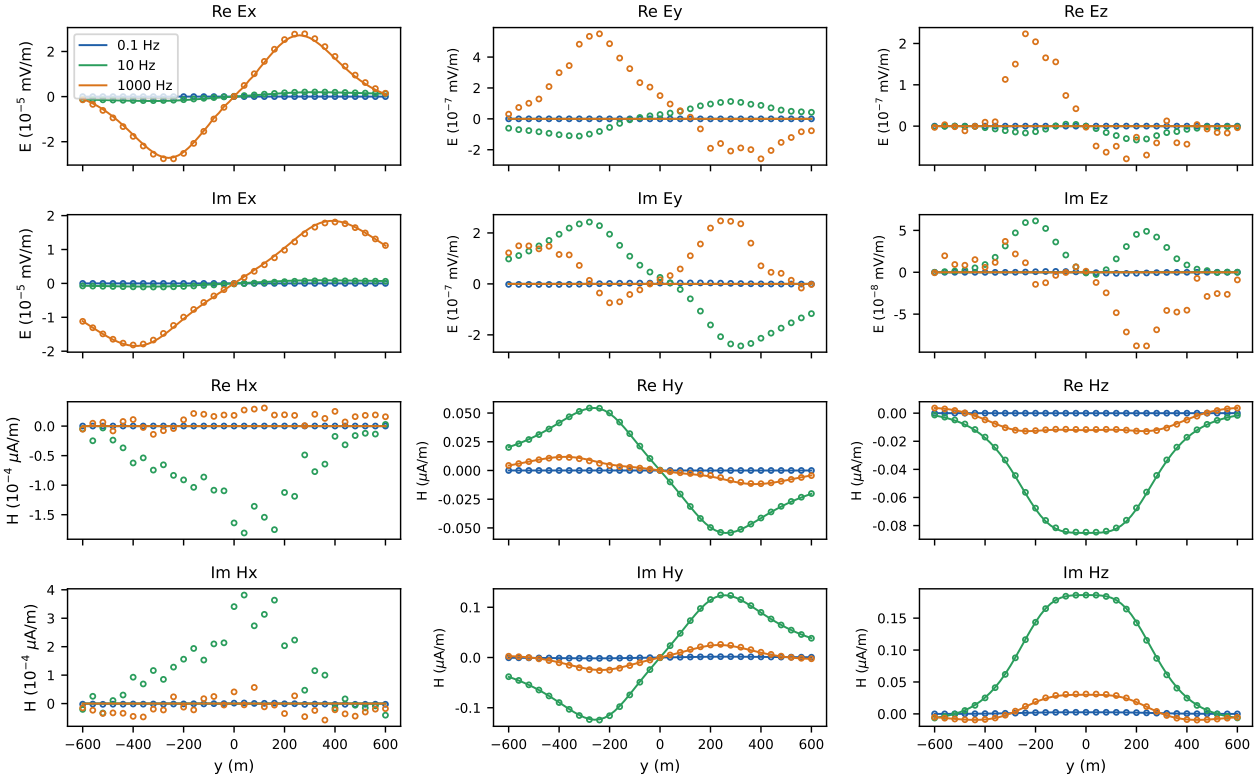


Figure 24: T4_loop_large, y-profile, sphere anomaly $V(\text{sphere}) - V(\text{no sphere})$: series (lines) and elfe3D run with sphere minus reference run on the same mesh (circles).

9 Deviations

Per component two columns, in per cent: total field $|V_{\mathrm{elfe}} - V_{\mathrm{series}}| / \max |V_{\mathrm{series}}|$ (median over both profiles), and anomaly $\max |A_{\mathrm{elfe}} - A_{\mathrm{series}}| / \max |A_{\mathrm{series}}|$ with the size of the anomaly $\max |A_{\mathrm{series}}| / \max |V_{\mathrm{series}}|$ in brackets (anomalies below 1 % of the field: size only; –: the component vanishes in the series, E_z of T4).

case	f	E_x		E_y		E_z		H_x		H_y		H_z	
T1_wire_inline	0.1	0.065	3.6 (1.6)	0.29	3 (23)	0.52	5.7 (5.6)	0.2	(0.06)	0.33	(0.42)	0.16	3.9 (3.7)
T1_wire_inline	10	0.077	3.5 (1.5)	0.37	2.9 (21)	0.51	5.7 (5.6)	0.18	(0.06)	0.16	(0.38)	0.39	4.1 (3.8)
T1_wire_inline	1000	0.24	(0.22)	0.71	(0.21)	0.42	(0.67)	0.59	(0.026)	0.22	(0.039)	1.8	5.4 (1.8)
T2_wire_broadside	0.1	0.22	(0.33)	0.13	(0.81)	0.68	5 (5.2)	0.31	(0.17)	0.19	(0.02)	0.23	(0.18)
T2_wire_broadside	10	0.25	(0.35)	0.092	2.9 (1)	0.48	5.1 (5.1)	0.043	(0.23)	0.22	(0.062)	0.14	(0.27)
T2_wire_broadside	1000	0.68	(0.065)	0.21	(0.22)	0.59	(0.83)	0.3	(0.036)	0.42	(0.032)	0.49	(0.31)
T3_loop_offset	0.1	0.34	(0.5)	0.2	3.1 (1.9)	0.29	6 (5.9)	0.62	(0.029)	2.7	(0.031)	0.05	(0.0013)
T3_loop_offset	10	0.35	(0.57)	0.16	3.1 (1.6)	0.31	6 (5.6)	0.31	(0.71)	0.68	(0.37)	0.035	(0.097)
T3_loop_offset	1000	0.66	(0.17)	0.053	(0.12)	0.51	(0.87)	0.099	(0.023)	0.44	(0.35)	0.14	(0.059)
T4_loop_large	0.1	0.053	(0.00029)	0.087	(0.00029)	–	(–)	4.2	(0.077)	1.5	(0.077)	0.75	(0.00062)
T4_loop_large	10	0.075	(0.026)	0.056	(0.026)	–	(–)	0.34	(0.52)	0.16	(0.52)	0.63	(0.056)
T4_loop_large	1000	0.18	(0.063)	0.2	(0.063)	–	(–)	0.093	(0.041)	0.07	(0.041)	1.2	(0.1)